
Mathematical Statistics

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

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Introduction

Probability begins with a law and deduces the behavior of the data. A coin is declared fair, and the theory returns the distribution of the number of heads in n tosses, the rate at which the proportion settles to $1/2$, and the Gaussian shape of the fluctuation around it. Statistics begins at the other end. A coin has been tossed 1000 times and has come up heads 612 times, and the question is what the coin is. No theorem of the previous volume answers it, because every one of them takes the law as input. Inference is that inverse problem, and the first thing to say about it is that it has no solution in the sense the reader is used to. The data are consistent with many laws, and they always will be.

The formal move is to replace a single probability measure by a family. A *statistical model* is a family $\{P_\theta: \theta \in \Theta\}$ of candidate laws on the space where the data lives, exactly one of which is asserted to have produced the observation, and the object of inference is the index θ . Everything from *Probability* [5] remains available and acquires that index: an expectation $\mathbb{E}_\theta$ for each candidate, a variance Var_θ , an almost-sure statement that has to be read separately at each θ . A procedure is therefore judged by a whole function of θ rather than by a number, and functions are only partially ordered. That partial order is why the subject carries a list of optimality criteria where a reader might expect one.

What that costs is visible in the smallest example. Let $X_1, \dots, X_n$ be independent $\mathcal{N}(\mu, 1)$ variables with μ unknown, and measure an estimator by its mean squared error. The sample mean $\bar{X}$ has mean squared error $1/n$, the same at every μ . The rule that ignores the data and reports 0 has mean squared error μ^2 , which is zero at $\mu = 0$ and smaller than $1/n$ throughout $|\mu| < 1/\sqrt{n}$. Neither rule is better than the other. No rule at all has smallest mean squared error at every μ , since for each μ_0 the constant rule $\delta \equiv \mu_0$ cannot be beaten at μ_0 . So the problem “find the estimator with the smallest error” has no solution, and looking harder will not produce one. Every optimality theory in this book is a repair of that failure, and the repairs come in two kinds. One restricts the competition to a class from which the constant rules are absent, which is what unbiasedness does for estimators and what unbiasedness and invariance do for tests. The other replaces the risk function by a single number, either its average against a prior on Θ or its worst value over Θ . Chapter 1 sets out both, and the rest of the book works out what they give for estimation and for testing.

A second idea has no counterpart in the previous volume either. A statistic is a function $T(X)$ of the data, and passing from X to $T(X)$ can only lose information in the sense of σ -algebras, since

$\sigma(T(X))$ is contained in $\sigma(X)$. Against a family, the loss can be exactly nothing. The statistic T is *sufficient* when the conditional law of X given T is the same measure for every θ , so that what has been discarded carries no trace of which candidate is in force. In the Gaussian model above, $\bar{X}$ is sufficient, and the n observations can be replaced by that one number without loss for any purpose the model can express. Sufficiency cannot be stated for a single measure. It is a property of the family, and chapter 2 develops it alongside the exponential families, in which a sufficient statistic of fixed dimension exists at every sample size.

The subject arose from the inverse problem in a concrete form: several imperfect measurements of one quantity, and the need to report a single value. Legendre published the method of least squares in 1805, which prompted Gauss to state that he had been using it in astronomical orbit-fitting since 1795. Between them they formulated the linear model in the parametric form ?? returns to, and what they were after was its unknown parameters.¹

Through the nineteenth century and into the twentieth, the standard route from data to a conclusion ran through a prior distribution on the unknown, under the name *inverse probability*. The theory in this book was built by people setting out to do without one: Fisher through the 1920s and 1930s, Neyman and Pearson from 1933, and Wald in 1950. Fisher arrived at sufficiency in 1920 and, in a paper of 1922, named it and stated the factorization criterion that section 2.1 proves. In 1934 he identified the one-parameter exponential families as the only regular families admitting a one-dimensional sufficient statistic, a result Darmois extended to higher dimensions the following year; section 2.4 develops the families themselves.²

The measure-theoretic form of these results came later, and it is the form used here. Neyman rediscovered the factorization criterion in 1935, and in 1949 Halmos and Savage proved it for an arbitrary dominated family by applying the Radon-Nikodym theorem, which is how section 2.1 states it. Rao in 1945 and Blackwell in 1947 showed that conditioning an estimator on a sufficient statistic does not increase its squared-error risk, and Lehmann and Scheffé in 1950 added the completeness that makes the resulting estimator unique. Testing acquired its optimality theory from Neyman and Pearson, who introduced the unbiasedness that ?? needs for two-sided problems, and whose 1933 paper worked out the optimal properties of the one- and two-sample normal-theory tests as its principal applications.³

What makes estimation and testing one subject rather than two is Wald's decision theory, set out in 1950 and adopted as the frame of chapter 1. An inference is a decision, a loss function prices a wrong one, and the risk is the expected loss under each candidate law. Wald introduced the minimax principle and least favorable distributions in close parallel with the theory of games, and admissibility along with them.

Two later results fix where this book stops. In 1956 Stein showed that the sample mean of a Gaussian vector in three or more dimensions is inadmissible, and ?? constructs an estimator whose risk is strictly smaller at every parameter value at once. From 1979 Efron's bootstrap replaced a sampling distribution that cannot be computed by one resampled from the data, which is where ?? leaves the reader.⁴ The frame has outlasted the period that produced it. The risk function of section 1.2 is the quantity statistical learning minimizes, and the estimator of ?? improves on the sample mean by

¹The priority dispute is recounted in Efron and Hastie, *Computer Age Statistical Inference* [6], Section 1.3, and the attribution of the linear model in Lehmann and Romano, *Testing Statistical Hypotheses* [12], Section 7.11.

²Lehmann and Casella, *Theory of Point Estimation* [11], Sections 1.10.2, 1.10.3 and 5.9.1.

³Lehmann and Casella, *Theory of Point Estimation* [11], Section 1.10.2, and Lehmann and Romano, *Testing Statistical Hypotheses* [12], Section 4.11.

⁴Lehmann and Casella, *Theory of Point Estimation* [11], Section 5.9.1, and Casella and Berger, *Statistical Inference* [3], Chapter 10.

pulling it toward a fixed point, which is what a penalized estimator does. Both are starting points in Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning* [7].

The book is written for a reader who has met probability at the level of *Probability* [5]: probability spaces, random variables and independence, expectation and the inequalities that go with it, the modes of convergence, the laws of large numbers, the central limit theorem, characteristic functions, and conditional expectation. No statistics is assumed. Those limit theorems are used here as stated, with their proofs carried by that volume, and densities of a dominated family rest on the Radon-Nikodym theorem of *Measure Theory* [4]. ?? additionally uses elementary linear algebra: orthogonal projection, rank, and quadratic forms.

Goal

Every volume in this series is titled *Everything You Need to Know About* its subject, and “everything” is contentious in mathematical statistics, where one department’s single core course is another’s two-year sequence. The bound set for this book is the following. By the end of it, the reader should be able to:

1. Take a question stated informally and set it up as a decision problem, naming the model, the action space, the loss and the risk, and say why no rule minimizes risk uniformly and which repair of that failure is in use.
2. Work through a model they are handed: find a sufficient statistic and decide whether it is complete, compute the Fisher information, construct the uniformly minimum variance unbiased estimator and the maximum-likelihood estimator, and say whether the information bound is attained and why.
3. Construct a test and a confidence set for a stated hypothesis, name what optimality the construction has and what it lacks, and pass between the two by inverting one into the other.
4. Read the standard graduate literature. Lehmann and Casella, *Theory of Point Estimation* [11], Lehmann and Romano, *Testing Statistical Hypotheses* [12], and van der Vaart, *Asymptotic Statistics* [14], should all be readable, and an asymptotic argument in the last of them should be followable closely enough to see which regularity condition each step uses.

Part I builds the apparatus. Chapter 1 fixes the statistical model, prices a wrong decision with a loss function, averages it into the risk, and reaches the impossibility above together with its two repairs, the Bayes average and the minimax worst case. Chapter 2 reduces the data: sufficiency and the factorization criterion, minimal sufficiency and ancillarity, completeness and Basu’s theorem, and the exponential families, the models in which the optimality theory of the next two Parts is exact.

Part II takes the first of the two inferential acts, producing a value offered as the parameter itself. Chapter 3 follows the decision-theoretic route as far as it goes: the bias-variance split of mean squared error, the Fisher information and the Cramér-Rao bound, and the Rao-Blackwell and Lehmann-Scheffé theorems, which produce the best unbiased estimator and say when it sits on the bound. ?? gives up exact optimality and maximizes the likelihood instead, a recipe that applies to every model and whose justification waits for the asymptotics. ?? puts a prior on the parameter and reports the posterior, which turns estimation into a computation rather than an optimization, and closes on the James-Stein estimator.

?? does the same for the second act, deciding between two statements about θ . ?? defines tests, the two error probabilities and power, proves the Neyman-Pearson lemma for a simple hypothesis against a simple alternative, extends it to one-sided problems through monotone likelihood ratios, and reaches two-sided problems through unbiasedness and invariance. ?? takes the generalized likelihood ratio as the general-purpose method, constructs confidence sets, and proves the duality that converts a family of tests into a confidence set and back. It ends on the p -value, its uniform null distribution, and the readings of it that the distribution does not support.

?? is where the limit theorems of *Probability* [5] do the work. Exact optimality is available only in special models, and in general the justification for the maximum-likelihood estimator and the likelihood-ratio test is a large-sample one. ?? proves consistency and $\sqrt{n}$ -asymptotic normality with the inverse Fisher information as asymptotic variance, both resting directly on the laws of large numbers and the central limit theorem, and develops the delta method that carries asymptotic normality through a smooth transformation. ?? does the same for testing, with the Wald, score and likelihood-ratio statistics, the chi-square limit of Wilks, and the chi-square tests of fit and of independence.

?? ends on the computable. ?? is the Gaussian linear model, where the sufficiency, estimation, testing and distribution theory of the whole book meet on one object: least squares read as an orthogonal projection, the Gauss-Markov theorem, and the t , F and chi-square laws coming out of Cochran's theorem. ?? steps past the parametric assumption with the empirical distribution and the Glivenko-Cantelli theorem, the plug-in principle, the bootstrap and kernel density estimation, and stops where statistical learning begins.

Two models run through all five Parts: the Gaussian location-scale model $\mathcal{N}(\mu, \sigma^2)$ and the Bernoulli model, the second as the first instance of an exponential family. Each is carried from the reduction by sufficiency through estimation, testing, asymptotics and the linear model, so that the successive layers of theory arrive on objects already familiar. Exercises close every section and are worked in full.

Notation and Conventions

The probabilistic notation follows *Probability* [5]. The conventions particular to this book are collected here.

- A *statistical model* is written $\{P_\theta : \theta \in \Theta\}$ on a measurable space $(\mathcal{X}, \mathcal{B})$, with X the data and x a realized value. Every probabilistic operator carries the candidate as a subscript: $\mathbb{P}_\theta$, $\mathbb{E}_\theta$, Var_θ , Cov_θ . Sampling n times from one candidate is written $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta$.
- A *decision rule* is δ , taking values in an action space $\mathcal{A}$. The *loss* is $L(\theta, a)$ and the *risk* is $R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))]$. The loss is a plain L and the likelihood is $\mathcal{L}$, and the two are kept distinct throughout on purpose.
- The *likelihood* is $\mathcal{L}(\theta) = \mathcal{L}(\theta; x)$, its logarithm $\ell(\theta) = \log \mathcal{L}(\theta)$, and the Fisher information $\mathcal{I}(\theta)$. An estimator of θ is $\hat{\theta}$, the sample mean is $\bar{X}$ and the sample variance S^2 .
- Modes of convergence use the arrows $\xrightarrow{p}$, $\xrightarrow{\text{a.s.}}$ and $\xrightarrow{d}$, for convergence in probability, almost surely, and in distribution. Independence is $\perp\!\!\!\perp$ and the conditioning bar is $|$.

- Standard laws use upright operator names: $\text{Ber}(p)$, $\text{Bin}(n, p)$, $\text{Poi}(\lambda)$, $\text{Unif}(a, b)$, $\text{Exp}(\lambda)$, $\text{Gamma}(\alpha, \beta)$ and $\text{Beta}(\alpha, \beta)$. The Gaussian law is $\mathcal{N}(\mu, \sigma^2)$, with $\mathcal{N}(0, 1)$ standard.
- Function declarations use $f: X \rightarrow Y$, and set-builder and conditioning both use the bar $|$. Structural divisions are referenced by name (Part I, and so on), never by a hardcoded numeral. The single hyphen is the only dash in the text.

Omitted Material

No book of this length covers everything that has been called mathematical statistics, and several subjects a reader might look for here are absent by decision. Each is named below with the reason and with somewhere to go.

1. *Statistical and high-dimensional learning*. Prediction rather than inference, and the regime in which the number of parameters grows with the sample size, need a volume of their own. What they take from here is the decision-theoretic frame of chapter 1. See Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning* [7], and Wainwright, *High-Dimensional Statistics* [16].
2. *Empirical process theory*. Uniform limit theorems over classes of functions, Donsker classes, and the modern treatment of M -estimation need weak convergence in metric spaces, which is past what *Probability* [5] carries. ?? uses the Glivenko-Cantelli theorem and nothing stronger. See van der Vaart and Wellner, *Weak Convergence and Empirical Processes* [15].
3. *Le Cam's asymptotic theory*. Local asymptotic normality, contiguity, limits of experiments, and the convolution and local asymptotic minimax theorems are the general form of what ?? states in a special case, and the subject needs its own book. See Le Cam, *Asymptotic Methods in Statistical Decision Theory* [2], and van der Vaart, *Asymptotic Statistics* [14], Chapters 7 to 9.
4. *Dependent data*. Every model in this book is one of independent observations, possibly with covariates. Dropping independence changes the limit theory and the models together, which is a subject of its own size. See Brockwell and Davis, *Time Series: Theory and Methods* [1].
5. *Causal inference*. Estimating the effect of an intervention needs a formalism for what an intervention is, and that formalism is not a parameter in a model of the observed data. See Imbens and Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences* [9].
6. *Computational Bayesian statistics*. ?? develops the Bayesian answer as a decision problem and stops before the algorithms that make a posterior computable outside the conjugate families. Markov chain Monte Carlo and its relatives are a computational subject. See Robert and Casella, *Monte Carlo Statistical Methods* [13].
7. *Robust estimation and permutation tests*. Influence functions, breakdown points, and inference that conditions on the observed sample have narrow application on a first pass through the theory, and each is a subject on its own. See Huber and Ronchetti, *Robust Statistics* [8], and Lehmann and Romano, *Testing Statistical Hypotheses* [12], Section 15.2.

Part I

Models and Sufficiency

This first Part constructs the statistical model and the two ideas that organize everything after it. Chapter 1 casts inference as a decision problem, in which a loss function prices a wrong answer and the resulting risk becomes the quantity every method is designed to make small. Chapter 2 then isolates the part of the data that carries the information about the unknown law, the *sufficient* statistic, and identifies the exponential families, where that reduction is exact at every sample size.

Statistical Models and the Decision-Theoretic Frame

Deciding which member of a family of probability laws produced the data is not yet a mathematical question. It becomes one once there is a statement of what a good answer is, and what a wrong one costs.

This chapter gives that statement in the language of decision theory. Section 1.1 fixes the statistical model, the sample drawn from it, and the statistics computed from the sample. Section 1.2 prices the cost of a wrong decision with a loss function and averages it into the risk, the function of the unknown parameter that every method in this book is designed to keep small. Section 1.3 confronts the problem that no decision rule has smallest risk everywhere at once, and names the two ways around it: restrict attention to a reasonable subclass of rules, or collapse the risk function to a single number. Section 1.4 develops the second route in its two forms, the Bayesian average of the risk against a prior and the minimax worst case, and shows how the two are related.

1.1 Statistical Models and Statistics

A measure is no longer given. It is one of a family of candidates, and the data is what is used for choosing among them. That family is the object this section makes precise. Before any estimator, test, or confidence interval can be discussed, one must say what a statistical problem *is*: what the data are, what laws are in contention, and what functions of the data are available to reason with. These are the statistical model (the sample, and the statistic).

Definition 1.1.1: Statistical Model

A *statistical model* (or *statistical experiment*) consists of the following data:

1. A measurable space $(\mathcal{X}, \mathcal{B})$, the *sample space*, whose points are the possible values of the observed data. The data is a random variable X taking values in $\mathcal{X}$; a *realized value* is written $x \in \mathcal{X}$.
2. A family

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

of probability measures on $(\mathcal{X}, \mathcal{B})$, indexed by a *parameter* θ ranging over a *parameter space* Θ . Each P_θ is a candidate law for X : the model asserts that $X \sim P_\theta$ for exactly one, unknown, $\theta \in \Theta$.

We shall often say “fix a statistical model $\{P_\theta : \theta \in \Theta\}$ on $(\mathcal{X}, \mathcal{B})$ ”.

The model is *parametric* if Θ can be taken to be a subset of a finite-dimensional Euclidean space, $\Theta \subseteq \mathbb{R}^k$ for some $k < \infty$, and *nonparametric* otherwise^a

^athe index set is infinite-dimensional, for instance the set of all densities, or all continuous distribution functions, on the line.

Three features of the definition deserve emphasis. First, the model gives a whole family $\mathcal{P}$, and the point of inference is that the true index θ is unknown. The subscript on P_θ (and on the associated expectation $\mathbb{E}_\theta$ and probability $\mathbb{P}_\theta$) is the bookkeeping that keeps track of which candidate one is reasoning about. The second is that the model lives on the sample space $\mathcal{X}$, which is distinct from the underlying probability space Ω . One may always take $\Omega = \mathcal{X}$ with X the identity map, so that the sample space *is* the probability space and the data *is* the outcome. This is the canonical construction, and it is why a model can be specified by naming $\mathcal{X}$ and the family of laws alone. The third is the parametric/nonparametric divide: it is a divide in the finiteness of the description, and every parametric chapter of this book takes $\Theta \subseteq \mathbb{R}^k$, with the infinite-dimensional case postponed to the closing nonparametrics chapter (??).

The distinction between a parameter and the objects computed from data is the organizing distinction of the subject. A *parameter* is a feature of the unknown law, a function of θ . It is not observed and cannot be evaluated from the data, because it depends on the unknown P_θ . Everything the reader will learn to construct is a function of X . Keeping straight what depends on θ (and so is unknown) versus what depends on X (and so is computable) is important, since the two are easily confounded.

Before turning to what one computes from the data, it is important to determine whether the label θ is unambiguous. The model attaches a parameter to each law, but nothing so far forbids two different labels from naming the same law. When that happens the data cannot distinguish the two:

Definition 1.1.2: Identifiability

The parametrization $\theta \mapsto P_\theta$ of a statistical model (definition 1.1.1) is *identifiable* if the map is injective:

$$P_{\theta_1} = P_{\theta_2} \implies \theta_1 = \theta_2 \quad \text{for all } \theta_1, \theta_2 \in \Theta$$

Equivalently, distinct parameter values induce distinct laws for the data: $\theta_1 \neq \theta_2$ implies $P_{\theta_1} \neq P_{\theta_2}$.

Identifiability is the minimal sanity condition on a parametrization. If it fails, two parameter values $\theta_1 \neq \theta_2$ give the same law $P_{\theta_1} = P_{\theta_2}$, so X has the same distribution under both. By contrast, when identifiability holds, the law determines the parameter and the map $\theta \mapsto P_\theta$ may in principle be inverted, which is the ground on which estimation stands. A failure is easy to arrange:

Example 1.1.3: A Non-Identifiable Parametrization

Take the data to be a single observation $X \sim \mathcal{N}(\mu, \sigma^2)$ on $\mathcal{X} = \mathbb{R}$, but parametrize the family by $\theta = (\alpha, \beta) \in \mathbb{R}^2 = \Theta$ through the mean $\mu = \alpha + \beta$, with $\sigma^2 = 1$ fixed. Then

$$P_{(\alpha, \beta)} = \mathcal{N}(\alpha + \beta, 1)$$

depends on (α, β) only through the sum $\alpha + \beta$. The two parameter values $(\alpha, \beta) = (0, 1)$ and $(\alpha, \beta) = (1, 0)$ give the same law $\mathcal{N}(1, 1)$, so the parametrization is not identifiable: the data can report the value of $\alpha + \beta$ but can never split it into its summands. The remedy is to reparametrize by the identifiable quantity $\mu = \alpha + \beta$ itself, collapsing the redundant coordinate.

The same phenomenon appears without a contrived reparametrization. A *two-component mixture*

$$P_{(p, \mu_1, \mu_2)} = p\mathcal{N}(\mu_1, 1) + (1 - p)\mathcal{N}(\mu_2, 1)$$

with $p \in (0, 1)$ and $\mu_1, \mu_2 \in \mathbb{R}$ is unchanged under the relabelling $(p, \mu_1, \mu_2) \mapsto (1 - p, \mu_2, \mu_1)$ that swaps the two components: the two parameter triples name one and the same law. This *label switching* is a failure of identifiability, irreparable without a convention, in mixture models, resolved in practice by imposing a strict ordering such as $\mu_1 < \mu_2$ to pick one representative from each swap-pair with $\mu_1 \neq \mu_2$. The inequality must be strict: on the diagonal $\mu_1 = \mu_2 = \mu$ the triples (p, μ, μ) all name the single law $\mathcal{N}(\mu, 1)$, and no ordering convention can separate them.

The model as defined governs a single observation X . In almost every application there are many observations gathered under conditions arranged to be identical and non-interfering, and the mathematical idealization of “identical and non-interfering” is that the observations are independent and identically distributed (iid):

Definition 1.1.4: Random Sample and the Sampling Model

Fix a statistical model $\{P_\theta: \theta \in \Theta\}$ on $(\mathcal{X}_1, \mathcal{B}_1)$ for a single observation. A *random sample of size n* (or *sample size n*) from P_θ is a vector

$$X = (X_1, \dots, X_n), \quad X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta$$

of independent and identically distributed random variables, each with law P_θ . The data then lives on the product sample space $\mathcal{X} = \mathcal{X}_1^n$ with the product σ -algebra $\mathcal{B} = \mathcal{B}_1^{\otimes n}$, and its law is the n -fold product measure [4]

$$P_\theta^{\otimes n} = \underbrace{P_\theta \otimes \dots \otimes P_\theta}_n$$

the unique measure on $(\mathcal{X}, \mathcal{B})$ under which the coordinates are independent with common law P_θ [5, Independence as a Product Law]. The family $\{P_\theta^{\otimes n}: \theta \in \Theta\}$ is again a statistical model in the sense of definition 1.1.1, the *sampling model* or *n -fold i.i.d. model*, now for the data $X = (X_1, \dots, X_n)$.

Independence makes the joint law factor,

$$P_\theta^{\otimes n}(A_1 \times \dots \times A_n) = \prod_{i=1}^n P_\theta(A_i)$$

so that when P_θ has a density p_θ with respect to a common dominating measure the joint density is the product $\prod_{i=1}^n p_\theta(x_i)$, and this product is the *likelihood*, arguably the most-used object in the book. Identical distribution means every observation carries information about the same θ , so accumulating observations accumulates evidence, and it is precisely because the X_i are i.i.d. that the law of large numbers and the central limit theorem of *Probability* [5] apply to their averages, delivering the large-sample behavior that justifies the methods when exact answers are out of reach.

As the parameter is unknown and unobservable, what one holds and manipulates is the data, and every quantity computed from the data is a *statistic*:

Definition 1.1.5: Statistic

Let X be the data of a statistical model on $(\mathcal{X}, \mathcal{B})$. A *statistic* is a measurable map

$$T: (\mathcal{X}, \mathcal{B}) \longrightarrow (\mathcal{T}, \mathcal{C})$$

into a measurable space $(\mathcal{T}, \mathcal{C})$, whose definition does *not* depend on the parameter θ . Its value $T = T(X)$ is a random variable computable from the data.

The one requirement that gives the definition its content is the clause that T must not depend on θ : a statistic is something evaluable. A parameter such as the mean $\mathbb{E}_\theta[X_1]$ is a function of θ and cannot be evaluated without knowing the law, whereas the sample mean $\bar{X} = n^{-1} \sum_i X_i$ is a function of X and can be evaluated. The two are constantly paired, a statistic computed from the data to stand in for a parameter it estimates, and the entire theory of estimation is the study of how well the computable side approximates the unknown side. As a random variable, a statistic $T(X)$ has a distribution, and that distribution does depend on θ (it is the pushforward of P_θ under T), which

is what makes a statistic informative about θ even though its rule of computation is θ -free. That distribution is called the *sampling distribution* of T , and computing it is a recurring task.

Example 1.1.6: The Sample Mean, Sample Variance, and Order Statistics

The three statistics met most often are the sample mean, the sample variance, and the order statistics. Each summarizes the sample in a different way, and each is a statistic in the precise sense just defined. Let $X = (X_1, \dots, X_n)$ be a random sample (definition 1.1.4) with the X_i real-valued.

1. The sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is a statistic: its computation involves only the data. When the common law has finite mean $\mathbb{E}_\theta[X_1] = \mu(\theta)$ and variance $\text{Var}_\theta(X_1) = \sigma^2(\theta)$, linearity of expectation and independence give

$$\mathbb{E}_\theta[\bar{X}] = \mu(\theta), \quad \text{Var}_\theta(\bar{X}) = \frac{\sigma^2(\theta)}{n}$$

The mean of $\bar{X}$ is the population mean $\mu(\theta)$, a parameter, and the two are named by the same letter in elementary courses precisely because $\bar{X}$ is built to estimate $\mu(\theta)$, but they sit on opposite sides of the parameter/statistic line.

2. The sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is a statistic, and the divisor $n-1$ (rather than n) is chosen so that $\mathbb{E}_\theta[S^2] = \sigma^2(\theta)$ exactly, for every θ with finite variance. A short computation records why the correction is needed: expanding the square and using $\mathbb{E}_\theta[\sum_i (X_i - \bar{X})^2] = \sum_i \text{Var}_\theta(X_i) - n \text{Var}_\theta(\bar{X}) = n\sigma^2(\theta) - \sigma^2(\theta) = (n-1)\sigma^2(\theta)$, so dividing by $n-1$ (not n) is what makes $\mathbb{E}_\theta[S^2] = \sigma^2(\theta)$. The divisor- n version has expectation $\frac{n-1}{n}\sigma^2(\theta)$, understating the variance.

3. The order statistics $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ are the sample values sorted into increasing order, and each $X_{(k)}$ is a statistic. The extremes $X_{(1)} = \min_i X_i$ and $X_{(n)} = \max_i X_i$ and the median (the middle order statistic, or the average of the two middle ones when n is even) are the most-used, and the sample maximum will be the natural statistic for the uniform model below.

Each of these summaries throws away information: $\bar{X}$ collapses the whole sample to a single number, $(\bar{X}, S^2)$ to two, and the full order-statistic vector $(X_{(1)}, \dots, X_{(n)})$ discards the labels that say which observation was which. Which summaries lose nothing about θ , and which lose everything but the irrelevant information, is the subject of sufficiency in chapter 2. For now the point is only that these are statistics.

The models that recur the rest of the book are the following three, illustrating sufficiency, estimation, testing, and asymptotics in turn. Two are the standing running examples of the volume, the Bernoulli model and the Gaussian location-scale model. The third, the uniform model, is the standing *irregular* foil, the example on which the smooth theory of the other two breaks down:

Example 1.1.7: The Three Running Models

Each of the following specifies a sample space, a parameter space, and a natural statistic. Each is presented for a single observation, with the i.i.d. sample model $P_\theta^{\otimes n}$ (definition 1.1.4) understood.

1. *The Bernoulli model* $\text{Ber}(p)$. The sample space is $\mathcal{X}_1 = \{0, 1\}$, one bit recording a success or failure, and the parameter is the success probability $p \in \Theta = [0, 1]$, with $P_p(\{1\}) = p$ and $P_p(\{0\}) = 1 - p$. It is the model for coin flips, defective/non-defective items, and yes/no responses. For a sample of size n the natural statistic is the total number of successes $T = \sum_{i=1}^n X_i$, equivalently (up to the factor n) the sample mean $\bar{X}$, whose law is $\text{Bin}(n, p)$, so $\mathbb{E}_p[T] = np$ and $\text{Var}_p(T) = np(1 - p)$. The parametrization $p \mapsto \text{Ber}(p)$ is identifiable, since $p = P_p(\{1\})$ is recovered from the law.
2. *The Gaussian location-scale model* $\mathcal{N}(\mu, \sigma^2)$. The sample space is $\mathcal{X}_1 = \mathbb{R}$, and the parameter is $\theta = (\mu, \sigma^2) \in \Theta = \mathbb{R} \times (0, \infty)$, a location μ and a scale σ^2 , and the density with respect to Lebesgue measure is

$$p_{\mu, \sigma^2}(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The natural statistic for a sample is the pair $(\bar{X}, S^2)$ of sample mean and sample variance, which will turn out to summarize the sample without loss. The parametrization is identifiable, since μ and σ^2 are the mean and variance of the law. This is the basic model of the whole subject: the central limit theorem makes it the limiting model for averages, and the linear model of ?? is Gaussian throughout.

3. *The uniform model* $\text{Unif}(0, \theta)$. The sample space is $\mathcal{X}_1 = (0, \infty)$ (each observation falls in $(0, \theta)$, but the *support* moves with the parameter), and the parameter is the upper endpoint $\theta \in \Theta = (0, \infty)$, and the density is $p_\theta(x) = \theta^{-1} \mathbf{1}\{0 < x < \theta\}$. The natural statistic for a sample is the sample maximum $X_{(n)} = \max_i X_i$, which cannot exceed θ and creeps up toward it as n grows. The parametrization is identifiable, since $\theta = \sup\{x : p_\theta(x) > 0\}$ is the right endpoint of the support. This model is the standing foil: because its support depends on θ , the smoothness assumptions behind the Cramér-Rao bound and the asymptotic theory fail, and the maximum $X_{(n)}$, not the mean, is the statistic that matters.

Note that the Bernoulli model is discrete and one-dimensional, the simplest place to see every idea first; the Gaussian model is continuous and two-dimensional, smooth in its parameter and the model in which the exact distribution theory of the linear model is available in closed form; the uniform model is continuous but *irregular*, its support hinging on θ , and it is kept in view to demonstrate the regularity hypotheses of the general theory.

Exercise 1.1.1

1. Consider the model in which $X \sim \text{Poi}(\lambda)$ on $\mathcal{X} = \{0, 1, 2, \dots\}$, parametrized by $\lambda \in \Theta = (0, \infty)$. Show that this parametrization is identifiable. Then exhibit a reparametrization of the same family of laws that is *not* identifiable, and say which quantity the data can and cannot recover.
2. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ and let $T = \sum_{i=1}^n X_i$ be the number of successes. Working from the product law $P_p^{\otimes n}$, show that for each binary vector $x = (x_1, \dots, x_n) \in \{0, 1\}^n$ with $\sum_i x_i = t$ one has $P_p^{\otimes n}(\{x\}) = p^t(1 - p)^{n-t}$, and deduce that $T \sim \text{Bin}(n, p)$.

3. Prove the identity used in example 1.1.6(2): for a random sample with finite variance σ^2 ,

$$\mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = (n-1)\sigma^2$$

Conclude that $\mathbb{E}[S^2] = \sigma^2$ and that the divisor- n estimator $\hat{\sigma}^2 = n^{-1} \sum_i (X_i - \bar{X})^2$ has $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$.

4. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and $M = X_{(n)} = \max_i X_i$. Compute the distribution function $P_\theta(M \leq t)$ for $0 \leq t \leq \theta$, differentiate to obtain the density of M , and evaluate $\mathbb{E}_\theta[M]$. Is M larger or smaller than θ on average, and by how much?
5. Decide, for each of the following functions of a random sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, whether it is a statistic in the sense of definition 1.1.5, and justify: (i) $\bar{X}$; (ii) $(\bar{X} - \mu)/\sigma$; (iii) S^2 ; (iv) $\sum_i (X_i - \bar{X})^2$; (v) $X_{(1)}$.
6. In the two-component mixture of example 1.1.3, prove that the map $(p, \mu_1, \mu_2) \mapsto (1-p, \mu_2, \mu_1)$ leaves the law of X unchanged, and explain why restricting the parameter space to $\{(p, \mu_1, \mu_2) : \mu_1 \leq \mu_2\}$ restores identifiability (assuming $\mu_1 \neq \mu_2$).

1.2 Loss, Risk, and the Risk Function

The previous section fixed what a statistical problem consists of: data X whose law P_θ is one of a family $\{P_\theta : \theta \in \Theta\}$ (definition 1.1.1), and statistics $T(X)$ computed from the data without reference to the unknown θ (definition 1.1.5). What it did not fix is what one is trying to *do* with the data. Estimate θ ? Estimate some function of θ ? Decide between two hypotheses about it? Each of these is a different task, and until the task is named, there is no way to say that one procedure is better than another.

Every such task is a decision made under a cost. One commits to an action, the true state of nature θ then determines how much that action costs, and because θ is unknown the cost is a random variable whose average is the figure of merit. This section builds that frame in three steps: a loss function that prices actions against states, a decision rule that turns data into an action, and the risk that averages the loss the rule incurs. The risk is a function of θ , not a single number, and the fact that it rarely orders two rules completely is the tension that drives the rest of the chapter.

The starting point is to name the set of available actions and to price each one against each possible state of nature. For estimating a real parameter the actions are the candidate values, and the price is how far the guess lands from the truth, measured by some rule chosen in advance.

Definition 1.2.1: Action Space and Loss Function

A *decision problem* for the statistical model $\{P_\theta : \theta \in \Theta\}$ (definition 1.1.1) consists of an *action space* $\mathcal{A}$, the set of decisions available to the statistician, together with a *loss function*

$$L: \Theta \times \mathcal{A} \rightarrow [0, \infty), \quad (\theta, a) \mapsto L(\theta, a)$$

with $L(\theta, a) \geq 0$ for all θ and a . The value $L(\theta, a)$ is the cost incurred by taking action a when the true state of nature is θ . Throughout, Θ and $\mathcal{A}$ are Borel subsets of Euclidean spaces, each carrying its Borel σ -algebra, and L is jointly measurable in (θ, a) .

In an *estimation problem* for a real parameter one takes $\mathcal{A} = \Theta \subseteq \mathbb{R}$ (the actions are the candidate values of θ). When Θ is a proper subset of $\mathbb{R}$ it is often convenient to allow a larger action space, $\mathcal{A} = \mathbb{R}$ or $\mathcal{A} = \overline{\Theta}$, so that rules which can overshoot the parameter set are still competitors and can be compared with those that cannot; example 1.3.4 turns on exactly that comparison. The three standard losses are the *squared-error loss*

$$L(\theta, a) = (\theta - a)^2$$

the *absolute-error loss* $L(\theta, a) = |\theta - a|$, and, for a problem where an estimate counts only as right or wrong, the *0-1 loss* $L(\theta, a) = \mathbb{1}\{a \neq \theta\}$.

The loss function encodes the goal of the analysis as sharply as the model encodes its probabilistic content. Nonnegativity is a convention, not a restriction: a loss is a penalty for being wrong, calibrated so that a perfect action costs nothing. For estimation the three losses of the definition sit on a spectrum of how harshly large errors are punished. Squared error grows quadratically, so a single gross miss dominates the cost, and its expectation, worked out below, is the mean squared error that runs through all of estimation theory. Absolute error grows only linearly and so tolerates the occasional large deviation, which makes it the loss under which the median rather than the mean is optimal (a computation carried out in theorem 1.4.4 below). The 0-1 loss ignores the size of an error entirely and records only its occurrence. It is the natural loss for a discrete parameter and, transported to the testing problem, becomes the accounting of Type I and Type II errors of ???. The choice among them is part of the modeling, made before any data are seen.

The choice of loss is not confined to estimation. A testing problem, developed in ???, has action space $\mathcal{A} = \{0, 1\}$, the two verdicts “accept the null” and “reject the null”, and a loss that charges for each kind of wrong verdict. A problem may also price errors asymmetrically, when underestimating a quantity costs more than overestimating it or the reverse. The frame accommodates all of these, and the estimation losses are the instances the reader will meet most often, and they are the ones carried through this section.

Example 1.2.2: The Three Estimation Losses on a Bernoulli Parameter

Take the Bernoulli model $\text{Ber}(p)$ of example 1.1.7, so $\Theta = [0, 1]$, and consider estimating the success probability p , with action space $\mathcal{A} = [0, 1]$. Suppose the truth is $p = 0.5$ and two analysts report the guesses $a_1 = 0.6$ and $a_2 = 0.9$.

Under squared-error loss the costs are

$$L(0.5, a_1) = (0.5 - 0.6)^2 = 0.01, \quad L(0.5, a_2) = (0.5 - 0.9)^2 = 0.16$$

so the second guess, four times farther from the truth, is penalized sixteenfold: quadratic loss punishes the gross miss disproportionately. Under absolute-error loss the costs are $|0.5 - 0.6| = 0.1$

and $|0.5 - 0.9| = 0.4$, a ratio of four, matching the ratio of the errors themselves. Under 0-1 loss both guesses cost 1, since neither equals 0.5 exactly, and the loss cannot distinguish a near miss from a wild one. The squared-error and absolute losses agree on the ordering of the two actions and disagree sharply on how much worse the second is, while the 0-1 loss ties them; that disagreement is exactly what selecting a loss function decides in advance.

A loss prices an action, but a statistician does not choose an action directly, since the data choose it. The bridge is a rule that reads the data and returns an action, and naming that rule is the second ingredient of the frame.

Definition 1.2.3: Decision Rule

For a decision problem with model $\{P_\theta : \theta \in \Theta\}$, sample space $\mathcal{X}$, and action space $\mathcal{A}$, a *nonrandomized decision rule* is a measurable map

$$\delta: \mathcal{X} \rightarrow \mathcal{A}$$

assigning to each possible data value $x \in \mathcal{X}$ the action $\delta(x)$ to be taken when $X = x$ is observed. In an estimation problem, where $\mathcal{A} = \Theta$, a decision rule is called an *estimator* and is often written $\hat{\theta} = \hat{\theta}(X) = \delta(X)$.

A *randomized decision rule* allows the action to depend on an auxiliary randomization: for each x it specifies a probability distribution $\delta(\cdot|x)$ on $\mathcal{A}$, and the action is drawn from $\delta(\cdot|x)$ after $X = x$ is observed. A nonrandomized rule is the special case where $\delta(\cdot|x)$ is the point mass at $\delta(x)$.

A decision rule is a complete plan of action: it says what to do for every data value that might arise, decided before the data are in hand. An estimator is the special case where the action taken is itself a guess at the parameter, so the sample mean $\bar{X}$, the sample median, and the maximum $\max_i X_i$ are all estimators, three different rules δ competing to solve the same estimation problem. Randomized rules enlarge the class by permitting a coin flip to break ties. They are a technical convenience, indispensable for the clean statement of the optimal-testing theory in ??, where a randomized test achieves a prescribed error rate exactly. In estimation under a convex loss on an interval action space they give nothing: averaging over the auxiliary randomization never increases the risk defined below, so an optimal rule can always be taken nonrandomized. For this reason the development here works with nonrandomized rules, reintroducing randomization only when the testing theory needs it.

Because a decision rule δ acts on the random data X , the action $\delta(X)$ is itself random, and so is the loss $L(\theta, \delta(X))$ it incurs. For a fixed rule this loss is a single number only after the data are seen; before the data are seen it is a random variable, and its distribution depends on which θ governs X . A rule cannot be judged by the loss at one realization, which is unknowable in advance and varies from sample to sample. The figure of merit is the average loss, and averaging the random loss under P_θ produces the central object of the chapter.

Definition 1.2.4: Risk Function

Let δ be a decision rule for the model $\{P_\theta : \theta \in \Theta\}$ under loss L . The *risk* of δ at θ is the expected loss under P_θ ,

$$R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))]$$

where $\mathbb{E}_\theta$ denotes expectation when $X \sim P_\theta$ (definition 1.1.1). The *risk function* of δ is the map

$$\theta \mapsto R(\theta, \delta), \quad \theta \in \Theta$$

assigning to each candidate state of nature the average cost the rule incurs there.

The point to hold onto is that the risk is a function of θ , not a number. Each rule δ carries an entire curve $\theta \mapsto R(\theta, \delta)$ over the parameter space, recording its average performance against every candidate law at once. Comparing two rules means comparing two functions, and two functions on Θ need not be comparable: one rule may have smaller risk for some θ and larger for others. This is the structural fact the chapter builds on. The expectation itself is the expectation of [5]: for a discrete model $R(\theta, \delta) = \sum_x L(\theta, \delta(x)) P_\theta(X = x)$, and for a model with density f_θ it is $R(\theta, \delta) = \int L(\theta, \delta(x)) f_\theta(x) dx$, the integral of the loss against the law of the data. Nothing in the definition is new probability. What is new is reading the expectation as a function of the unknown θ and treating that function as the object to be made small.

The risk is the yardstick, but for the most-used loss it collapses to a quantity with a name and a decomposition already familiar from probability. Under squared-error loss the risk of an estimator is its mean squared error, and the mean squared error splits into a variance term and a bias term. This decomposition separates the two distinct ways an estimator can be wrong: it can be systematically off-center, or it can be erratic, and the two defects trade against each other.

Proposition 1.2.5: Mean Squared Error and the Bias-Variance Decomposition

Let $\hat{\theta} = \hat{\theta}(X)$ be an estimator of a real parameter θ with $\mathbb{E}_\theta[\hat{\theta}^2] < \infty$ for every $\theta \in \Theta$, and take the squared-error loss $L(\theta, a) = (\theta - a)^2$. Then the risk of $\hat{\theta}$ is its *mean squared error*

$$\text{MSE}(\theta, \hat{\theta}) = R(\theta, \hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2]$$

Defining the *bias* of $\hat{\theta}$ at θ by

$$\text{Bias}(\theta, \hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta$$

the mean squared error decomposes as

$$\text{MSE}(\theta, \hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{Bias}(\theta, \hat{\theta})^2 \quad (1.1)$$

where $\text{Var}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \mathbb{E}_\theta[\hat{\theta}])^2]$ is the variance of $\hat{\theta}$ under P_θ . In particular an *unbiased* estimator, one with $\text{Bias}(\theta, \hat{\theta}) = 0$ for all θ , has $\text{MSE}(\theta, \hat{\theta}) = \text{Var}_\theta(\hat{\theta})$.

Proof:

That the risk equals the mean squared error is immediate from the definitions: under the loss $L(\theta, a) = (\theta - a)^2$,

$$R(\theta, \hat{\theta}) = \mathbb{E}_\theta[L(\theta, \hat{\theta}(X))] = \mathbb{E}_\theta[(\theta - \hat{\theta})^2] = \text{MSE}(\theta, \hat{\theta})$$

which is finite because $\mathbb{E}_\theta[\hat{\theta}^2] < \infty$. It remains to establish (1.1).

Write $\mu = \mathbb{E}_\theta[\hat{\theta}]$ for the mean of the estimator under P_θ , so that $\text{Bias}(\theta, \hat{\theta}) = \mu - \theta$, a constant (it does not depend on the data). Insert and remove μ inside the square and expand:

$$(\hat{\theta} - \theta)^2 = ((\hat{\theta} - \mu) + (\mu - \theta))^2 = (\hat{\theta} - \mu)^2 + 2(\hat{\theta} - \mu)(\mu - \theta) + (\mu - \theta)^2$$

Take $\mathbb{E}_\theta$ of both sides and use linearity of expectation ([5, Foundational Properties of Expectation]). The three terms are handled in turn. The first term gives $\mathbb{E}_\theta[(\hat{\theta} - \mu)^2] = \text{Var}_\theta(\hat{\theta})$ by the definition of variance. In the second term the factor $(\mu - \theta)$ is a constant and factors out, leaving

$$2(\mu - \theta) \mathbb{E}_\theta[\hat{\theta} - \mu] = 2(\mu - \theta)(\mathbb{E}_\theta[\hat{\theta}] - \mu) = 0$$

since $\mathbb{E}_\theta[\hat{\theta}] = \mu$, and the cross term vanishes, which is the point of centering at the mean. The third term $(\mu - \theta)^2 = \text{Bias}(\theta, \hat{\theta})^2$ is constant and equals its own expectation. Summing the three contributions,

$$\mathbb{E}_\theta[(\hat{\theta} - \theta)^2] = \text{Var}_\theta(\hat{\theta}) + \text{Bias}(\theta, \hat{\theta})^2$$

which is (1.1). The unbiased case follows by setting $\text{Bias}(\theta, \hat{\theta}) = 0$, as we sought to show.

The decomposition says that an estimator's average squared error is the sum of two independent liabilities. The variance $\text{Var}_\theta(\hat{\theta})$ measures the spread of the estimator around its own mean, how much it jitters from sample to sample; the squared bias $\text{Bias}(\theta, \hat{\theta})^2$ measures how far that mean sits from the target θ , a systematic offset that no amount of data averaged the wrong way will remove. An estimator can fail by being erratic (large variance) or by being consistently off-center (large bias), and (1.1) shows these are the only two ways it can fail under squared-error loss. The identity is also the reason unbiasedness is not the last word: an estimator can accept a little bias to give a large reduction in variance and so come out ahead in mean squared error, which is exactly the trade the shrinkage estimators of ?? exploit. The bias-variance decomposition returns in section 3.1.

The abstract identity becomes concrete on the two running models, where every term is computable in closed form.

Example 1.2.6: The Sample Mean in the Bernoulli and Gaussian Models

In both running models the sample mean $\bar{X} = n^{-1} \sum_{i=1}^n X_i$ is the obvious estimator, and its mean squared error is read off from proposition 1.2.5 once its mean and variance are known.

The Bernoulli model. Let $X_1, \dots, X_n$ be an i.i.d. sample from $\text{Ber}(p)$, so $\mathbb{E}_p[X_i] = p$ and $\text{Var}_p(X_i) = p(1-p)$ ([5]), and estimate p by $\bar{X}$. By linearity $\mathbb{E}_p[\bar{X}] = p$, so $\bar{X}$ is unbiased and $\text{Bias}(p, \bar{X}) = 0$. Since the X_i are independent, $\text{Var}_p(\bar{X}) = n^{-2} \sum_{i=1}^n \text{Var}_p(X_i) = p(1-p)/n$, and the decomposition (1.1) gives

$$\text{MSE}(p, \bar{X}) = \text{Var}_p(\bar{X}) + 0 = \frac{p(1-p)}{n}$$

The risk curve $p \mapsto p(1-p)/n$ is an inverted parabola, largest at $p = \frac{1}{2}$, where the Bernoulli variable is hardest to pin down, and vanishing at the endpoints $p \in \{0, 1\}$, where the outcome is deterministic.

The Gaussian model. Let $X_1, \dots, X_n$ be an i.i.d. sample from $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known, and estimate the mean μ by $\bar{X}$. Again $\mathbb{E}_\mu[\bar{X}] = \mu$, so $\bar{X}$ is unbiased, and by independence $\text{Var}_\mu(\bar{X}) = \sigma^2/n$,

whence

$$\text{MSE}(\mu, \bar{X}) = \frac{\sigma^2}{n}$$

Here the risk does not depend on μ at all: the risk function is the constant σ^2/n . This is the signature of a location parameter, where shifting the truth shifts the estimator by the same amount and leaves the error distribution unchanged. A constant risk function is the flattest possible curve, and this flatness is what makes $\bar{X}$ so hard to beat in the Gaussian location model, a point sharpened in section 3.2 and again in ??.

Both sample means are unbiased, so their mean squared error is pure variance and decreases like $1/n$: more data sharpens the estimate at the standard rate. That $\bar{X}$ is unbiased makes it a natural benchmark, but it does not make it optimal. The risk function is a curve over Θ , and a competing estimator can dip below $\bar{X}$'s risk curve on part of Θ while rising above it elsewhere. When that happens neither rule dominates, and the ordering of rules by risk is only partial. The next example exhibits this crossing explicitly.

Example 1.2.7: Competing Risk Curves: Shrinking the Sample Mean

Stay in the Bernoulli model $\text{Ber}(p)$ with an i.i.d. sample of size n , where $\bar{X}$ has risk $R(p, \bar{X}) = p(1-p)/n$ by example 1.2.6. Compare it with a biased competitor that pulls the sample mean toward the center value $\frac{1}{2}$. Concretely, take the estimator

$$\hat{p} = \frac{\sum_{i=1}^n X_i + \frac{1}{2}\sqrt{n}}{n + \sqrt{n}} = \frac{\sqrt{n}}{n + \sqrt{n}} \cdot \frac{1}{2} + \frac{n}{n + \sqrt{n}} \cdot \bar{X}$$

a weighted average of the sample mean and the pinned value $\frac{1}{2}$. This is the posterior mean under a symmetric Beta prior, an instance of the conjugate computation of section 1.4. Here it is treated purely as a rival estimator whose risk curve is to be compared with that of $\bar{X}$.

Write $c = n/(n + \sqrt{n})$ for the weight on $\bar{X}$, so $\hat{p} = c\bar{X} + (1-c)/2$. Its mean is $\mathbb{E}_p[\hat{p}] = cp + (1-c)/2$, giving the bias

$$\text{Bias}(p, \hat{p}) = \mathbb{E}_p[\hat{p}] - p = (1-c)\left(\frac{1}{2} - p\right)$$

which vanishes at $p = \frac{1}{2}$ and grows as p moves toward the endpoints. Its variance is $\text{Var}_p(\hat{p}) = c^2 \text{Var}_p(\bar{X}) = c^2 p(1-p)/n$, so by (1.1)

$$R(p, \hat{p}) = c^2 \frac{p(1-p)}{n} + (1-c)^2 \left(\frac{1}{2} - p\right)^2$$

Because $c < 1$, the first term is strictly smaller than $\bar{X}$'s risk $p(1-p)/n$: shrinking toward $\frac{1}{2}$ reduces variance everywhere on $(0, 1)$. The price is the second term, the squared bias, which is zero at $p = \frac{1}{2}$ and largest at the endpoints. The two risk curves therefore cross. Near $p = \frac{1}{2}$ the variance saving dominates and $\hat{p}$ has the smaller risk; near $p = 0$ or $p = 1$ the bias penalty dominates and $\bar{X}$ has the smaller risk.

To see the crossing numerically, take $n = 4$, so $\sqrt{n} = 2$, $c = 4/6 = 2/3$, and $1 - c = 1/3$. Then

$$R(p, \bar{X}) = \frac{p(1-p)}{4}, \quad R(p, \hat{p}) = \frac{4}{9} \cdot \frac{p(1-p)}{4} + \frac{1}{9} \left(\frac{1}{2} - p\right)^2$$

At $p = \frac{1}{2}$ the shrinkage estimator wins decisively: $R(\frac{1}{2}, \bar{X}) = \frac{1}{16} = 0.0625$ against $R(\frac{1}{2}, \hat{p}) = \frac{4}{9} \cdot \frac{1}{16} = \frac{1}{36} \approx 0.0278$, less than half. At the endpoint $p = 0$ the sample mean wins: $R(0, \bar{X}) = 0$ while

$R(0, \hat{p}) = \frac{1}{9} \cdot \frac{1}{4} = \frac{1}{36} \approx 0.0278$, since the biased rule reports a value near $\frac{1}{2}$ even when the truth is 0. Solving $R(p, \hat{p}) = R(p, \bar{X})$ locates the crossover: the equation reduces to $-\frac{5}{9} \cdot \frac{p(1-p)}{4} + \frac{1}{9}(\frac{1}{2} - p)^2 = 0$, that is $4(\frac{1}{2} - p)^2 = 5p(1-p)$, a quadratic $9p^2 - 9p + 1 = 0$ with roots $p = \frac{1}{2}(1 \pm \sqrt{5/9}) = \frac{1}{2}(1 \pm \frac{\sqrt{5}}{3})$, numerically $p \approx 0.127$ and $p \approx 0.873$. For p between these roots the shrinkage estimator has smaller risk; outside them the sample mean does.

Neither estimator dominates the other: each is better on part of the parameter space and worse on the rest, and no observation of the data can tell the analyst which region the true p lies in. This is the obstacle that section 1.3 confronts. The risk function orders rules only partially, and to select a single rule the theory must either restrict the field of candidates or collapse each risk curve to one number, the two escape routes taken up in the remainder of the chapter.

The lesson of the crossing curves is that “smallest risk” is not by itself a usable criterion, because the minimum is taken pointwise in θ and different rules attain it at different θ . Every subsequent decision-theoretic notion in this book is a response to this failure. Section 1.3 keeps the risk function whole and asks only for rules that are not beaten outright, the admissible ones, and finds that admissibility alone still leaves many rules on the table. Section 1.4 collapses the risk function to a single number, by averaging it against a prior (the Bayes principle) or by recording its worst-case value over θ (the minimax principle), and each choice does pick out a rule. The loss, the decision rule, and the risk function assembled here are the vocabulary in which all of that is said.

Exercise 1.2.1

1. In the estimation setting $\mathcal{A} = \Theta = \mathbb{R}$, suppose the true parameter is $\theta = 2$ and an estimator returns $a = 5$. Compute the squared-error, absolute-error, and 0-1 losses. Then suppose a second estimator returns $a = 3$, and recompute all three losses and state, for each loss, the ratio of the two costs. Explain in one sentence why the three losses disagree on how much worse the estimate $a = 5$ is.
2. Let $X_1, \dots, X_n$ be an i.i.d. sample from the Poisson law $\text{Poi}(\lambda)$, for which $\mathbb{E}_\lambda[X_i] = \text{Var}_\lambda(X_i) = \lambda$, and estimate λ by $\bar{X}$. Show that $\bar{X}$ is unbiased and compute its mean squared error via proposition 1.2.5. Sketch the shape of the risk function $\lambda \mapsto R(\lambda, \bar{X})$ and contrast it with the Bernoulli risk curve of example 1.2.6.
3. For an i.i.d. sample from $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, consider the two estimators of σ^2

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Using $\mathbb{E}_\mu[S^2] = \sigma^2$ (so S^2 is unbiased) and $\text{Var}(S^2) = 2\sigma^4/(n-1)$, first write down $\text{MSE}(\sigma^2, S^2)$. Then, noting $\hat{\sigma}^2 = \frac{n-1}{n}S^2$, compute the bias and variance of $\hat{\sigma}^2$ and its mean squared error, and determine which of the two has the smaller MSE.

4. Consider the constant rule $\delta \equiv \theta_0$ that ignores the data and always reports the fixed value $\theta_0 \in \Theta$, in an estimation problem under squared-error loss. Compute its risk function $\theta \mapsto R(\theta, \delta)$. At which θ is its risk zero, and what does this say about comparing δ with any nonconstant estimator? (This computation underlies the no-uniform-best result of section 1.3.)
5. Return to the shrinkage estimator $\hat{p} = c\bar{X} + (1-c)/2$ of example 1.2.7 in the $\text{Ber}(p)$ model, but now treat the weight $c \in [0, 1]$ as a free tuning constant rather than fixing $c = n/(n + \sqrt{n})$.

Write $R(p, \hat{p})$ as a function of c and p , and for the fixed value $p = \frac{1}{2}$ find the weight c that minimizes the risk. Interpret the answer.

6. Let δ be a randomized estimator that, on observing $X = x$, returns $\hat{\theta} = \delta(x) + \varepsilon$, where ε is an independent mean-zero randomization with variance $\tau^2 > 0$, and let $\delta_0(x) = \delta(x)$ be its nonrandomized core. Under squared-error loss, show that $R(\theta, \delta) = R(\theta, \delta_0) + \tau^2 \geq R(\theta, \delta_0)$ for every θ , so the randomization strictly increases the risk. (This is the concrete form of the claim that randomization gives nothing in estimation under a convex loss on an interval action space.)

1.3 Admissibility and the Search for Optimal Rules

The risk function $R(\theta, \delta)$ of definition 1.2.4 assigns to each decision rule a whole function of the unknown parameter, not a single number. If risk were a number, “best” would mean “smallest” and the problem would be a minimization. Because risk is a function on Θ , two rules can be compared at each θ separately, and a rule that wins at one value of θ may lose at another. This section makes the comparison precise, records the obstacle it runs into, and names the two ways around it.

The comparison is unambiguous whenever one rule is at least as good as another everywhere and strictly better somewhere. Then there is no reason to keep the worse rule, and the language of *domination* captures this situation.

Definition 1.3.1: Domination

Let δ_1 and δ_2 be decision rules (definition 1.2.3) for a statistical problem with risk function R . The rule δ_1 *dominates* δ_2 if

$$R(\theta, \delta_1) \leq R(\theta, \delta_2) \quad \text{for every } \theta \in \Theta$$

and the inequality is strict for at least one $\theta_0 \in \Theta$:

$$R(\theta_0, \delta_1) < R(\theta_0, \delta_2)$$

Domination is the comparison on which every reasonable observer agrees. If δ_1 dominates δ_2 , then whatever the true θ , the rule δ_1 costs no more on average, and for some parameter values it costs strictly less. No weighting of the possible values of θ could prefer δ_2 : it is beaten uniformly and beaten strictly somewhere. This is a partial order on rules, and only partial: most pairs of rules are incomparable, each winning on part of the parameter space. A dominated rule, though, is unambiguously discardable.

Definition 1.3.2: Admissible and Inadmissible Rules

A decision rule δ is *inadmissible* if some rule dominates it (definition 1.3.1), that is, if there exists a rule δ' with $R(\theta, \delta') \leq R(\theta, \delta)$ for all θ and strict inequality for some θ . A rule is *admissible* if it is not inadmissible: no rule dominates it. A rule δ^* is *uniformly best* within a class $\mathcal{D}$ of rules if $\delta^* \in \mathcal{D}$ and

$$R(\theta, \delta^*) \leq R(\theta, \delta) \quad \text{for every } \theta \in \Theta \text{ and every } \delta \in \mathcal{D}$$

Admissibility is a minimal requirement. It does not say a rule is good; it says only that the rule is not transparently bad, in that no other rule beats it everywhere. An admissible rule can have enormous risk at every θ , provided no single competitor improves on it across the whole parameter space at once. The constant rule that ignores the data and always returns a fixed value θ_0 is a case in point: its risk is zero at θ_0 and typically large elsewhere, yet because no other rule can match that zero at θ_0 while also doing at least as well everywhere else, the constant rule is often admissible. Admissibility weeds out the wasteful rules and nothing more: it is a floor, not a target.

A uniformly best rule would be the ideal: a single δ^* whose risk curve lies at or below every competitor's curve at every θ simultaneously. Such a rule, if it existed, would leave nothing to choose and nothing to trade off. The distinction between the two definitions is that admissibility is almost always achievable and almost never unique, while uniform optimality is what one wants and almost never has. The next result explains why.

The mechanism is clearest in the smallest case. Fix any target value θ_0 in the parameter space and consider the rule that throws away the data and reports θ_0 no matter what is observed. Under squared-error loss this rule has risk $(\theta - \theta_0)^2$, which is zero when $\theta = \theta_0$. No estimator that uses the data can have risk zero at θ_0 unless it equals θ_0 with probability one under P_{θ_0} , and such an estimator can hardly remain reasonable across the whole family. So at θ_0 the useless constant rule beats every serious estimator. Since θ_0 was arbitrary, every point of the parameter space is claimed by some constant rule, and no single rule can be best at all of them at once.

Proposition 1.3.3: No Uniformly Best Rule

Consider a statistical problem with parameter space Θ containing at least two points, under squared-error loss $L(\theta, a) = (\theta - a)^2$, and take $\mathcal{D}$ to be the class of all decision rules $\delta: \mathcal{X} \rightarrow \mathbb{R}$. Suppose that for each $\theta \in \Theta$ there is a rule of risk zero at θ (for instance the constant rule $\delta_\theta \equiv \theta$, which has $R(\theta, \delta_\theta) = 0$), and that there exist two distinct $\theta_1, \theta_2 \in \Theta$ at which no single rule has risk zero simultaneously. Then no rule in $\mathcal{D}$ is uniformly best: there is no $\delta^* \in \mathcal{D}$ with $R(\theta, \delta^*) \leq R(\theta, \delta)$ for all θ and all $\delta \in \mathcal{D}$.

Proof:

Suppose, for contradiction, that δ^* were uniformly best in $\mathcal{D}$. Take the two points $\theta_1 \neq \theta_2$ named in the hypothesis.

Step 1: The optimum is forced to zero risk at every parameter value. For each $\theta \in \Theta$ the constant rule $\delta_\theta \equiv \theta$ lies in $\mathcal{D}$ and has

$$R(\theta, \delta_\theta) = \mathbb{E}_\theta[(\theta - \theta)^2] = 0$$

Since δ^* is uniformly best, its risk at θ can be no larger than that of any competitor evaluated at θ , and in particular no larger than $R(\theta, \delta_\theta) = 0$. Risk is nonnegative because the loss is nonnegative, so

$$0 \leq R(\theta, \delta^*) \leq R(\theta, \delta_\theta) = 0$$

forcing $R(\theta, \delta^*) = 0$. The parameter θ was arbitrary, so $R(\theta, \delta^*) = 0$ for every $\theta \in \Theta$.

Step 2: Zero risk at two points is impossible. Applying the conclusion of Step 1 at θ_1 and θ_2 gives $R(\theta_1, \delta^*) = 0$ and $R(\theta_2, \delta^*) = 0$, so the single rule δ^* has risk zero at two distinct parameter values. This is exactly what the hypothesis forbids, a contradiction. Hence no uniformly best rule exists in $\mathcal{D}$, completing the proof.

The proof turns on a tension that recurs. To be best at θ_1 a rule must tune itself to θ_1 , most sharply by ignoring the data and announcing θ_1 , giving zero risk there and disaster elsewhere; to be best at θ_2 it must do the opposite. No rule can do both, because zero risk at a point under squared-error loss pins the rule to that point, and no rule is pinned to two points at once. The hypothesis asks only for *one* pair of parameter values that no rule can be right about at once, which is why it is stated that way rather than as a claim about every pair. The distinction is not idle. In the Bernoulli model with $\Theta = [0, 1]$ the sample mean $\bar{X}$ does have risk zero at two points, namely $p = 0$ and $p = 1$, where the data are deterministic and $\bar{X}$ recovers p exactly; this is the vanishing at the endpoints already visible in the risk curve of example 1.2.6. The hypothesis survives because it needs only one bad pair, and $p = \frac{1}{3}, \frac{1}{2}$ is one: risk zero at p under squared-error loss forces $\delta^* \equiv p$ on all of $\{0, 1\}^n$, every point having positive mass, and no rule is identically $\frac{1}{3}$ and identically $\frac{1}{2}$ at once. The running models of example 1.1.7 all satisfy the hypothesis in this form. The risk function does not order the rules totally, so “minimize the risk” is not well-posed, and admissibility, which only removes dominated rules, leaves a large class of mutually incomparable survivors. Some further principle is needed to choose among them.

The proposition rules out a uniformly best rule; it does not say that inadmissible rules are rare. They are common, and recognizing one is often the first sign that a proposed estimator can be improved for free. The next example gives two, each an estimator that wastes information a competitor can recover: one by respecting a constraint the estimator ignores, one by correcting a systematic error.

Example 1.3.4: Two Inadmissible Estimators

Both parts exhibit an estimator that a concrete competitor dominates (definition 1.3.1), with the risks compared directly.

1. *Truncation to a known range.* Take the Gaussian location model of example 1.1.7 with the variance σ^2 known, and suppose it is known in addition that the mean lies in a bounded interval, $\theta = \mu \in [a, b]$, so $\Theta = [a, b]$. The sample mean $\bar{X}$ is the natural estimator, with squared-error risk

$$R(\theta, \bar{X}) = \mathbb{E}_\theta[(\bar{X} - \theta)^2] = \text{Var}_\theta(\bar{X}) = \frac{\sigma^2}{n}$$

constant in θ . But $\bar{X}$ can fall outside $[a, b]$, and when it does it reports a value the parameter is known not to take. Define the truncated estimator

$$\hat{\theta}(X) = \begin{cases} a & \text{if } \bar{X} < a \\ \bar{X} & \text{if } a \leq \bar{X} \leq b \\ b & \text{if } \bar{X} > b \end{cases}$$

the projection of $\bar{X}$ onto the interval $[a, b]$. For every $\theta \in [a, b]$ and every possible value of X , projecting a point onto the interval that already contains θ moves it no farther from θ : $|\hat{\theta}(X) - \theta| \leq |\bar{X} - \theta|$ pointwise, because the nearest-point projection onto a convex set is a contraction toward any point of that set. Squaring and taking expectations,

$$R(\theta, \hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2] \leq \mathbb{E}_\theta[(\bar{X} - \theta)^2] = R(\theta, \bar{X})$$

for all $\theta \in [a, b]$. The inequality is strict whenever $P_\theta(\bar{X} \notin [a, b]) > 0$, which holds for every θ since the Gaussian charges every interval with positive probability, so on the event $\{\bar{X} \notin [a, b]\}$ the truncation strictly reduces the squared error, and that event has positive probability. Thus $\hat{\theta}$ dominates $\bar{X}$, and $\bar{X}$ is inadmissible on $\Theta = [a, b]$: the estimator that can name an impossible value is beaten by the one that cannot.

2. *The wrong divisor in the sample variance.* Take an i.i.d. sample $X_1, \dots, X_n$ from $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, and estimate σ^2 . Compare the two standard divisors,

$$S_c^2 = c \sum_{i=1}^n (X_i - \bar{X})^2, \quad W = \sum_{i=1}^n (X_i - \bar{X})^2$$

for a constant $c > 0$ to be chosen, so $c = \frac{1}{n-1}$ gives the unbiased sample variance and $c = \frac{1}{n}$ the maximum-likelihood estimator. The sum of squares W satisfies $W/\sigma^2 \sim \chi_{n-1}^2$, the Gaussian sampling theory this book proves in ??, so

$$\mathbb{E}_\theta[W] = (n-1)\sigma^2, \quad \text{Var}_\theta(W) = 2(n-1)\sigma^4$$

using $\mathbb{E}[\chi_{n-1}^2] = n-1$ and $\text{Var}(\chi_{n-1}^2) = 2(n-1)$. The squared-error risk of $S_c^2 = cW$ is then

$$\begin{aligned} R(\theta, S_c^2) &= \mathbb{E}_\theta[(cW - \sigma^2)^2] = c^2 \mathbb{E}_\theta[W^2] - 2c\sigma^2 \mathbb{E}_\theta[W] + \sigma^4 \\ &= c^2 (\text{Var}_\theta(W) + \mathbb{E}_\theta[W]^2) - 2c\sigma^2(n-1)\sigma^2 + \sigma^4 \\ &= c^2(2(n-1) + (n-1)^2)\sigma^4 - 2c(n-1)\sigma^4 + \sigma^4 \\ &= \sigma^4[(n-1)(n+1)c^2 - 2(n-1)c + 1] \end{aligned}$$

a quadratic in c with positive leading coefficient, minimized where its derivative $2(n-1)(n+1)c - 2(n-1)$ vanishes, at

$$c^* = \frac{1}{n+1}$$

Because the risk is σ^4 times a strictly convex parabola in c , any $c \neq c^*$ gives strictly larger risk at every θ (the factor $\sigma^4 > 0$ never changes sign). In particular the unbiased choice $c = \frac{1}{n-1}$ and the maximum-likelihood choice $c = \frac{1}{n}$ both differ from $c^* = \frac{1}{n+1}$, so each is dominated by $S_{c^*}^2 = \frac{1}{n+1} \sum_i (X_i - \bar{X})^2$ and each is inadmissible under squared-error loss. The estimator that corrects for bias is not the one that minimizes mean squared error.

The two parts illustrate the two textures of inadmissibility. In the first, the improvement is qualitative: the original estimator does something visibly foolish, reporting values outside the known range, and the fix is to stop doing it. In the second, the improvement is quantitative and less intuitive: the unbiased sample variance is not foolish in any obvious way, yet a different constant multiple has uniformly smaller mean squared error because shrinking the estimate slightly toward zero trades a little bias for a larger reduction in variance. This second phenomenon is the seed of shrinkage, and it recurs in a stronger form when the same bias-variance trade is played across several coordinates at once, which is the James-Stein estimator, which shows the ordinary sample mean itself to be inadmissible in three or more dimensions, developed in ??. The lesson is that inadmissibility is easy to produce, admissibility is a weak requirement, and neither, by proposition 1.3.3, singles out one rule.

There are exactly two routes out of this impasse. The obstacle is that the risk function is a function, not a number: it compares rules at every θ at once and so leaves most of them incomparable. A route out must therefore either shrink the field of rules until a best one emerges, or shrink the risk function to a number that can be minimized outright.

The first route *restricts the class of rules*. If uniform optimality is unattainable over all rules, one may impose a side condition that rules out the degenerate competitors, chiefly the data-ignoring

constant rules that spoiled proposition 1.3.3, and search for a uniformly best rule within the smaller class that remains. Three such conditions organize the estimation theory of this book. *Unbiasedness* requires $\mathbb{E}_\theta[\delta(X)] = \theta$ for all θ , which forbids a rule from pinning itself to one parameter value and so eliminates the constant rules. Within the unbiased class a uniformly minimum-variance estimator often does exist, and its construction through sufficiency is the subject of chapter 3. *Invariance* requires the rule to transform compatibly with symmetries of the problem, such as shifts of a location parameter or rescalings of a scale parameter, and cuts the field down to those rules that respect the structure the model already has. *Sufficiency* reduces the data itself to its informative core before any rule is applied. Chapter 2 shows that any rule is matched, risk for risk and under any loss, by one that reads only a sufficient statistic together with a parameter-free randomization, so attention may be restricted to such rules without loss, and section 3.3 sharpens the match to a strict improvement under squared-error loss for rules not already almost surely equal to a function of the sufficient statistic.

The second route *summarizes the risk* by collapsing the function $\theta \mapsto R(\theta, \delta)$ to a single number, after which “smallest” regains its meaning and the search becomes an ordinary minimization. There are two natural summaries. The *Bayes* summary averages the risk against a prior distribution π on the parameter space, forming $r(\pi, \delta) = \int R(\theta, \delta) \pi(d\theta)$, and asks for the rule of smallest average risk, which weights the parameter values by how plausible the prior deems them. The *minimax* summary takes the worst case, $\sup_\theta R(\theta, \delta)$, and asks for the rule whose worst-case risk is smallest, guarding against the least favorable parameter value with no prior commitment about which value that is. Both summaries reduce the partial order of risk functions to a total order on numbers, so both make “best” well-posed, and both connect back to admissibility in ways that recover much of what the uniform criterion could not deliver. These are the subject of section 1.4, and the Bayesian route is developed in full in ??.

Exercise 1.3.1

1. Under squared-error loss, show that the constant rule $\delta \equiv \theta_0$ is admissible in the Gaussian location model with $\Theta = \mathbb{R}$ and σ^2 known. (Argue that any dominating rule would have to match its zero risk at θ_0 , and use proposition 1.3.3 reasoning to derive a contradiction from a rule that also does at least as well elsewhere.)
2. Prove that domination (definition 1.3.1) is a strict partial order on the set of decision rules of a given problem, taken up to risk-function equality: it is irreflexive and transitive, and no two rules dominate each other. Give an example of two rules, neither of which dominates the other.
3. In the truncation example (example 1.3.4, part 1), the claim that $\hat{\theta}$ strictly improves on $\bar{X}$ relied on $|\hat{\theta}(X) - \theta| \leq |\bar{X} - \theta|$ pointwise for $\theta \in [a, b]$. Prove this pointwise inequality directly from the definition of $\hat{\theta}$ by checking the three cases $\bar{X} < a$, $a \leq \bar{X} \leq b$, and $\bar{X} > b$, and identify on which of the three the inequality is strict.
4. For the sample-variance family $S_c^2 = c \sum_i (X_i - \bar{X})^2$ of example 1.3.4 (part 2), verify by direct computation that the risk-minimizing constant is $c^* = \frac{1}{n+1}$, and compute the ratio $R(\theta, S_{1/(n-1)}^2) / R(\theta, S_{1/(n+1)}^2)$ of the unbiased estimator’s risk to the optimal one’s. Confirm that the ratio exceeds 1 and does not depend on θ .
5. A biased estimator can dominate an unbiased one, which is why unbiasedness is a restriction and not a free improvement. For a single observation $X \sim \mathcal{N}(\theta, 1)$, compare the unbiased

estimator $\delta_0(X) = X$ of θ with the shrunken estimator $\delta_\lambda(X) = \lambda X$ for a constant $0 < \lambda < 1$, under squared-error loss. Compute $R(\theta, \delta_\lambda)$ as a function of θ and λ , and show that no δ_λ with $\lambda < 1$ dominates δ_0 (so shrinkage helps near $\theta = 0$ but hurts for large $|\theta|$), illustrating that the incomparability of proposition 1.3.3 persists even between two very simple rules.

6. Give an example, in the Bernoulli model of example 1.1.7, of two distinct estimators of p that are each admissible yet neither dominates the other, and explain in one sentence why their existence is consistent with proposition 1.3.3.

1.4 The Bayes and Minimax Principles

The risk function $\theta \mapsto R(\theta, \delta)$ of definition 1.2.4 is a function on the parameter space, not a number, and this prevents it from ordering decision rules: two rules whose risk curves cross cannot be ranked, each better on part of Θ and worse on the rest, and neither dominating the other in the sense of definition 1.3.1. The escape route named at the close of section 1.3 is to collapse the whole curve to a single number. There are two canonical ways to do it. One weights the parameter values by a probability distribution and reports the weighted average risk; the other ignores the weighting and reports the worst value the risk attains. The first is the Bayes principle, the second the minimax principle. They are not competing philosophies so much as two functionals applied to the same risk function, and the point where they touch, a rule whose risk is constant across Θ , is where the section ends.

The averaging route requires a distribution on the parameter to average against. Treating the unknown θ as itself the value of a random variable is the move that separates Bayesian from frequentist inference, and it converts the family $\{P_\theta\}$ into a joint law on parameter and data. Once θ carries a distribution, the data update it: the distribution of θ after seeing X is a conditional distribution, and computing it is an application of the conditional-expectation machinery of the prerequisite volume.

Definition 1.4.1: Prior and Posterior

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model dominated by a σ -finite measure μ on the sample space $\mathcal{X}$, with densities $p(x|\theta) = dP_\theta/d\mu$ (the Radon-Nikodym derivatives, guaranteed to exist by the Radon-Nikodym theorem [4]). A *prior* is a probability distribution π on Θ , regarded as the law of θ before the data are seen. Assume throughout this section that π has a density, again written $\pi(\theta)$, with respect to a σ -finite reference measure ν on Θ , and that $(\theta, x) \mapsto p(x|\theta)$ is jointly measurable. The prior and the model then define a joint law on (θ, X) with density $p(x|\theta)\pi(\theta)$ against $\nu \otimes \mu$, and the σ -finiteness is what lets the order of integration be exchanged below ([4, Fubini-Tonelli Theorem]). The *posterior* is the conditional distribution $\pi(\cdot|X)$ of θ given X , and Bayes' rule for densities gives

$$\pi(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{m(x)}, \quad m(x) = \int_{\Theta} p(x|\theta')\pi(\theta')\nu(d\theta') \quad (1.2)$$

where the normalizing denominator $m(x)$ is the *marginal* (or *prior predictive*) density of X . The posterior is defined for every x with $m(x) > 0$, which is P -almost every x under the joint law.

Every ingredient of (1.2) has already been built. The densities $p(x|\theta)$ are Radon-Nikodym derivatives of the dominated family [4]. The posterior is a conditional distribution of one coordinate of a joint law given the other, and its existence and near-uniqueness as a regular conditional law is the conditional-expectation theory of the probability volume, whose existence theorem covers a real-valued target [5, Existence of Regular Conditional Distributions] and so applies directly when $\Theta \subseteq \mathbb{R}$. Nothing below needs the general case: the posterior is used only through the density formula (1.2), which constructs it explicitly whenever the model is dominated. Bayes' rule (1.2) is what that abstract conditional distribution becomes when both model and prior are given by densities. The numerator $p(x|\theta) \pi(\theta)$ is the joint density of (θ, X) , the denominator $m(x)$ is the X -marginal, and their ratio is the conditional density of θ given X . The posterior carries all the information the data give about θ : it starts from the prior $\pi(\theta)$ and reweights it by the likelihood $p(x|\theta)$ of the observed data, so parameter values that render the data probable gain posterior mass and those that render it improbable lose it.

Read (1.2) as proportionality in θ with x held fixed. The denominator $m(x)$ does not depend on θ , so

$$\pi(\theta|x) \propto p(x|\theta) \pi(\theta)$$

is often the fastest route to the posterior: compute the product of likelihood and prior, recognize its functional form in θ , and normalize. When the prior is chosen so that the posterior lies in the same parametric family, the family is called *conjugate* to the model, and the normalization can be read off without integrating. The running example is the prototype, and ?? develops conjugacy in full.

With a prior in hand, the risk function can be averaged against it.

Definition 1.4.2: Bayes Risk

Let π be a prior on Θ and δ a decision rule with risk function $R(\cdot, \delta)$ (definition 1.2.4). The *Bayes risk* of δ with respect to π is the average risk

$$r(\pi, \delta) = \int_{\Theta} R(\theta, \delta) \pi(d\theta) = \int_{\Theta} \mathbb{E}_{\theta}[L(\theta, \delta(X))] \pi(d\theta) \quad (1.3)$$

a single nonnegative number attached to each rule.

Averaging the risk curve against π replaces the partial order of definition 1.3.1 by a total order: any two rules can now be compared through the numbers $r(\pi, \delta)$, and any rule attaining the smallest such number is a best rule. The price is that the ranking depends on π . A rule that is Bayes for one prior need not be Bayes for another, and the choice of prior encodes which regions of Θ the analyst cares to do well on. This dependence is not a defect but the mechanism by which prior information enters the inference.

Definition 1.4.3: Bayes Rule

A decision rule δ_{π} is a *Bayes rule* with respect to the prior π if it minimizes the Bayes risk over all rules,

$$r(\pi, \delta_{\pi}) = \inf_{\delta} r(\pi, \delta)$$

the infimum ranging over all (measurable) decision rules $\delta: \mathcal{X} \rightarrow \mathcal{A}$. The common value $r(\pi) = \inf_{\delta} r(\pi, \delta)$ is the *Bayes risk of the problem* under π .

A Bayes rule is defined by an optimization over the space of all decision rules, unwieldy to search

directly. The next theorem reduces this global search to a family of independent pointwise problems, one for each possible value of the data. Its engine is Fubini's theorem, which lets the double integral defining the Bayes risk, over θ against the prior and over X against P_θ , be reorganized as an integral over X of a posterior average over θ . Once the order of integration is swapped, minimizing the Bayes risk amounts to minimizing the inner integrand, the posterior expected loss, separately at each observed x .

Theorem 1.4.4: Bayes Rules Minimize Posterior Risk

Suppose the Bayes risk $r(\pi, \delta)$ is finite for some rule, and for each $x \in \mathcal{X}$ define the *posterior expected loss* of an action $a \in \mathcal{A}$ by

$$\rho(a|x) = \mathbb{E}[L(\theta, a)|X = x] = \int_{\Theta} L(\theta, a) \pi(\theta|x) \nu(d\theta) \quad (1.4)$$

Assume that for m -almost every x the infimum of $a \mapsto \rho(a|x)$ over $\mathcal{A}$ is attained, and that a minimizing action can be chosen so that $x \mapsto \delta_\pi(x)$ is measurable. Then δ_π is a Bayes rule with respect to π , and conversely every Bayes rule minimizes the posterior expected loss for almost every x . In particular, if $\Theta \subseteq \mathcal{A} \subseteq \mathbb{R}$ with $\mathcal{A}$ convex and closed, then under squared-error loss $L(\theta, a) = (\theta - a)^2$ the Bayes rule is the posterior mean $\delta_\pi(x) = \mathbb{E}[\theta|X = x]$, and under absolute loss $L(\theta, a) = |\theta - a|$ it is a posterior median. The containment $\Theta \subseteq \mathcal{A}$, with $\mathcal{A}$ convex and closed, is what puts these in $\mathcal{A}$: for a two-point $\mathcal{A}$ the posterior mean is not an available action, and the Bayes rule is its projection onto $\mathcal{A}$ instead.

Proof:

Write P for the joint law of (θ, X) determined by the prior and the model, and m for the marginal law of X under P .

Step 1: Rewrite the Bayes risk as an average of posterior risk. By (1.3) the Bayes risk is the expectation of $L(\theta, \delta(X))$ under the joint law,

$$r(\pi, \delta) = \int_{\Theta} \int_{\mathcal{X}} L(\theta, \delta(x)) p(x|\theta) \mu(dx) \pi(\theta) \nu(d\theta) = \mathbb{E}_P[L(\theta, \delta(X))]$$

The integrand is nonnegative and jointly measurable in (θ, x) (L is jointly measurable by definition 1.2.1, and δ and the density are measurable), so Tonelli's theorem permits the order of integration to be exchanged, and where $r(\pi, \delta) < \infty$ Fubini's theorem justifies the same exchange with the integrand of either sign [4, Fubini-Tonelli Theorem]. Integrating first over θ against the posterior and then over x against the marginal gives the *tower factorization*

$$r(\pi, \delta) = \int_{\mathcal{X}} \left(\int_{\Theta} L(\theta, \delta(x)) \pi(\theta|x) \nu(d\theta) \right) m(x) \mu(dx) = \int_{\mathcal{X}} \rho(\delta(x)|x) m(x) \mu(dx) \quad (1.5)$$

which expresses $r(\pi, \delta)$ as the m -average of the posterior expected loss $\rho(\delta(x)|x)$ incurred by the action the rule takes at x . This is the abstract law of iterated expectation $r(\pi, \delta) = \mathbb{E}[\mathbb{E}[L(\theta, \delta(X))|X]]$ made explicit through densities.

Step 2: Minimize pointwise. The factorization (1.5) exhibits $r(\pi, \delta)$ as an integral, against the fixed measure $m(x) \mu(dx)$, of a term $\rho(\delta(x)|x)$ that the rule controls only through its value $\delta(x)$ at that single point. The value chosen at x affects the integrand at x and nowhere else, so the

integral is made as small as possible by making each integrand as small as possible. For each x set

$$\rho^*(x) = \inf_{a \in \mathcal{A}} \rho(a|x)$$

Then $\rho(\delta(x)|x) \geq \rho^*(x)$ for every rule δ and every x , hence $r(\pi, \delta) \geq \int_{\mathcal{X}} \rho^*(x) m(x) \mu(dx)$ for every δ . If δ_π attains the pointwise infimum, meaning $\rho(\delta_\pi(x)|x) = \rho^*(x)$ for m -almost every x , then its Bayes risk equals this lower bound, so δ_π minimizes $r(\pi, \cdot)$ and is a Bayes rule. Conversely, if some rule δ has $\rho(\delta(x)|x) > \rho^*(x)$ on a set of positive m -measure, then $r(\pi, \delta)$ strictly exceeds the lower bound and δ cannot be Bayes, so every Bayes rule attains $\rho^*(x)$ for almost every x . Attainment and measurable selection are hypotheses of the theorem rather than consequences of continuity: a continuous function on a non-compact $\mathcal{A}$ need not attain its infimum, and measurable selection is a theorem in its own right. All three running models satisfy them.^a

Step 3: The two estimation losses. Take $\Theta \subseteq \mathcal{A} \subseteq \mathbb{R}$ with $\mathcal{A}$ convex and closed and specialize the pointwise problem. The posterior is a distribution on Θ and $\mathcal{A}$ is a closed interval containing Θ , so the posterior mean and every posterior median lie in $\mathcal{A}$. Under squared-error loss the posterior expected loss is $\rho(a|x) = \mathbb{E}[(\theta - a)^2 | X = x]$, a quadratic in a whose derivative $2(a - \mathbb{E}[\theta | X = x])$ vanishes only at $a = \mathbb{E}[\theta | X = x]$, and since the quadratic opens upward and its minimizer lies in $\mathcal{A}$, this is the unique minimizer over $\mathcal{A}$. The Bayes rule is therefore the posterior mean $\delta_\pi(x) = \mathbb{E}[\theta | X = x]$. Under absolute loss $\rho(a|x) = \mathbb{E}[|\theta - a| | X = x]$, and for a real random variable θ with distribution the posterior, the map $a \mapsto \mathbb{E}[|\theta - a|]$ is minimized over $\mathbb{R}$ exactly at the medians of that distribution; the medians lie in $\mathcal{A}$, so any posterior median is a Bayes rule. Both specializations follow from the pointwise principle of Step 2, completing the proof.

^aattainment is assumed explicitly by [10, Theorem 7.1(b)]; the measurability caveat is van der Vaart's [14, §10.3], which notes that otherwise the statements are to be read relative to outer probabilities.

The theorem makes Bayesian estimation computable and explains why the posterior, not the prior, is the object one reports. It reduces a search over the infinite-dimensional space of decision rules to a single one-parameter minimization, performed after the data arrive, over the action space: one waits for x , forms the posterior, and minimizes the posterior expected loss in a , never contemplating rules as functions. The choice of loss then selects the posterior summary. Squared-error loss returns the posterior mean, absolute loss the posterior median, and, as ?? records, 0-1 loss on a discrete parameter returns the posterior mode. Each familiar point estimate is the Bayes rule for a particular loss, read off from the common posterior.

One structural consequence links this section back to section 1.3. Because a Bayes rule minimizes the average risk, it is hard for another rule to beat it everywhere: if δ' dominated δ_π in the sense of definition 1.3.1, then δ' would have strictly smaller Bayes risk whenever the prior charges the region where the domination is strict, contradicting that δ_π is Bayes. This is the seed of the theorem, developed in ??, that Bayes rules are, under mild conditions, admissible (definition 1.3.2): the Bayes construction is one of the main sources of admissible rules.

The second scalarization discards the prior and asks instead for protection against the worst that the parameter can do. Where the Bayes principle averages the risk curve, the minimax principle reports its supremum, and a minimax rule is one whose highest risk is as low as possible.

Definition 1.4.5: Minimax Rule

The *maximum risk* of a decision rule δ is $\sup_{\theta \in \Theta} R(\theta, \delta)$, the worst risk it incurs over the parameter space. A rule δ^* is *minimax* if it minimizes the maximum risk,

$$\sup_{\theta \in \Theta} R(\theta, \delta^*) = \inf_{\delta} \sup_{\theta \in \Theta} R(\theta, \delta)$$

over all decision rules δ . The common value on the right is the *minimax risk* of the problem.

The minimax rule is the pessimist's choice: it fixes attention on whatever value of θ would make a given rule perform worst, and chooses the rule for which this worst case is least bad. It gives a guarantee that holds uniformly over Θ , but tailors the rule to a parameter value that may be far from the truth. Unlike the Bayes rule it needs no prior, which is both its appeal (no subjective input) and its burden (the supremum can be dominated by an implausible corner of Θ). Computing a minimax rule directly means solving a minimization of a supremum, rarely tractable in closed form. The next result converts the minimax problem into a Bayes problem whenever the two principles agree, which they do when a Bayes rule has a flat risk curve.

The idea is a short game-theoretic argument. A rule's Bayes risk against any prior is an average of its risk values, so it never exceeds the maximum risk. If a rule has constant risk, its average risk against every prior equals that constant, and if in addition the rule is Bayes for some prior, then no rule can beat it on that prior's average, hence none on the worst case either. The prior that certifies this concentrates the difficulty of the problem, and is called least favorable.

Proposition 1.4.6: Constant-Risk Bayes Rules are Minimax

Let δ_π be a Bayes rule with respect to a prior π , and suppose its risk function is constant on Θ ,

$$R(\theta, \delta_\pi) = c \quad \text{for all } \theta \in \Theta$$

Then δ_π is minimax, and π is a *least favorable prior*: it maximizes the Bayes risk of the problem, $r(\pi') \leq r(\pi)$ for every prior π' .

Proof:

Step 1: The Bayes rule is minimax. Let δ be any competing rule. Its Bayes risk against π is an average of its risk values, so it is bounded above by its maximum risk,

$$r(\pi, \delta) = \int_{\Theta} R(\theta, \delta) \pi(d\theta) \leq \int_{\Theta} \left(\sup_{\theta'} R(\theta', \delta) \right) \pi(d\theta) = \sup_{\theta' \in \Theta} R(\theta', \delta)$$

the last equality because the supremum is a constant in the variable of integration and π is a probability measure. Since δ_π is Bayes for π , its Bayes risk is the smallest, so $r(\pi, \delta_\pi) \leq r(\pi, \delta)$. On the other hand the constant-risk hypothesis makes δ_π 's Bayes risk equal to its own maximum risk,

$$r(\pi, \delta_\pi) = \int_{\Theta} c \pi(d\theta) = c = \sup_{\theta \in \Theta} R(\theta, \delta_\pi)$$

Chaining these gives, for every rule δ ,

$$\sup_{\theta} R(\theta, \delta_\pi) = r(\pi, \delta_\pi) \leq r(\pi, \delta) \leq \sup_{\theta} R(\theta, \delta)$$

so the maximum risk of δ_π is no larger than that of any δ . Hence δ_π minimizes the maximum risk and is minimax.

Step 2: The prior is least favorable. For any prior π' the Bayes risk of the problem under π' satisfies

$$r(\pi') = \inf_{\delta} r(\pi', \delta) \leq r(\pi', \delta_\pi) = \int_{\Theta} c \pi'(d\theta) = c = r(\pi, \delta_\pi) = r(\pi)$$

where the first inequality holds because $r(\pi')$ is the infimum over rules and δ_π is one competing rule, the middle equalities use the constant risk of δ_π , and the last equality identifies c as the Bayes risk of the problem under π (the Bayes rule attains the infimum). Thus $r(\pi') \leq r(\pi)$ for every π' , so π maximizes the Bayes risk of the problem and is least favorable, as we sought to show.

The proposition is the standard route to minimax rules. A prior whose Bayes risk is the largest available is called least favorable because Nature, choosing θ adversarially, can do no worse to the statistician than to draw it from π , and the statistician's best reply, the Bayes rule against that hardest prior, is the minimax rule. The flat risk curve is the certificate tying the two together: it forces the average and the maximum of the risk to coincide, so that optimizing the average (a tractable Bayes computation) also optimizes the maximum (the intractable minimax objective). The recipe is to guess a prior making the Bayes risk constant in θ , compute the Bayes rule, and verify the risk is flat. When it is, the rule is minimax and the prior is least favorable for free. The running example carries out this recipe.

Example 1.4.7: The Beta-Bernoulli Bayes Estimator

Return to the Bernoulli model (example 1.1.7): $X_1, \dots, X_n$ are i.i.d. $\text{Ber}(p)$ with $p \in \Theta = [0, 1]$ unknown, and the object is to estimate p under squared-error loss. The sufficient statistic is the total number of successes $S = \sum_{i=1}^n X_i$, which is $\text{Bin}(n, p)$.

The conjugate prior and the posterior. Take a $\text{Beta}(\alpha, \beta)$ prior on p , with density $\pi(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$ on $(0, 1)$, where the hyperparameters $\alpha, \beta > 0$ are fixed. The likelihood of the data is $p(x|p) \propto p^S(1-p)^{n-S}$, so by (1.2) the posterior is proportional in p to

$$\pi(p|x) \propto p^S(1-p)^{n-S} \cdot p^{\alpha-1}(1-p)^{\beta-1} = p^{\alpha+S-1}(1-p)^{\beta+n-S-1}$$

which is the density of a $\text{Beta}(\alpha + S, \beta + n - S)$ distribution. The Beta family is conjugate to the Bernoulli model: the posterior is again a Beta, with the successes S added to α and the failures $n - S$ added to β . The prior behaves like α prior successes and β prior failures, and the data add the observed counts.

The posterior-mean estimator. By theorem 1.4.4 the Bayes rule under squared-error loss is the posterior mean. A $\text{Beta}(a, b)$ variable has mean $a/(a+b)$, so

$$\hat{p}_\pi = \mathbb{E}[p|X] = \frac{\alpha + S}{\alpha + \beta + n} \quad (1.6)$$

This is a weighted average of the prior mean $\alpha/(\alpha + \beta)$ and the sample proportion $\bar{X} = S/n$:

$$\hat{p}_\pi = \frac{\alpha + \beta}{\alpha + \beta + n} \cdot \frac{\alpha}{\alpha + \beta} + \frac{n}{\alpha + \beta + n} \cdot \bar{X}$$

with weights proportional to the prior sample size $\alpha + \beta$ and the actual sample size n . As $n \rightarrow \infty$ the data dominate and $\hat{p}_\pi$ pulls toward $\bar{X}$; for small n the prior shrinks the estimate toward its mean, which is why $\hat{p}_\pi$ is a *shrinkage* estimator.

Risk comparison with the sample mean. The risk of $\hat{p}_\pi$ under squared-error loss is its mean squared error, which by the bias-variance decomposition (proposition 1.2.5) is $\text{Var}_p(\hat{p}_\pi) + \text{Bias}_p(\hat{p}_\pi)^2$. Writing $\hat{p}_\pi = (\alpha + S)/(\alpha + \beta + n)$ and using $\mathbb{E}_p S = np$, $\text{Var}_p S = np(1-p)$,

$$\mathbb{E}_p \hat{p}_\pi = \frac{\alpha + np}{\alpha + \beta + n}, \quad \text{Var}_p \hat{p}_\pi = \frac{np(1-p)}{(\alpha + \beta + n)^2}$$

so, writing $N = \alpha + \beta + n$,

$$R(p, \hat{p}_\pi) = \frac{np(1-p)}{N^2} + \left(\frac{\alpha + np}{N} - p \right)^2 = \frac{np(1-p) + (\alpha - (\alpha + \beta)p)^2}{N^2} \quad (1.7)$$

Compare this with the risk of the unbiased sample proportion $\bar{X}$, which is $R(p, \bar{X}) = \text{Var}_p \bar{X} = p(1-p)/n$. The two risk curves cross: near the prior mean $p = \alpha/(\alpha + \beta)$ the shrinkage estimator $\hat{p}_\pi$ has smaller risk (its variance is reduced by the factor $(n/N)^2 < 1$ and its bias vanishes there), while near the endpoints $p \rightarrow 0$ or $p \rightarrow 1$ the bias term in (1.7) makes it larger. Neither dominates the other in the sense of definition 1.3.1, which is the phenomenon section 1.3 anticipated: the averaging and worst-case principles exist precisely to break such ties.

The least favorable prior. The prior can be tuned to flatten the risk curve (1.7). Its numerator $np(1-p) + (\alpha - (\alpha + \beta)p)^2$ is a quadratic in p , and it is constant in p exactly when the coefficients of p^2 and of p both vanish, that is when $(\alpha + \beta)^2 = n$ and $2\alpha(\alpha + \beta) = n$, which together force the symmetric choice

$$\alpha = \beta = \frac{\sqrt{n}}{2}$$

With these hyperparameters $N = \alpha + \beta + n = n + \sqrt{n}$, and the numerator collapses: $np(1-p) + (\frac{\sqrt{n}}{2} - \sqrt{n}p)^2 = \frac{n}{4}$ for every p , so

$$R(p, \hat{p}_\pi) = \frac{n/4}{(n + \sqrt{n})^2} = \frac{1}{4(1 + \sqrt{n})^2} \quad (1.8)$$

which does not depend on p . The Bayes rule for the $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ prior therefore has constant risk, so by proposition 1.4.6 it is minimax for estimating p under squared-error loss, and the $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ prior is least favorable. Explicitly the minimax estimator is

$$\hat{p}_{\text{mm}} = \frac{S + \sqrt{n}/2}{n + \sqrt{n}}$$

a shrinkage of $\bar{X}$ toward $1/2$ by an amount that vanishes as $n \rightarrow \infty$. Its constant risk (1.8) is below the maximum risk $\sup_p p(1-p)/n = 1/(4n)$ of $\bar{X}$, so trading a little bias for reduced variance strictly improves the worst case. The full development of conjugacy, posterior computation, and the admissibility of these rules is the subject of ??.

The example is the section in miniature. A conjugate prior turns the posterior computation into pattern-matching on densities, the posterior mean is the Bayes estimator for squared-error loss, that estimator shrinks the sample proportion toward the prior mean, and one distinguished prior flattens the risk and delivers the minimax rule. It also displays the tension between the two principles: the minimax estimator protects the worst case uniformly but is a peculiar object, biased and tuned to an adversary, and for most plausible values of p the plain sample proportion or a differently-tuned

Bayes rule does better. Which principle to invoke is a modeling decision, not a theorem.

Exercise 1.4.1

1. In the Gaussian location model (example 1.1.7), let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\theta, \sigma^2)$ with σ^2 known, and place a $\mathcal{N}(\mu_0, \tau^2)$ prior on θ . Using the proportionality form of (1.2), show that the posterior is $\mathcal{N}(\mu_n, \tau_n^2)$ with

$$\mu_n = \frac{\frac{n}{\sigma^2} \bar{X} + \frac{1}{\tau^2} \mu_0}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}, \quad \tau_n^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2} \right)^{-1}$$

and identify the Bayes estimator of θ under squared-error loss.

2. Let θ have a distribution F with $\mathbb{E}|\theta| < \infty$ (for instance a posterior). Prove that $a \mapsto \mathbb{E}|\theta - a|$ is minimized at every median of F , and deduce the absolute-loss claim of theorem 1.4.4. (Hint: for $a < b$ compute $\mathbb{E}|\theta - b| - \mathbb{E}|\theta - a| = \int_a^b (2F(t) - 1) dt$.)
3. Suppose Θ is finite and $L(\theta, a) = \mathbb{1}\{a \neq \theta\}$ is the 0-1 loss on $\mathcal{A} = \Theta$. Show that the posterior expected loss of an action a is $1 - \pi(a|x)$, and conclude that the Bayes rule chooses the posterior mode $\arg \max_{\theta} \pi(\theta|x)$.
4. For the Bernoulli model with n observations, verify the claim of example 1.4.7 by direct algebra: substitute $\alpha = \beta = \sqrt{n}/2$ into (1.7) and show the risk equals the constant (1.8). Then check numerically at $n = 1$ that the minimax risk $1/(4(1 + \sqrt{n})^2)$ is smaller than the maximum risk $1/(4n)$ of $\bar{X}$.
5. Prove the following partial converse structure used implicitly in proposition 1.4.6: if a rule δ has $\sup_{\theta} R(\theta, \delta) = r(\pi, \delta)$ for some prior π and δ is Bayes for π , then δ is minimax and π is least favorable, without assuming the risk is constant. (This isolates the equalization property from the constant-risk sufficient condition.)
6. Take $\Theta \subseteq \mathcal{A} \subseteq \mathbb{R}$ with $\mathcal{A}$ convex and closed, and assume $r(\pi, \delta_0) < \infty$ for some rule δ_0 . Show that a Bayes rule under squared-error loss is then essentially unique: if δ_{π} and δ'_{π} are both Bayes for the same prior π , then $\delta_{\pi}(X) = \delta'_{\pi}(X)$ for m -almost every value of X . Then explain why the finiteness assumption cannot be dropped, by describing what happens when every rule has infinite Bayes risk. (Use theorem 1.4.4 and the strict convexity of the squared-error posterior risk in the action.)

Sufficiency, Exponential Families, and Ancillarity

A sample of size n is a cumbersome object, and most of what it records is irrelevant to the parameter. In the model $\text{Ber}(p)$ the full sample $(X_1, \dots, X_n)$ says which trials succeeded, but only the count $\sum_i X_i$ bears on p . The pattern of successes is noise. A *sufficient* statistic captures everything the data say about θ , so that discarding the rest loses nothing. This chapter makes the idea precise and draws out its consequences.

Section 2.1 defines sufficiency through the conditional distribution of the data given the statistic and proves the factorization theorem, the working test for it. Section 2.2 finds the coarsest sufficient statistic and, opposite to it, the ancillary statistics that carry no information about θ . Section 2.3 isolates completeness, the property that makes the unbiased estimator built on a sufficient statistic unique, and proves Basu's theorem. Section 2.4 identifies the exponential families, where a sufficient statistic of fixed dimension exists however large the sample, and where the theory of the later chapters is sharpest.

2.1 Sufficiency and the Factorization Theorem

Fix an i.i.d. sample $X = (X_1, \dots, X_n)$ from a model $\{P_\theta : \theta \in \Theta\}$. This section makes “carries everything the data say about θ ” precise through the conditional distribution of the data given a statistic, and gives the working test for it, the factorization theorem.

A statistic $T = T(X)$ (definition 1.1.5) partitions the sample space into the level sets $\{T = t\}$. Recording T retains which cell X landed in and forgets the position within the cell. The statistic loses nothing about θ when that within-cell position is uninformative, and the sharp form of “uninformative” is that its law is the same for every θ .

2.1.1 Convention: Standard Borel Throughout From here on, wherever sufficiency is defined, proved, or used, the sample space $(\mathcal{X}, \mathcal{B})$ and the codomain $(\mathcal{T}, \mathcal{C})$ of every statistic are assumed *standard Borel*: Borel subsets of complete separable metric spaces, as every Euclidean space with its Borel σ -algebra is, and as every model in this book is. The assumption is needed twice over: it is what grants the regular conditional distributions that definition 2.1.2 requires ([5, Existence of Regular Conditional Distributions] proves the real-valued case, records in prose that the statement is really one about standard Borel spaces, and that book's warning box shows regularity can genuinely fail beyond them; the full standard-Borel development is a recorded gap in the series and this chapter uses the extension as a standing assumption), and it is what licenses reading sufficiency off the level sets of T in theorem 2.2.2: for an exotic codomain σ -algebra the level sets can carry strictly less information than $\sigma(T)$, and both results below fail as stated.

Definition 2.1.2: Sufficient Statistic

Let X be the data of a statistical model $\{P_\theta : \theta \in \Theta\}$ and let $T = T(X)$ be a statistic. Then T is *sufficient* for the model (or *sufficient for θ*) if the conditional distribution of X given $T = t$,

$$P_\theta(X \in A | T = t)$$

does not depend on θ : for each measurable A there is a version of the conditional probability, and a common regular conditional distribution $P(X \in \cdot | T = t)$, that is one and the same for every $\theta \in \Theta$.

The conditional distribution and, in the general case, the regular conditional distribution of X given T are the ones built in *Probability* through the Radon-Nikodym theorem [5]. In the discrete case they are the elementary ratios below. Sufficiency asks that this object be θ -free. When it is, one may picture generating the data in two stages: first draw T from its θ -dependent law, then fill in X from the fixed conditional distribution $P(X \in \cdot | T = t)$, a draw that consults no unknown parameter. Every trace of θ has been forced into the first stage. Given T , the residual randomness is θ -free, so a statistician who is handed $T = t$ and then simulates the second stage on a computer holds data with exactly the law of the original X : the discarded information was reconstructible without knowing θ .

The definition is exact but awkward to check, since it asks about a conditional distribution for every θ at once. In $\text{Ber}(p)$ it can be checked directly. With $T = \sum_i X_i$ and $t \in \{0, \dots, n\}$, for any $x \in \{0, 1\}^n$ with $\sum_i x_i = t$,

$$P_p(X = x | T = t) = \frac{P_p(X = x)}{P_p(T = t)} = \frac{p^t(1-p)^{n-t}}{\binom{n}{t} p^t(1-p)^{n-t}} = \frac{1}{\binom{n}{t}}$$

using $T \sim \text{Bin}(n, p)$ and that every such x has the same mass. The ratio is p -free: given t successes, the data is a uniform draw from the $\binom{n}{t}$ arrangements of those successes, whatever p may be. So $T = \sum_i X_i$ is sufficient for $\text{Ber}(p)$. Checking each model this way is laborious. The factorization theorem replaces the conditional-distribution test with an algebraic one on the density.

We work with densities against a common dominating measure. Suppose every P_θ has a density (mass function) $f_\theta = dP_\theta/d\nu$ with respect to a single σ -finite measure ν that does not depend on θ (counting measure in the discrete case, Lebesgue measure for the continuous models). Such a family is *dominated*. The joint density of the sample at x is written $f_\theta(x)$.

Theorem 2.1.3: Fisher-Neyman Factorization Theorem

Let $\{P_\theta : \theta \in \Theta\}$ be dominated by ν , with densities $f_\theta = dP_\theta/d\nu$, and let $T = T(X)$ be a statistic. Then T is sufficient for θ (definition 2.1.2) if and only if there exist measurable functions $g_\theta \geq 0$ on the range of T and $h \geq 0$ on the sample space, with h not depending on θ , such that

$$f_\theta(x) = g_\theta(T(x)) h(x) \quad (2.1)$$

for every $\theta \in \Theta$, for ν -almost every x (the exceptional null set may depend on θ).

The criterion says sufficiency is exactly the algebraic property that θ and x interact only through $T(x)$: in the density, θ touches the data through the single channel $g_\theta(T(x))$, while the remaining factor $h(x)$ carries no θ . To read a statistic off a model, one writes the joint density, and whatever combination of the data the parameter multiplies against is a sufficient statistic. The proof in the discrete case is a direct computation with conditional masses; the continuous case is taken up after the proof.

Proof:

Take ν counting measure on a countable sample space $\mathcal{X}$, so $f_\theta(x) = P_\theta(X = x)$ and the conditional probabilities are elementary ratios. Write $\mathcal{X}_t = \{x : T(x) = t\}$ for the level sets of T .

Step 1: Sufficiency implies factorization. Suppose T is sufficient. For x with $T(x) = t$ and $P_\theta(T = t) > 0$,

$$P_\theta(X = x) = P_\theta(T = t) P_\theta(X = x | T = t)$$

since $\{X = x\} \subseteq \{T = t\}$. By sufficiency the second factor is θ -free. Call it $h(x) := P(X = x | T = t)$, a function of x alone (the value $t = T(x)$ is determined by x). Set $g_\theta(t) := P_\theta(T = t)$. Then

$$f_\theta(x) = P_\theta(X = x) = g_\theta(T(x)) h(x)$$

which is (2.1). For x with $P_\theta(T = t) = 0$ one has $P_\theta(X = x) = 0$ as well, and the identity reads $0 = 0$. Assigning $h(x)$ any fixed value there leaves it θ -free. This gives the factorization.

Step 2: Factorization implies sufficiency. Suppose conversely that $f_\theta(x) = g_\theta(T(x)) h(x)$ holds. Summing over the level set $\mathcal{X}_t$, and using that $g_\theta(T(x)) = g_\theta(t)$ is constant on it,

$$P_\theta(T = t) = \sum_{x \in \mathcal{X}_t} f_\theta(x) = g_\theta(t) \sum_{x \in \mathcal{X}_t} h(x)$$

Fix t with $P_\theta(T = t) > 0$. Then $\sum_{x \in \mathcal{X}_t} h(x) > 0$. For any x with $T(x) = t$,

$$P_\theta(X = x | T = t) = \frac{P_\theta(X = x)}{P_\theta(T = t)} = \frac{g_\theta(t) h(x)}{g_\theta(t) \sum_{y \in \mathcal{X}_t} h(y)} = \frac{h(x)}{\sum_{y \in \mathcal{X}_t} h(y)}$$

The factor $g_\theta(t)$ cancels, and what remains involves only h and the level set $\mathcal{X}_t$, not θ . Whenever $P_\theta(T = t) > 0$ the conditional distribution of X given $T = t$ is the same for every θ , so T is sufficient, as we sought to show.

The cancellation of $g_\theta(t)$ is the whole mechanism. Sufficiency holds because conditioning on T divides the joint density by exactly the factor that carried θ , leaving the θ -free profile h over each level set. The two halves also expose the anatomy of a factorization: g_θ can be taken as the law of T and h as

the conditional law of X given T , though the theorem requires no such interpretation of the factors, only that they exist.

The continuous case carries the same content but cannot divide by $P_\theta(X = x) = 0$. The cancellation now happens at the level of Radon-Nikodym densities: dividing the P_θ -density on a fiber $\{T = t\}$ by the factor carrying θ recovers a fiber law free of θ , the mechanism of Step 2 again. The genuinely harder half is the converse, that sufficiency produces a single θ -free factor h serving every θ at once: for a general dominated family this is the theorem of Halmos and Savage, and its proof replaces ν by a privileged dominating measure, a countable mixture $\sum_i c_i P_{\theta_i}$ equivalent to the whole family. That proof is not reproduced in this book; the general case is established, from the Radon-Nikodym theorem up, in [12, §2.6]. The upshot is unchanged: (2.1) is the operative criterion, in both directions, in every model in this book, discrete or continuous.

With the criterion in hand, reading a sufficient statistic off a model is a matter of inspecting where the parameter meets the data in the joint density. The three running models illustrate the whole range: a discrete family, a two-parameter smooth family, and the irregular family whose support moves with θ .

Example 2.1.4: Sufficient Statistics in the Three Running Models

Each model is the i.i.d. sample version of example 1.1.7. In each the factorization (2.1) is exhibited explicitly.

1. *Bernoulli* $\text{Ber}(p)$. For $x \in \{0, 1\}^n$ the joint mass is

$$f_p(x) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{\sum_i x_i} (1-p)^{n-\sum_i x_i} = \underbrace{\left(\frac{p}{1-p}\right)^{T(x)} (1-p)^n}_{g_p(T(x))} \cdot \underbrace{1}_{h(x)}$$

with $T(x) = \sum_i x_i$; the odds-ratio form divides by $1-p$, so read it for $0 < p < 1$ and use the middle expression $g_p(t) = p^t (1-p)^{n-t}$ directly at the endpoints, which keeps the factorization valid on all of $\Theta = [0, 1]$. The parameter reaches the data only through the count T , so $T = \sum_i X_i$ is sufficient for p , recovering by theorem 2.1.3 the conditional-distribution computation above. Here $h \equiv 1$: the whole density is a function of T .

2. *Gaussian* $\mathcal{N}(\mu, \sigma^2)$. For $x \in \mathbb{R}^n$ the joint density against Lebesgue measure is

$$f_{\mu, \sigma^2}(x) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

Expanding $\sum_i (x_i - \mu)^2 = \sum_i x_i^2 - 2\mu \sum_i x_i + n\mu^2$ shows the data enter the exponent only through the pair $T(x) = (\sum_i x_i, \sum_i x_i^2)$:

$$f_{\mu, \sigma^2}(x) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{n\mu^2}{2\sigma^2} + \frac{\mu}{\sigma^2} \sum_i x_i - \frac{1}{2\sigma^2} \sum_i x_i^2\right)}_{g_{\mu, \sigma^2}(T(x))} \cdot \underbrace{1}_{h(x)}$$

so $T = (\sum_i X_i, \sum_i X_i^2)$ is sufficient for (μ, σ^2) . Equivalently $(\bar{X}, S^2)$ is sufficient, being a one-to-one function of T , which is why the pair $(\bar{X}, S^2)$ of example 1.1.7 summarizes the Gaussian sample without loss.

3. *Uniform* $\text{Unif}(0, \theta)$. The single-observation density is $p_\theta(x) = \theta^{-1} \mathbb{1}\{0 < x < \theta\}$, so for $x \in (0, \infty)^n$,

$$f_\theta(x) = \prod_{i=1}^n \theta^{-1} \mathbb{1}\{0 < x_i < \theta\} = \theta^{-n} \mathbb{1}\left\{\max_i x_i < \theta\right\} \mathbb{1}\left\{\min_i x_i > 0\right\}$$

because all n observations lie below θ exactly when their maximum does. With $T(x) = \max_i x_i$,

$$f_\theta(x) = \underbrace{\theta^{-n} \mathbb{1}\{T(x) < \theta\}}_{g_\theta(T(x))} \cdot \underbrace{\mathbb{1}\{\min_i x_i > 0\}}_{h(x)}$$

so $T = \max_i X_i = X_{(n)}$ is sufficient for θ . The parameter meets the data only through the indicator $\mathbb{1}\{T < \theta\}$; the positivity factor h is θ -free. This is the model where sufficiency and the elementary theory part ways: the sufficient statistic is the maximum, not the mean, precisely because θ sits in the support rather than in a smooth density.

In each case the factorization is read off by inspection, and the sufficient statistic is the vector of data combinations the parameter multiplies against: the count for Bernoulli, the pair $(\sum x_i, \sum x_i^2)$ for the Gaussian, the maximum for the uniform. The uniform model warns against a common expectation. One might guess the sample mean is always the informative summary, but here the mean discards what matters and the maximum keeps it. Sufficiency is a property of how the parameter enters the model, not a fixed recipe.

Everything so far treats sufficiency as data reduction. Its decision-theoretic force is that this reduction is free: no rule that uses the whole sample beats every rule that uses only T . This is the claim that a sufficient statistic loses no information, and it is the seed of the improvement theorems of chapter 3.

Proposition 2.1.5: Sufficiency Loses No Information

Let $T = T(X)$ be sufficient for $\{P_\theta : \theta \in \Theta\}$ (definition 2.1.2), and let $L(\theta, a) \geq 0$ be any loss function. For every decision rule $\delta(X)$ there is a randomized rule δ^* depending on the data only through T whose risk matches that of δ :

$$R(\theta, \delta^*) = R(\theta, \delta) \quad \text{for every } \theta \in \Theta$$

The rule δ^* is constructed by drawing an auxiliary $\tilde{X}$ from the conditional distribution $P(X \in \cdot | T)$ and returning $\delta(\tilde{X})$. The draw uses no unknown parameter.

Proof:

Given the value $T = t$, draw $\tilde{X}$ from the conditional distribution $P(X \in \cdot | T = t)$, using a source of randomness independent of the data, and set $\delta^* := \delta(\tilde{X})$. The draw is possible without knowing θ precisely because T is sufficient: the conditional distribution $P(X \in \cdot | T = t)$ is one fixed law, the same for every θ , so δ^* is a legitimate statistic (a θ -free randomized function of T).

Fix θ . Conditionally on $T = t$, the auxiliary $\tilde{X}$ has law $P(X \in \cdot | T = t) = P_\theta(X \in \cdot | T = t)$, the last equality by sufficiency, which is the same as the conditional law of the original data X given

$T = t$ under P_θ . Hence $\tilde{X}$ has the same law as X under P_θ : for every measurable A ,

$$P_\theta(\tilde{X} \in A) = \int P(X \in A | T = t) dP_\theta^T(t) = \int P_\theta(X \in A | T = t) dP_\theta^T(t) = P_\theta(X \in A)$$

where P_θ^T is the law of T under P_θ and the last step is the tower property [5, Tower Property]. So $\tilde{X} \sim X$ under P_θ , and therefore $\delta(\tilde{X}) \sim \delta(X)$. The losses have equal expectation:

$$R(\theta, \delta^*) = \mathbb{E}_\theta[L(\theta, \delta(\tilde{X}))] = \mathbb{E}_\theta[L(\theta, \delta(X))] = R(\theta, \delta)$$

for every θ , which is the claim, completing the proof.

Any rule built on the full sample is thus matched, risk for risk, by one built on T alone together with a parameter-free coin toss. Nothing an analyst could extract from the discarded part of the data improves on T , since the discarded part can be regenerated from T by a computation that never consults θ . Reducing to a sufficient statistic loses nothing.

The construction does more than preserve risk when the loss is convex. Averaging out the auxiliary randomness (replacing $\delta(\tilde{X})$ by its conditional mean given T) can only shrink risk, by Jensen's inequality. That sharpening is the Rao-Blackwell theorem, taken up with the estimation theory of section 3.3. Here the point is the weaker but unconditional one: sufficiency never costs risk, so the search for good rules may be confined to functions of a sufficient statistic without loss.

Exercise 2.1.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ on $\{0, 1, 2, \dots\}$. Write the joint mass function and use theorem 2.1.3 to show that $T = \sum_i X_i$ is sufficient for λ . Then verify sufficiency directly by computing $P_\lambda(X = x | T = t)$ and confirming it is λ -free. (You may use that $\sum_i X_i \sim \text{Poi}(n\lambda)$.)
2. Show that the identity map $T(X) = X$ (the full sample) and any one-to-one function of a sufficient statistic are sufficient. Deduce that sufficiency alone does not single out a “best” reduction, and name the additional requirement (developed in section 2.2) that does.
3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$ with density $f_\theta(x) = \theta e^{-\theta x} \mathbb{1}\{x > 0\}$, $\theta > 0$. Exhibit a factorization (2.1) and identify a one-dimensional sufficient statistic.
4. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(\alpha, \beta)$ with both endpoints unknown, $\theta = (\alpha, \beta)$, $\alpha < \beta$. Write the joint density and use the factorization criterion to show that $T = (X_{(1)}, X_{(n)}) = (\min_i X_i, \max_i X_i)$ is sufficient. Explain why neither endpoint statistic alone suffices.
5. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma_0^2)$ with σ_0^2 known and μ the only unknown. Show from the factorization that $T = \sum_i X_i$ (equivalently $\bar{X}$) is sufficient for μ , and contrast with the two-dimensional sufficient statistic $(\sum_i X_i, \sum_i X_i^2)$ of example 2.1.4 when σ^2 is also unknown.
6. For any i.i.d. sample $X_1, \dots, X_n$ from a model on the line, show that the vector of order statistics $(X_{(1)}, \dots, X_{(n)})$ is sufficient for θ . (Interpret the joint density's symmetry in the coordinates.) What information about the sample has been discarded, and why is it θ -free?

2.2 Minimal Sufficiency and Ancillarity

Sufficiency has a slack the factorization theorem does not remove: many statistics are sufficient, and some carry more clutter than others. In the Bernoulli model the count $\sum_i X_i$ is sufficient, but so is the whole sample $(X_1, \dots, X_n)$, and so is the pair $(\sum_i X_i, X_1)$, which appends an irrelevant coordinate. All three lose no information about p ; the first is the tightest. This section finds the tightest sufficient statistic, the one that cannot be reduced further without sacrificing sufficiency, and gives a mechanical test for it. It then turns to the opposite extreme, statistics whose distribution ignores θ entirely and so say nothing about it on their own.

A sufficient statistic T compresses the data, and coarser statistics compress more. If $T = \varphi(U)$ for some sufficient U , then T is a further reduction of U : it merges values of U that φ cannot tell apart. The reduction that merges as much as possible, while keeping sufficiency, is the one below which no information survives to be discarded.

Definition 2.2.1: Minimal Sufficient Statistic

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model with data X . A sufficient statistic T (definition 2.1.2) is *minimal sufficient* if, for every sufficient statistic U , there is a measurable function φ with

$$T = \varphi(U) \quad P_\theta\text{-a.s. for every } \theta \in \Theta$$

that is, T is a function of every other sufficient statistic.

A minimal sufficient statistic is the coarsest sufficient reduction of the data. Being a function of every sufficient U means T can be computed from any U , so U never merges two data points that T resolves. Equivalently, T induces the coarsest partition of the sample space that a sufficient statistic can induce. Two minimal sufficient statistics are functions of each other, hence in one-to-one correspondence almost surely, so “the” minimal sufficient statistic is well-defined up to relabelling. What the definition does not give is a way to recognize a minimal sufficient statistic, or even to know one exists. Quantifying over all sufficient U is not something one checks directly. The next result converts the condition into a test that reads only the density.

The idea is to compare the density at two data points. If the ratio $f_\theta(x)/f_\theta(y)$ does not depend on θ , then x and y give proportional likelihoods, with a proportionality constant the same under every candidate law: no observation could favor θ differently depending on which of x, y was seen. A minimal sufficient statistic should merge exactly such pairs, and no others.

Theorem 2.2.2: Lehmann-Scheffé Minimality Criterion

Let $\{P_\theta : \theta \in \Theta\}$ be dominated, with densities f_θ with respect to a common σ -finite measure, and fix the sample space to the set $\mathcal{X}_0 = \{x : f_\theta(x) > 0 \text{ for some } \theta\}$, assumed measurable, where the data can land. Suppose T is a statistic such that, for all $x, y \in \mathcal{X}_0$,

$$\frac{f_\theta(x)}{f_\theta(y)} \text{ is constant in } \theta \quad \Longleftrightarrow \quad T(x) = T(y) \quad (2.2)$$

the left side read multiplicatively: there is a constant $c(x, y)$ with $f_\theta(x) = c(x, y) f_\theta(y)$ for every $\theta \in \Theta$. Then T is minimal sufficient.

The criterion says T separates the data into exactly the level sets on which the likelihood ratio is θ -free. The reverse direction of (2.2) ($T(x) = T(y)$ *only* when the ratio is constant) forces T to be sufficient, by collapsing only likelihood-equivalent points; the forward direction ($T(x) = T(y)$ whenever the ratio is constant) forces T to be minimal, by collapsing every likelihood-equivalent pair. As with theorem 2.1.3, the proof is given in the discrete case; the continuous case is taken up after the proof.

Proof:

Take ν counting measure on a countable sample space with the σ -algebra of all subsets, as in the proof of theorem 2.1.3. Two features of this setting are used freely: the only ν -null set is the empty set, so the factorization (2.1) of a sufficient statistic holds at every point, not just ν -almost everywhere; and every function is measurable, so the h , g_θ and φ constructed below meet the measurability that theorem 2.1.3 and definition 2.2.1 require. Assume (2.2). The argument has two halves, matching the two implications: first that T is sufficient, then that it is a function of every sufficient statistic.

Step 1: T is sufficient. Let $\mathcal{T} = T(\mathcal{X}_0)$ be the set of values T takes, and for each value $t \in \mathcal{T}$ choose once and for all a representative point $x_t \in \mathcal{X}_0$ with $T(x_t) = t$. This choice fixes a map $x \mapsto x_{T(x)}$ sending each x to the representative of its own T -value. By construction $T(x_{T(x)}) = T(x)$, so x and $x_{T(x)}$ share a T -value, and by the reverse direction of (2.2) the ratio $f_\theta(x)/f_\theta(x_{T(x)})$ is constant in θ . Call that constant $h(x)$, a function of x alone. Then

$$f_\theta(x) = f_\theta(x_{T(x)}) h(x)$$

Set $g_\theta(t) = f_\theta(x_t)$, a function of θ and of $t = T(x)$ only. The display reads $f_\theta(x) = g_\theta(T(x)) h(x)$, the factorization of theorem 2.1.3, so T is sufficient.

Step 2: T is a function of every sufficient statistic. Let U be any sufficient statistic. By theorem 2.1.3 its density factors as

$$f_\theta(x) = \tilde{g}_\theta(U(x)) \tilde{h}(x)$$

for some $\tilde{g}_\theta \geq 0$ and $\tilde{h} \geq 0$. Suppose $U(x) = U(y)$ for two points $x, y \in \mathcal{X}_0$. Since $y \in \mathcal{X}_0$, some θ has $f_\theta(y) = \tilde{g}_\theta(U(y)) \tilde{h}(y) > 0$, so $\tilde{h}(y) > 0$. Then for every θ ,

$$f_\theta(x) = \tilde{g}_\theta(U(x)) \tilde{h}(x) = \tilde{g}_\theta(U(y)) \tilde{h}(y) \cdot \frac{\tilde{h}(x)}{\tilde{h}(y)} = \frac{\tilde{h}(x)}{\tilde{h}(y)} f_\theta(y)$$

since the factors $\tilde{g}_\theta(U(x))$ and $\tilde{g}_\theta(U(y))$ coincide. The multiplier $\tilde{h}(x)/\tilde{h}(y)$ is free of θ , so the ratio is constant in θ , and by the forward direction of (2.2) this forces $T(x) = T(y)$. Thus $U(x) = U(y)$ implies $T(x) = T(y)$: the value of U determines the value of T . Define φ on the range of U by $\varphi(u) = T(x)$ for any x with $U(x) = u$. The implication just proved makes φ well-defined, and $T = \varphi(U)$ on $\mathcal{X}_0$. Hence T is a function of U , and since U was an arbitrary sufficient statistic, T is minimal sufficient, as we sought to show.

The discrete setting did real work. In a general dominated family the factorization of U holds only ν -almost everywhere, with an exceptional set that can depend on θ , so Step 2 cannot evaluate it at a chosen pair (x, y) ; the maps built by choosing representatives are not automatically measurable; and $\mathcal{X}_0$, an uncountable union of measurable sets, is measurable only because the statement assumes it. The repairs pass through the countable equivalent mixture of Halmos and Savage, and they deliver $T = \varphi(U)$ only P_θ -almost surely, which is all definition 2.2.1 asks. That development is

not reproduced in this book: [11, §1.6] proves the countable case and reduces most uncountable families to it, and [10, §3.4] states the criterion for dominated families and locates the remaining measure-theoretic difficulties. The upshot is unchanged: (2.2) is the operative test in every model in this book, discrete or continuous.

The proof leaves a recipe. To find a minimal sufficient statistic, form the likelihood ratio $f_\theta(x)/f_\theta(y)$, determine which pairs (x, y) make it free of θ , and read off the statistic that has those pairs as its level sets. The equivalence relation “ $x \sim y$ iff the ratio is θ -free” is exactly the partition of a minimal sufficient statistic, and any statistic realizing that partition serves. Existence is a byproduct: the relation always exists, and any labelling of its classes is minimal sufficient (when it is a statistic, which for the dominated models below it is). The criterion also settles minimality of the sufficient statistics already in hand, all of which turn out minimal.

Example 2.2.3: Minimal Sufficient Statistics in the Running Models

The criterion (2.2) is applied to each running model (example 1.1.7) with data $X = (X_1, \dots, X_n)$.

1. *The Bernoulli model* $\text{Ber}(p)$, $0 < p < 1$. The mass function of a binary vector $x \in \{0, 1\}^n$ is $f_p(x) = p^{s(x)}(1-p)^{n-s(x)}$ where $s(x) = \sum_i x_i$. For two vectors x, y ,

$$\frac{f_p(x)}{f_p(y)} = p^{s(x)-s(y)}(1-p)^{s(y)-s(x)} = \left(\frac{p}{1-p}\right)^{s(x)-s(y)}$$

The odds $p/(1-p)$ range over $(0, \infty)$ as p ranges over $(0, 1)$, so this ratio is constant in p if and only if the exponent $s(x) - s(y)$ vanishes, that is $s(x) = s(y)$. The level sets of the ratio-equivalence are the level sets of $T = \sum_i X_i$, so $T = \sum_i X_i$ is minimal sufficient. The full sample and the count are both sufficient, but only the count is minimal.

2. *The Gaussian model* $\mathcal{N}(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$. The joint density is

$$f_\theta(x) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2\right) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_i x_i^2 + \frac{\mu}{\sigma^2} \sum_i x_i - \frac{n\mu^2}{2\sigma^2}\right)$$

The log-ratio for two samples x, y is

$$\log \frac{f_\theta(x)}{f_\theta(y)} = -\frac{1}{2\sigma^2} \left(\sum_i x_i^2 - \sum_i y_i^2 \right) + \frac{\mu}{\sigma^2} \left(\sum_i x_i - \sum_i y_i \right)$$

which depends on $\theta = (\mu, \sigma^2)$ through the coefficients $1/\sigma^2$ and μ/σ^2 . As (μ, σ^2) varies these two coefficients vary freely, so the expression is constant in θ (namely zero) for all θ if and only if both bracketed differences vanish: $\sum_i x_i = \sum_i y_i$ and $\sum_i x_i^2 = \sum_i y_i^2$. The ratio-equivalence classes are the level sets of $T = (\sum_i X_i, \sum_i X_i^2)$, equivalently of the pair $(\bar{X}, S^2)$ of sample mean and variance, which is therefore minimal sufficient. Its dimension is two whatever the sample size, matching the two free parameters.

3. *The uniform model* $\text{Unif}(0, \theta)$, $\theta > 0$. The joint density is $f_\theta(x) = \theta^{-n} \mathbb{1}\{0 < x_{(1)}\} \mathbb{1}\{x_{(n)} < \theta\}$, with $x_{(1)} = \min_i x_i$ and $x_{(n)} = \max_i x_i$. On the sample space where all coordinates are positive, the ratio for two samples with all coordinates in the positive reals is

$$\frac{f_\theta(x)}{f_\theta(y)} = \frac{\mathbb{1}\{x_{(n)} < \theta\}}{\mathbb{1}\{y_{(n)} < \theta\}}$$

wherever the denominator is nonzero. As θ increases past $\max(x_{(n)}, y_{(n)})$ both indicators turn on and the ratio is 1; but for θ between $x_{(n)}$ and $y_{(n)}$ (when these differ) exactly one indicator is on and the ratio is 0 or undefined. The ratio is thus constant in θ precisely when $x_{(n)} = y_{(n)}$, so the maximum $T = X_{(n)} = \max_i X_i$ is minimal sufficient. The whole sample is sufficient, and even the order statistics are, but the single number $X_{(n)}$ suffices minimally.

The Bernoulli and Gaussian cases share a feature the uniform case lacks. In the first two the minimal sufficient statistic has a fixed dimension, one and two respectively, no matter how large n grows, so the sample of n numbers compresses to a summary of bounded size. This is the signature of an exponential family (section 2.4), where the count of sufficient coordinates equals the number of parameters. The uniform model has a one-dimensional minimal sufficient statistic too, but for a different reason, and it is a member of no exponential family. The dimension of the minimal sufficient statistic is a first diagnostic for how much a model can be reduced.

Minimal sufficiency squeezes the data down to its informative core. The opposite notion isolates a piece of the data that is informative about nothing: a statistic whose distribution is the same under every candidate law. Such a statistic varies with the data but not with θ , so its observed value records where the sample landed without narrowing which law it came from.

Definition 2.2.4: Ancillary Statistic

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model with data X . A statistic $A = A(X)$ is *ancillary* if its distribution under P_θ does not depend on θ : the law of A is the same measure for every $\theta \in \Theta$. It is *first-order ancillary* if only its mean is θ -free, $\mathbb{E}_\theta[A] = c$ for a constant c not depending on θ .

An ancillary statistic carries no information about θ on its own. Since its sampling distribution is fixed across the model, observing A cannot shift belief between candidate laws: the value of A is as likely under one θ as under any other. Ancillarity is the exact complement of sufficiency in a sampling sense. A sufficient statistic retains everything about θ and its complement is θ -free noise; an ancillary statistic *is* θ -free noise, retaining nothing. First-order ancillarity relaxes the requirement to the mean alone, useful when the full distribution is hard to pin down but the expectation is not. A first-order ancillary need not be ancillary, since matching one moment across the model is weaker than matching the whole law. The interesting ancillaries arise in families with a group structure, where changing θ moves the data in a way some coordinate is blind to.

Example 2.2.5: Ancillaries in Location Families and the Gaussian Model

Each part exhibits a statistic whose sampling law does not move with the parameter.

1. *The range in a location family.* Let $X_i = \theta + \varepsilon_i$ for $i = 1, \dots, n$, where the errors $\varepsilon_1, \dots, \varepsilon_n \stackrel{\text{iid}}{\sim} F$ have a fixed law F not depending on θ , and $\theta \in \mathbb{R}$ is a location parameter. This is a *location family*: shifting θ translates the whole sample. The *range*

$$R = X_{(n)} - X_{(1)} = \max_i X_i - \min_i X_i$$

is ancillary. Writing $X_i = \theta + \varepsilon_i$, the common shift cancels in every difference:

$$R = \max_i (\theta + \varepsilon_i) - \min_i (\theta + \varepsilon_i) = (\theta + \max_i \varepsilon_i) - (\theta + \min_i \varepsilon_i) = \max_i \varepsilon_i - \min_i \varepsilon_i$$

so R equals the range of the errors, whose law is F -determined and free of θ . Hence the distribution of R is the same for every θ , and R is ancillary. The range measures the spread of

the sample, which a pure translation leaves untouched, so it says nothing about the location. The same argument makes any difference $X_i - X_j$, and any function of the centered sample $(X_1 - \bar{X}, \dots, X_n - \bar{X})$, ancillary in a location family.

2. *The difference $X_1 - X_2$ in the Gaussian location model.* Fix the variance and take $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known and $\mu \in \mathbb{R}$ the only parameter. This is the location family of part (1) with $F = \mathcal{N}(0, \sigma^2)$. The difference $D = X_1 - X_2$ satisfies

$$D = X_1 - X_2 = (\mu + \varepsilon_1) - (\mu + \varepsilon_2) = \varepsilon_1 - \varepsilon_2 \sim \mathcal{N}(0, 2\sigma^2)$$

a law free of μ (the mean is 0 and the variance is $\text{Var}(\varepsilon_1) + \text{Var}(\varepsilon_2) = 2\sigma^2$, both independent of μ), so D is ancillary for μ . The pair $(\bar{X}, D)$ splits the sample into a part that tracks μ , the mean $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$, and a part D blind to it. Once the variance is also unknown the picture changes: $D \sim \mathcal{N}(0, 2\sigma^2)$ still has a μ -free law but now depends on σ^2 , so D is no longer ancillary in the full location-scale model.

3. *The sample size under a mixed design.* Suppose the sample size N is itself random, drawn from a distribution fixed by the experimental protocol and not by θ , before $X_1, \dots, X_N \stackrel{\text{iid}}{\sim} P_\theta$ are observed. Then N is ancillary: its law is the protocol's, free of θ , so N alone reveals nothing about the parameter. It nonetheless governs the precision of any inference drawn from the sample, which is the point of the conditionality principle below.

Sufficiency and ancillarity mark the two poles of a statistic's relation to θ : one retains all the information, the other none. Between them sits the question of how they interact, and one guiding idea. The *conditionality principle* holds that inference should be carried out conditional on the observed value of an ancillary statistic. The reasoning is visible in the mixed-design example: the random sample size N says nothing about θ , but it fixes how much the data can say, so a report of precision that averages over the unrealized values of N misstates the precision of the sample seen. Conditioning on $A = a$ restores the relevant reference set, the outcomes that could have occurred given the ancillary configuration that did occur. The principle is not a theorem but a stance, and unconditional frequentist methods sometimes violate it. The tension it names recurs when likelihood and frequentist reasoning are compared in ???. The structural payoff of pairing the two notions comes next: Section 2.3 shows that when a sufficient statistic is complete it is independent of every ancillary, so the informative core and the θ -free noise separate cleanly.

Exercise 2.2.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$ with density $f_\theta(x) = \theta e^{-\theta x} \mathbb{1}\{x > 0\}$ for $\theta > 0$. Use the Lehmann-Scheffé criterion (theorem 2.2.2) to show that $T = \sum_i X_i$ is minimal sufficient. Which quantity of the sample does T record, and why is a single number enough however large n is?
2. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ with $\lambda > 0$. Show that $T = \sum_i X_i$ is minimal sufficient by computing the likelihood ratio for two samples x, y and identifying when it is free of λ .
3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(\theta, \theta + 1)$ with $\theta \in \mathbb{R}$, a location family on an interval of fixed length. Show that the pair $(X_{(1)}, X_{(n)})$ of sample minimum and maximum is minimal sufficient, and that the range $R = X_{(n)} - X_{(1)}$ is ancillary. Why does the minimal sufficient statistic here have dimension two, unlike $\text{Unif}(0, \theta)$ where it has dimension one?
4. In the Gaussian location-scale model $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown,

is the range $R = X_{(n)} - X_{(1)}$ ancillary? Is the ratio R/S (with S the sample standard deviation) ancillary? Justify each answer by tracking how the statistic transforms under $x_i \mapsto ax_i + b$.

5. Give a statistic that is first-order ancillary (definition 2.2.4) but not ancillary. Take $X \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 > 0$ the parameter and consider $A = \mathbb{1}\{X > 0\}$. Compute $\mathbb{E}_{\sigma^2}[A]$ and the full law of A , and explain which fails to be θ -free. (Here A is in fact ancillary. Then modify the model to $X \sim \mathcal{N}(\mu, 1)$ with μ the parameter and reconsider $A = \mathbb{1}\{X > 0\}$.)
6. In the location family of example 2.2.5(1), let $A = (X_1 - \bar{X}, \dots, X_n - \bar{X})$ be the vector of residuals. Prove A is ancillary. Then explain why A together with $\bar{X}$ recovers the whole sample, yet A alone says nothing about θ , and relate this to the split of the Gaussian sample in example 2.2.5(2).

2.3 Completeness and Basu's Theorem

Sufficiency reduces the data to a statistic that keeps all the information about θ ; minimal sufficiency (section 2.2) makes that reduction as coarse as possible. Neither property says anything about how the reduction interacts with unbiased estimation, and that is where the next ideas are needed. If T is sufficient and one seeks an unbiased estimator of a parameter that is a function of T , there may be many such estimators, or a wasteful one that can be improved. Completeness is the condition under which the ambiguity disappears: an unbiased function of T is then unique. The same condition, paired with sufficiency, forces a sharp probabilistic consequence, that T is independent of every statistic whose distribution ignores θ . This is Basu's theorem, and it turns independence proofs that would otherwise require a joint-density computation into one-line arguments.

Definition 2.3.1: Complete and Bounded-Complete Statistics

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model and let $T = T(X)$ be a statistic with values in $(\mathcal{T}, \mathcal{C})$. Write P_θ^T for the law of T under P_θ . The statistic T (or the family $\{P_\theta^T\}$ of its laws) is *complete* if the only measurable function $g : \mathcal{T} \rightarrow \mathbb{R}$ with $\mathbb{E}_\theta |g(T)| < \infty$ for all θ and satisfying

$$\mathbb{E}_\theta [g(T)] = 0 \quad \text{for all } \theta \in \Theta$$

is the zero function in the sense that $g(T) = 0$ holds P_θ -a.s. for every $\theta \in \Theta$. The statistic T is *bounded-complete* if the same implication holds under the added hypothesis that g is bounded.

The definition is a statement about the family of laws $\{P_\theta^T\}$, not about a single distribution. It asks that this family be rich enough to pin down a function from its integrals: if g integrates to zero against every member of the family, then g was already zero (as a random variable on T). Completeness is thus a condition of non-degeneracy on the parametrization, the model must move the law of T around enough that no nonzero $g(T)$ can average to zero everywhere at once. Bounded-completeness restricts the test class to bounded g and so is the weaker property of T : every complete statistic is bounded-complete, since the defining implication for the larger class of all g contains the implication for bounded g . The converse fails, and bounded-completeness is the property needed in several

places,¹ so the distinction is kept.

The reason completeness earns its place is the uniqueness it delivers. Suppose $g_1(T)$ and $g_2(T)$ are two unbiased estimators of the same quantity $\psi(\theta)$, so $\mathbb{E}_\theta[g_1(T)] = \mathbb{E}_\theta[g_2(T)] = \psi(\theta)$ for all θ . Their difference $g = g_1 - g_2$ satisfies $\mathbb{E}_\theta[g(T)] = 0$ for all θ , and completeness forces $g(T) = 0$ a.s., that is $g_1(T) = g_2(T)$ almost surely. An unbiased function of a complete statistic is therefore essentially unique. Paired with sufficiency, this is what proves the Lehmann-Scheffé theorem in section 3.3, where conditioning on a complete sufficient statistic produces the unique minimum variance unbiased estimator. The present section gives the completeness half of that machine and its independence corollary.

A concrete case makes the definition tangible before the general theory. The Bernoulli count is complete, and the argument is a fact about polynomials.

Example 2.3.2: Completeness of the Bernoulli Count

Let $X = (X_1, \dots, X_n)$ be a random sample from $\text{Ber}(p)$, $p \in \Theta = (0, 1)$, and let $T = \sum_{i=1}^n X_i$, so $T \sim \text{Bin}(n, p)$ (example 1.1.7). Fix a function $g: \{0, 1, \dots, n\} \rightarrow \mathbb{R}$ with $\mathbb{E}_p[g(T)] = 0$ for all $p \in (0, 1)$. Writing out the binomial expectation,

$$\mathbb{E}_p[g(T)] = \sum_{t=0}^n g(t) \binom{n}{t} p^t (1-p)^{n-t} = (1-p)^n \sum_{t=0}^n g(t) \binom{n}{t} \left(\frac{p}{1-p}\right)^t$$

Set $r = p/(1-p)$. As p ranges over $(0, 1)$, r ranges over all of $(0, \infty)$. Since $(1-p)^n \neq 0$, the vanishing of $\mathbb{E}_p[g(T)]$ for every p is equivalent to

$$\sum_{t=0}^n g(t) \binom{n}{t} r^t = 0 \quad \text{for all } r \in (0, \infty)$$

The left side is a polynomial in r of degree at most n , and a polynomial that vanishes on an interval is identically zero. Hence every coefficient $g(t) \binom{n}{t}$ is zero. As $\binom{n}{t} \neq 0$, this forces $g(t) = 0$ for $t = 0, 1, \dots, n$. Since T takes each of these values with positive probability under every $p \in (0, 1)$, the conclusion $g(T) = 0$ holds P_p -a.s. for all p , and T is complete.

The Bernoulli argument is the model for the whole class of one-parameter exponential families: completeness of the natural statistic reduces to the fact that its moment or Laplace transform determines the coefficients, so that a function integrating to zero against the whole one-parameter sweep must vanish. The Gaussian natural statistic is complete for the same structural reason, established once the exponential-family machinery is in place (section 2.4, theorem 2.4.3). The completeness of $(\sum_i X_i, \sum_i X_i^2)$ for $\mathcal{N}(\mu, \sigma^2)$ is recorded there.

Completeness alone does not connect the statistic to the rest of the data. The link comes from pairing it with sufficiency and with the opposite of sufficiency, the ancillary statistics of section 2.2, whose laws carry no information about θ . The next theorem is the payoff.

Before the statement, the mechanism is worth setting out, because it explains why the two hypotheses are exactly the ones needed. Let T be sufficient and V ancillary. To show $T \perp\!\!\!\perp V$ it suffices to show that the conditional law of V given T equals the unconditional law of V . Ancillarity makes the unconditional law $P_\theta(V \in A)$ free of θ ; sufficiency makes the conditional law $P_\theta(V \in A|T)$ free of θ as well, since V is a function of the data and the conditional distribution of the data given T is

¹it is what Basu's theorem uses below, applied to a bounded function of the complete statistic.

θ -free. Both sides of the equation one wants are therefore θ -free functions of T , and their difference is a θ -free function of T whose expectation vanishes for every θ . Completeness closes the argument by forcing that difference to be zero. Each hypothesis discharges one obstacle, and completeness is what converts “mean zero for all θ ” into “zero”.

Theorem 2.3.3: Basu's Theorem

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model. If T is a boundedly complete sufficient statistic (definition 2.1.2, definition 2.3.1) and V is an ancillary statistic (definition 2.2.4), then T and V are independent under P_θ for every $\theta \in \Theta$.

Proof:

Fix a measurable set A in the range of V . The plan is to show that the conditional probability $P_\theta(V \in A|T)$ equals the constant $P_\theta(V \in A)$ almost surely. Independence of T and V then follows, because a conditional probability of an event about V that does not depend on the conditioning value of T is exactly the statement that T and V are independent.

Step 1: The unconditional probability is θ -free. Since V is ancillary, its law does not depend on θ (definition 2.2.4), so

$$P_\theta(V \in A) = \alpha_A$$

for a constant $\alpha_A \in [0, 1]$ that is the same for every θ .

Step 2: The conditional probability is θ -free. Because T is sufficient, the conditional distribution of the data X given T does not depend on θ (definition 2.1.2). The event $\{V \in A\}$ is an event determined by X (as $V = V(X)$ is a function of the data), so its conditional probability given T is computed from that θ -free conditional distribution. Hence there is a fixed measurable function φ_A , not depending on θ , with

$$P_\theta(V \in A|T) = \varphi_A(T) \quad P_\theta\text{-a.s., for every } \theta$$

Sufficiency is used precisely here: without it the conditional probability could carry θ through the conditioning, and no single φ_A would serve.

Step 3: Completeness pins the conditional to the unconditional. By the tower property (iterated expectation), for every θ ,

$$\mathbb{E}_\theta[\varphi_A(T)] = \mathbb{E}_\theta[P_\theta(V \in A|T)] = P_\theta(V \in A) = \alpha_A$$

the first equality by Step 2, the last by Step 1. Consider the function

$$g(T) = \varphi_A(T) - \alpha_A$$

which is bounded (it takes values in $[-1, 1]$, being a difference of a probability and a constant) and satisfies $\mathbb{E}_\theta[g(T)] = \alpha_A - \alpha_A = 0$ for every θ . Bounded-completeness of T (definition 2.3.1) now forces $g(T) = 0$ P_θ -a.s. for every θ , that is

$$P_\theta(V \in A|T) = \varphi_A(T) = \alpha_A = P_\theta(V \in A) \quad P_\theta\text{-a.s.}$$

Step 4: Conclude independence. The conditional probability of $\{V \in A\}$ given T equals its unconditional probability almost surely. For any measurable set B in the range of T , integrating over $\{T \in B\}$ and using the defining property of conditional expectation gives

$$P_\theta(V \in A, T \in B) = \mathbb{E}_\theta[\mathbb{1}_{\{T \in B\}} P_\theta(V \in A|T)] = \alpha_A P_\theta(T \in B) = P_\theta(V \in A) P_\theta(T \in B)$$

Since A and B were arbitrary, T and V are independent under P_θ , and as θ was arbitrary, this holds for every $\theta \in \Theta$, completing the proof.

Two features of the argument deserve comment. First, only bounded-completeness is used, applied to the bounded function $\varphi_A - \alpha_A$. This is why the theorem is stated with the weaker hypothesis, and why bounded-completeness was kept as a named notion. Second, the direction of the theorem is worth registering: it converts a qualitative pairing (a complete sufficient statistic and any ancillary) into hard independence, with no distributional computation. The usual route to independence is to compute a joint density and check it factors; Basu's theorem bypasses that entirely, trading the computation for the verification of completeness, sufficiency, and ancillarity, each of which is often immediate from the structure of the model. The classic illustration is the Gaussian, where the independence of the sample mean and sample variance falls out with no integration at all.

Example 2.3.4: Sample Mean and Variance in the Gaussian Model

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. The claim is that the sample mean $\bar{X}$ and the sample variance $S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2$ are independent. A direct proof diagonalizes the quadratic form or exhibits an orthogonal transformation; Basu's theorem gives it in a few lines, provided σ^2 is held fixed and only μ varies.

Fix $\sigma^2 = \sigma_0^2 > 0$ and regard the family as parametrized by the location $\mu \in \mathbb{R}$ alone, a $\mathcal{N}(\mu, \sigma_0^2)$ location model. In this one-parameter model:

1. $\bar{X}$ is a complete sufficient statistic. Sufficiency follows from the factorization theorem (theorem 2.1.3): with σ_0^2 fixed the density of the sample is $\propto \exp(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu)^2) = \exp(-\frac{1}{2\sigma_0^2} \sum_i x_i^2) \exp(\frac{\mu}{\sigma_0^2} \sum_i x_i - \frac{n\mu^2}{2\sigma_0^2})$, and the μ -dependent factor involves the data only through $\sum_i x_i$, equivalently $\bar{X}$. Completeness holds because $\bar{X} \sim \mathcal{N}(\mu, \sigma_0^2/n)$ is a one-parameter Gaussian location family, a full-rank exponential family in μ , so its natural statistic is complete (theorem 2.4.3).
2. S^2 is ancillary. The variance S^2 is a function of the deviations $X_i - \bar{X}$, and shifting every observation by a constant c leaves each deviation unchanged: replacing X_i by $X_i + c$ sends $\bar{X}$ to $\bar{X} + c$ and $X_i - \bar{X}$ to itself. Since the location model is generated from a fixed $\mathcal{N}(0, \sigma_0^2)$ shape by translation in μ , the law of S^2 is the same for every μ : it does not depend on the location parameter, so S^2 is ancillary (definition 2.2.4).

Basu's theorem (theorem 2.3.3) applied to the complete sufficient statistic $\bar{X}$ and the ancillary S^2 gives $\bar{X} \perp\!\!\!\perp S^2$ under $\mathcal{N}(\mu, \sigma_0^2)$, for every μ . The value σ_0^2 was arbitrary, so $\bar{X} \perp\!\!\!\perp S^2$ holds under $\mathcal{N}(\mu, \sigma^2)$ for every (μ, σ^2) : fixing σ^2 at each value in turn and letting μ vary covers the whole two-parameter model.

The maneuver in the last example is characteristic. Basu's theorem does not apply to $\bar{X}$ in the full two-parameter model, so one restricts to a one-parameter sub-model in which the target statistic *is* complete sufficient and the other statistic *is* ancillary, applies Basu there, and then observes that the independence conclusion did not depend on the frozen coordinate. The independence of $\bar{X}$ and S^2 is a property of the joint law at each fixed (μ, σ^2) . Establishing it one σ^2 at a time loses nothing.

Completeness is a genuine restriction, not an automatic feature of sufficient statistics, and the cleanest way to see this is a family that carries a nonzero unbiased estimator of zero. Such a family is by definition not complete, and the uniqueness conclusion fails there.

Example 2.3.5: A Sufficient Statistic That Is Not Complete

Consider a single observation X from the location family $\{\text{Unif}(\theta, \theta + 1) : \theta \in \mathbb{R}\}$ on the line: given θ , X is uniform on the interval $(\theta, \theta + 1)$. Here X itself is (trivially) sufficient, being the whole data. It is not complete.

To exhibit the failure, seek a nonzero g with $\mathbb{E}_\theta[g(X)] = 0$ for all θ . Trigonometric functions are periodic with the period matched to the unit interval, so take $g(x) = \sin(2\pi x)$. Then, for every θ ,

$$\mathbb{E}_\theta[g(X)] = \int_\theta^{\theta+1} \sin(2\pi x) dx = \left[-\frac{\cos(2\pi x)}{2\pi} \right]_\theta^{\theta+1} = -\frac{\cos(2\pi(\theta+1)) - \cos(2\pi\theta)}{2\pi} = 0$$

because $x \mapsto \cos(2\pi x)$ has period 1: $\cos(2\pi(\theta+1)) = \cos(2\pi\theta)$. Thus $g(X) = \sin(2\pi X)$ is a nonzero function (it is nonzero on a set of positive probability under every θ) whose expectation vanishes identically in θ : an unbiased estimator of zero that is not zero. The statistic X is sufficient but not complete.

The consequence is a failure of uniqueness. If $\hat{\psi}(X)$ is any unbiased estimator of a parameter $\psi(\theta)$, then $\hat{\psi}(X) + c \sin(2\pi X)$ is a second unbiased estimator of the same ψ for every constant c , since the added term contributes zero to the expectation. There is a one-parameter family of distinct unbiased estimators, and no completeness argument is available to single one out. This is the precise sense in which completeness is what makes the Lehmann-Scheffé conclusion of section 3.3 possible: without it, “the” unbiased function of a sufficient statistic is not well defined.

The non-example also warns against a natural misreading of Basu's theorem. The theorem is an implication, not an equivalence: independence of a sufficient statistic from an ancillary is not guaranteed without completeness, and in a family like the one above, where completeness fails, the conclusion can fail too. Independence is not free from sufficiency and ancillarity alone.

Exercise 2.3.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$, $\lambda \in \Theta = (0, \infty)$, and $T = \sum_{i=1}^n X_i \sim \text{Poi}(n\lambda)$. Show that T is complete. (Suggestion: write $\mathbb{E}_\lambda[g(T)] = e^{-n\lambda} \sum_{t \geq 0} g(t)(n\lambda)^t / t!$ and use that a power series in λ vanishing on $(0, \infty)$ has all coefficients zero.)
2. Give an example of a statistic that is bounded-complete but not complete, or argue briefly why the two coincide for statistics taking finitely many values. (For the finite case: if T takes values in a finite set, every function $g(T)$ is bounded, so the two definitions have the same hypothesis and hence the same conclusion.)
3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and $M = X_{(n)} = \max_i X_i$, which is complete sufficient for θ (this may be assumed). Show that the ratio $R = X_{(1)}/X_{(n)}$ is ancillary, and conclude from Basu's theorem that M and R are independent. (Hint: $X_i = \theta U_i$ with $U_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$, so $R = U_{(1)}/U_{(n)}$ has a law free of θ .)
4. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ (variance known and equal to 1). Using that $\bar{X}$ is complete sufficient for μ and that the range $W = X_{(n)} - X_{(1)}$ is ancillary, prove $\bar{X} \perp\!\!\!\perp W$. Then explain why the same conclusion for $\mathcal{N}(\mu, \sigma^2)$ with σ^2 unknown does not follow from this argument directly, and how the two-parameter case is nonetheless recovered.
5. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ with density $\lambda e^{-\lambda x} \mathbb{1}\{x > 0\}$, and $T = \sum_i X_i$, which is complete

sufficient for λ (assume this). Show that each ratio X_i/T has a distribution not depending on λ , hence the vector $(X_1/T, \dots, X_n/T)$ is ancillary, and conclude that T is independent of $(X_1/T, \dots, X_n/T)$.

6. Building on example 2.3.5, suppose $X \sim \text{Unif}(\theta, \theta + 1)$ and one wants an unbiased estimator of θ . Verify that $\hat{\theta}(X) = X - \frac{1}{2}$ is unbiased, then use the incompleteness of X to write down a distinct unbiased estimator of θ , and comment on what property $\hat{\theta}(X) = X - \frac{1}{2}$ would need for it to be singled out as “best”.

2.4 Exponential Families

The general theory of the previous sections is hard to apply in most models, because the sufficient statistic grows with the sample size and completeness is delivered only by an ad hoc calculation. One class of models escapes both difficulties at once. In an exponential family a sufficient statistic of fixed dimension serves every sample size, its cumulant function generates the moments by differentiation, and completeness follows from a single analytic argument. The running Gaussian model is of this kind, and so is the Bernoulli model on the interior $0 < p < 1$ (the endpoint laws $p \in \{0, 1\}$ fail the common-support requirement, by the same support obstruction as example 2.4.6(6)); the uniform model is not of this kind at all, and the reason it fails is exactly the reason the general regularity theory of the later chapters excludes it.

Definition 2.4.1: Exponential Family

A statistical model $\{P_\theta : \theta \in \Theta\}$ dominated by a σ -finite measure μ , with densities $f_\theta = dP_\theta/d\mu$, is a *k-parameter exponential family* if there are functions $\eta_1, \dots, \eta_k : \Theta \rightarrow \mathbb{R}$ and $B : \Theta \rightarrow \mathbb{R}$, a statistic $T = (T_1, \dots, T_k) : \mathcal{X} \rightarrow \mathbb{R}^k$, and a nonnegative $h : \mathcal{X} \rightarrow [0, \infty)$, none depending on θ , such that

$$f_\theta(x) = h(x) \exp\left(\sum_{i=1}^k \eta_i(\theta) T_i(x) - B(\theta)\right) \quad (2.3)$$

for every θ , for μ -almost every x (the exceptional null set may depend on θ). The functions $\eta_i(\theta)$ are the *natural parameters* and T is the *natural statistic*.

Reparametrizing by $\eta = (\eta_1, \dots, \eta_k)$ directly puts the family in *natural* (or *canonical*) form. Set

$$A(\eta) = \log \int_{\mathcal{X}} h(x) e^{\eta^\top T(x)} d\mu(x)$$

and let

$$\Xi = \left\{ \eta \in \mathbb{R}^k : \int_{\mathcal{X}} h(x) e^{\eta^\top T(x)} d\mu(x) < \infty \right\} \quad (2.4)$$

be the *natural parameter space*, the set of η for which the integral converges. For $\eta \in \Xi$ the density in *canonical form* is

$$f_\eta(x) = h(x) \exp(\eta^\top T(x) - A(\eta)) \quad (2.5)$$

and $A(\eta)$ is the *log-partition function* (or *cumulant function*). The family is of *full rank* if the set of natural-parameter values indexing the family contains a nonempty open subset of $\mathbb{R}^k$ and neither the components $T_1, \dots, T_k$ nor the constant 1 satisfy a linear relation that holds $h d\mu$ -almost everywhere (equivalently, P_η -almost surely for any $\eta \in \Xi$). The first condition is a condition on the *parametrization*, not on Ξ : a curved family whose natural parameters trace a lower-dimensional set inside a large Ξ is not of full rank.

The exponent in (2.3) is linear in the natural statistic and separates the parameter from the data: $\eta_i(\theta)$ carries all the θ -dependence, $T_i(x)$ all the data-dependence, and they meet only in the products $\eta_i(\theta)T_i(x)$. This separation is what makes the class tractable, and it is already the factorization criterion of theorem 2.1.3 written out. The term $B(\theta)$, or $A(\eta)$ in canonical form, is not free: it is fixed by the requirement that f_θ integrate to one, so

$$e^{B(\theta)} = \int_{\mathcal{X}} h(x) \exp\left(\sum_i \eta_i(\theta) T_i(x)\right) d\mu(x)$$

and in canonical coordinates $B(\theta) = A(\eta(\theta))$ exactly. The passage from a general parametrization to the canonical one is a change of variables $\theta \mapsto \eta(\theta)$. The canonical form is the intrinsic object, and the natural parameter space Ξ of (2.4) is the largest set on which (2.5) defines a probability density. The dominating measure μ is counting measure for discrete models and Lebesgue measure for the continuous ones. That a dominated family has densities at all is the Radon-Nikodym theorem [4].

A worked instance grounds the definition before the theory. The Bernoulli model is the smallest exponential family, and putting it in canonical form already exhibits every ingredient.

Example 2.4.2: The Bernoulli Model in Canonical Form

Take a single observation $X \sim \text{Ber}(p)$ on $\mathcal{X} = \{0, 1\}$ with $p \in (0, 1)$, dominated by counting measure μ . The mass function is $f_p(x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$, and writing it as an exponential,

$$f_p(x) = \exp(x \log p + (1-x) \log(1-p)) = (1-p) \exp\left(x \log \frac{p}{1-p}\right)$$

This is (2.3) with $k = 1$: the natural statistic is $T(x) = x$, the carrier is $h(x) = 1$, the natural parameter is the *log-odds*

$$\eta = \log \frac{p}{1-p}$$

and $B(p) = -\log(1-p)$. Inverting $\eta = \log \frac{p}{1-p}$ gives $p = e^\eta / (1 + e^\eta)$, so $1-p = 1/(1 + e^\eta)$ and

$$A(\eta) = -\log(1-p) = \log(1 + e^\eta)$$

The map $p \mapsto \eta$ carries $(0, 1)$ onto all of $\mathbb{R}$, so the natural parameter space is $\Xi = \mathbb{R}$: the integral $\sum_{x \in \{0, 1\}} e^{\eta x} = 1 + e^\eta$ converges for every real η . The family is full rank, since T is not μ -a.e. constant. The canonical density is $f_\eta(x) = \exp(\eta x - \log(1 + e^\eta))$.

The natural statistic is sufficient, and under a mild condition it is also complete. Both facts follow from the exponential form, and both are why the estimation theory of the next chapter applies most cleanly here. Completeness is the harder half, and its proof rests on the uniqueness of Laplace transforms. The one-parameter case is given in full below, and the k -parameter case parallels it coordinate-wise.

Theorem 2.4.3: Sufficiency and Completeness of the Natural Statistic

Let $\{P_\eta : \eta \in \Xi\}$ be a k -parameter exponential family in canonical form (2.5) with natural statistic $T = (T_1, \dots, T_k)$.

1. T is sufficient for the family.
2. If the family is of full rank (in particular the set of parameter values indexing the family contains a nonempty open subset of $\mathbb{R}^k$), then T is complete.

Proof:

Sufficiency. The canonical density (2.5) factors as

$$f_\eta(x) = \underbrace{e^{\eta^\top T(x) - A(\eta)}}_{g_\eta(T(x))} \cdot \underbrace{h(x)}_{h(x)}$$

the first factor depending on x only through $T(x)$ and the second free of η . This is exactly the factorization $f_\eta(x) = g_\eta(T(x)) h(x)$ of theorem 2.1.3, so T is sufficient (definition 2.1.2). For an i.i.d. sample $X = (X_1, \dots, X_n)$ the joint density is the product

$$f_\eta^{\otimes n}(x) = \left(\prod_{j=1}^n h(x_j) \right) \exp\left(\eta^\top \sum_{j=1}^n T(x_j) - nA(\eta) \right)$$

which is again of canonical form with carrier $\prod_j h(x_j)$, natural statistic $\sum_{j=1}^n T(x_j)$, and cumulant function $nA(\eta)$. Its natural statistic $\sum_{j=1}^n T(x_j)$ has the same dimension k as for a single observation, whatever the sample size, so a sufficient statistic of fixed dimension serves every n .

Completeness, one-parameter case. Take $k = 1$, so the canonical density is $f_\eta(x) = h(x)e^{\eta T(x) - A(\eta)}$ and $\Xi \subseteq \mathbb{R}$ has nonempty interior; fix an open interval $(a, b) \subseteq \Xi$. Let $S = T(X)$ have law Q_η on $\mathbb{R}$, the pushforward of P_η under T . Because the exponent depends on x only through $T(x)$, the family $\{Q_\eta\}$ is itself the one-parameter exponential family

$$dQ_\eta(s) = e^{\eta s - A(\eta)} d\nu(s) \quad (2.6)$$

where ν is the pushforward of $h d\mu$ under T . Suppose $g: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\mathbb{E}_\eta[g(S)] = 0$ for every $\eta \in (a, b)$. Split $g = g^+ - g^-$ into positive and negative parts. Then for every $\eta \in (a, b)$,

$$\int_{\mathbb{R}} g^+(s) e^{\eta s} d\nu(s) = \int_{\mathbb{R}} g^-(s) e^{\eta s} d\nu(s)$$

both integrals finite (each is at most $e^{A(\eta)} \mathbb{E}_\eta[|g(S)|]$, which is finite by the integrability required of g in definition 2.3.1). Fix an interior point $\eta_0 \in (a, b)$ and define two finite measures on $\mathbb{R}$,

$$d\lambda^+(s) = g^+(s) e^{\eta_0 s} d\nu(s), \quad d\lambda^-(s) = g^-(s) e^{\eta_0 s} d\nu(s)$$

The displayed equality says that λ^+ and λ^- have equal total mass and, more, equal integrals against e^{ts} for every $t = \eta - \eta_0$ in the open interval $(a - \eta_0, b - \eta_0)$ around 0:

$$\int_{\mathbb{R}} e^{ts} d\lambda^+(s) = \int_{\mathbb{R}} e^{ts} d\lambda^-(s) \quad \text{for all } t \in (a - \eta_0, b - \eta_0)$$

The two sides are the bilateral Laplace transforms of the finite measures $\lambda^\pm$, and they agree on an open interval containing 0. A Laplace transform that converges on an open interval determines its measure uniquely. Each transform extends analytically to the vertical strip in $\mathbb{C}$ over that interval, so two that agree on a real segment of the strip agree throughout it, and in particular on the imaginary axis, where $t \mapsto \int_{\mathbb{R}} e^{its} d\lambda^\pm(s)$ are the Fourier transforms of $\lambda^\pm$. Evaluating at $t = 0$ gives equal total masses $m := \lambda^+(\mathbb{R}) = \lambda^-(\mathbb{R})$. If $m = 0$ both measures are zero and there is nothing to prove; otherwise $\lambda^\pm/m$ are probability laws on $\mathbb{R}$ with the same characteristic function, hence the same law by [5, Characteristic Functions Determine Laws], and multiplying back by m gives $\lambda^+ = \lambda^-$. (The normalization is needed because the cited uniqueness theorem is stated for probability laws, while $\lambda^\pm$ are only finite.) Since $e^{\eta_0 s} > 0$ everywhere, this forces $g^+ = g^-$ ν -a.e., that is $g(s) = 0$ for ν -a.e. s , and therefore $g(T) = 0$ P_η -a.s. for every η . Thus T is complete (definition 2.3.1).

Completeness, general case. For $k \geq 2$ the argument is identical in outline: $S = T(X)$ has a k -parameter exponential-family law of the form (2.6) with $\eta, s \in \mathbb{R}^k$, the hypothesis $\mathbb{E}_\eta[g(S)] = 0$ on the open set $\text{int } \Xi$ makes the k -dimensional Laplace transforms of $g^+ e^{\eta_0^\top s} d\nu$ and $g^- e^{\eta_0^\top s} d\nu$ agree on an open set, and analytic continuation in each coordinate carries that agreement to the purely imaginary points, where the two transforms are the Fourier transforms on $\mathbb{R}^k$. Normalizing by the common total mass exactly as above turns these into probability laws on $\mathbb{R}^k$ with equal characteristic functions, so they coincide by [5, Characteristic Functions Determine Laws in $\mathbb{R}^d$], giving $g = 0$ ν -a.e. Hence T is complete whenever $\text{int } \Xi \neq \emptyset$, as we sought to show.

The open-subset condition in full rank is not decoration. If the natural parameters are constrained to a lower-dimensional subset of $\mathbb{R}^k$, the Laplace transform is pinned down on too small a set to force uniqueness, and completeness can fail even though sufficiency does not. A standard instance is the *curved* exponential family $\mathcal{N}(\mu, \mu^2)$: the single scalar μ drives both natural coordinates, so the parameter image is a parabola with empty interior (full rank fails), and $(\sum X_i, \sum X_i^2)$ is sufficient but not complete. For exponential families, full rank is what delivers the completeness that Lehmann-Scheffé (section 3.3) uses; completeness can also arise outside the exponential frame entirely, as $\text{Unif}(0, \theta)$ shows (exercise 2.3.1 (3)). The independence conclusion of Basu's theorem (theorem 2.3.3) also runs through this completeness.

The next result explains why the cumulant function earns its name. In a canonical exponential family the moments of the natural statistic are read off from $A(\eta)$ by differentiation, with no separate integration: the gradient is the mean vector and the Hessian is the covariance matrix. This also makes the Fisher information (section 3.2) a derivative of A .

Proposition 2.4.4: The Log-Partition Function Generates Cumulants

Let $\{P_\eta: \eta \in \Xi\}$ be a canonical exponential family (2.5) with natural statistic T , and let $\eta_0 \in \text{int } \Xi$. Then A is infinitely differentiable at η_0 , and

$$\nabla A(\eta_0) = \mathbb{E}_{\eta_0}[T], \quad \nabla^2 A(\eta_0) = \text{Cov}_{\eta_0}(T) \quad (2.7)$$

In particular $\nabla^2 A(\eta_0)$ is positive semidefinite. Moreover the natural parameter space Ξ is convex, and A is convex on Ξ , strictly so when the family is of full rank.

Proof:

Write $Z(\eta) = \int_{\mathcal{X}} h(x) e^{\eta^\top T(x)} d\mu(x) = e^{A(\eta)}$ for the partition function, finite and positive on Ξ by (2.4). The claim is that Z may be differentiated under the integral sign at an interior point η_0 , and the identities follow.

Step 1: Differentiation under the integral sign. For η in a neighborhood of η_0 each partial derivative of the integrand $h(x) e^{\eta^\top T(x)}$ in η_i is $h(x) T_i(x) e^{\eta^\top T(x)}$. Because η_0 is interior to Ξ , there is $\delta > 0$ with the closed ball $\overline{B}(\eta_0, \delta) \subseteq \Xi$; set $\rho = \delta/(2\sqrt{k})$ and keep η in the cube $\|\eta - \eta_0\|_\infty \leq \rho$. On the cube, since $(\eta - \eta_0)^\top T(x) \leq \rho \sum_i |T_i(x)|$ and $e^{\rho|u|} \leq e^{\rho u} + e^{-\rho u}$, multiplying out the product over coordinates gives

$$e^{\eta^\top T(x)} \leq \sum_{s \in \{-1, 1\}^k} e^{(\eta_0 + \rho s)^\top T(x)}$$

with every $\eta_0 + \rho s$ in $\overline{B}(\eta_0, \delta)$. The factor $|T_i|$ folds in the same way: $|T_i| \leq C_\rho e^{\rho|T_i|} \leq C_\rho (e^{\rho T_i} + e^{-\rho T_i})$ for a constant C_ρ , and this shifts each exponent to $\eta_0 + \rho s \pm \rho e_i$, still in $\overline{B}(\eta_0, \delta) \subseteq \Xi$. So on the cube each such partial derivative is dominated in absolute value by a fixed finite sum of μ -integrable functions $C_\rho h e^{\zeta^\top T}$ with $\zeta \in \Xi$, of integral $C_\rho Z(\zeta) < \infty$. The dominated-convergence justification of differentiating under the integral is the standard one [4, Dominated Convergence Theorem (DCT)]. The same device dominates every order of derivative, a higher monomial in the T_i folding in with a smaller exponent in place of ρ , giving $Z \in C^\infty$ near η_0 with

$$\frac{\partial Z}{\partial \eta_i}(\eta) = \int_{\mathcal{X}} h(x) T_i(x) e^{\eta^\top T(x)} d\mu(x), \quad \frac{\partial^2 Z}{\partial \eta_i \partial \eta_j}(\eta) = \int_{\mathcal{X}} h(x) T_i(x) T_j(x) e^{\eta^\top T(x)} d\mu(x) \quad (2.8)$$

Step 2: The gradient is the mean. Since $A = \log Z$, the chain rule gives $\nabla A = \nabla Z/Z$. Dividing the first identity in (2.8) by $Z(\eta) = e^{A(\eta)}$ reconstitutes the canonical density $f_\eta = h e^{\eta^\top T - A}$ inside the integral,

$$\frac{\partial A}{\partial \eta_i}(\eta) = \frac{1}{Z(\eta)} \int_{\mathcal{X}} h(x) T_i(x) e^{\eta^\top T(x)} d\mu(x) = \int_{\mathcal{X}} T_i(x) f_\eta(x) d\mu(x) = \mathbb{E}_\eta[T_i]$$

so $\nabla A(\eta) = \mathbb{E}_\eta[T]$.

Step 3: The Hessian is the covariance. Differentiate $\partial_i A = Z^{-1} \partial_i Z$ once more in η_j , using $\partial_j(Z^{-1}) = -Z^{-2} \partial_j Z$:

$$\frac{\partial^2 A}{\partial \eta_i \partial \eta_j} = \frac{\partial_{ij} Z}{Z} - \frac{(\partial_i Z)(\partial_j Z)}{Z^2} = \mathbb{E}_\eta[T_i T_j] - \mathbb{E}_\eta[T_i] \mathbb{E}_\eta[T_j] = \text{Cov}_\eta(T_i, T_j)$$

the middle equality dividing the second identity in (2.8) by Z and recognizing the first-derivative ratios as $\mathbb{E}_\eta[T_i]$ and $\mathbb{E}_\eta[T_j]$ from Step 2. Hence $\nabla^2 A(\eta) = \text{Cov}_\eta(T)$, a covariance matrix and so positive semidefinite.

Step 4: Convexity. The convexity claims hold on all of Ξ and need no derivatives. For $\eta, \eta' \in \Xi$ and $t \in (0, 1)$, Hölder's inequality with exponents $1/t$ and $1/(1-t)$ gives

$$Z(t\eta + (1-t)\eta') = \int_{\mathcal{X}} (h e^{\eta^\top T})^t (h e^{\eta'^\top T})^{1-t} d\mu \leq Z(\eta)^t Z(\eta')^{1-t} < \infty$$

so $t\eta + (1-t)\eta' \in \Xi$, making Ξ convex, and taking logarithms makes A convex on Ξ . Equality in Hölder forces $h e^{\eta^\top T}$ and $h e^{\eta'^\top T}$ to be proportional μ -a.e., that is $(\eta - \eta')^\top T(x)$ constant for $h d\mu$ -almost every x , a linear relation the full-rank condition excludes unless $\eta = \eta'$. Thus A is strictly convex under full rank, completing the proof.

The two identities (2.7) turn a moment computation into a differentiation, and they explain the vocabulary: the successive derivatives of A at η are the cumulants of T under P_η , the mean and covariance being the first and second. Convexity of A is the geometric form of this. It also has an inferential payoff taken up later: the map $\eta \mapsto \nabla A(\eta) = \mathbb{E}_\eta[T]$ from natural parameters to mean parameters is, under full rank and on $\text{int } \Xi$, a diffeomorphism onto its image, and its inverse is what the maximum-likelihood estimator computes, since the likelihood equation in an exponential family sets the observed T equal to its expectation $\nabla A(\eta)$ (??).

The identity is quickest to verify where mean and variance are already known. The Poisson family provides the cleanest check: it forces mean and variance to coincide, and (2.7) reproduces this from a single second derivative.

Example 2.4.5: The Poisson Cumulant Check

Let $X \sim \text{Poi}(\lambda)$ on $\mathcal{X} = \{0, 1, 2, \dots\}$ with $\lambda > 0$, dominated by counting measure. The mass function $f_\lambda(x) = e^{-\lambda} \lambda^x / x!$ rewrites as

$$f_\lambda(x) = \frac{1}{x!} \exp(x \log \lambda - \lambda)$$

an exponential family with $h(x) = 1/x!$, natural statistic $T(x) = x$, natural parameter $\eta = \log \lambda$, and $B(\lambda) = \lambda$. In canonical coordinates $\lambda = e^\eta$, so

$$A(\eta) = \lambda = e^\eta, \quad \Xi = \mathbb{R}$$

the partition sum $\sum_{x \geq 0} e^{\eta x} / x! = \exp(e^\eta)$ converging for every real η . Now (2.7) gives, in one line each,

$$\mathbb{E}_\eta[T] = A'(\eta) = e^\eta = \lambda, \quad \text{Var}_\eta(T) = A''(\eta) = e^\eta = \lambda$$

so $\mathbb{E}_\lambda[X] = \text{Var}_\lambda(X) = \lambda$, the equidispersion of the Poisson law recovered without summing a series. The mean-equals-variance identity is the statement $A'' = A'$, which for $A(\eta) = e^\eta$ holds identically.

A single template thus captures the discrete and continuous models together. Collecting the running models in canonical form displays the pattern and, at its end, the model that breaks it.

Example 2.4.6: The Standard Models in Canonical Form

Each entry names the dominating measure, the carrier h , the natural statistic T , the natural parameter η , the cumulant function $A(\eta)$, and the natural parameter space Ξ . Throughout, a single observation is taken. The i.i.d. sample version replaces T by $\sum_j T(X_j)$ and A by nA , by theorem 2.4.3.

1. *Bernoulli* $\text{Ber}(p)$, $p \in (0, 1)$, counting measure. From example 2.4.2: $h(x) = 1$, $T(x) = x$, natural parameter the log-odds $\eta = \log \frac{p}{1-p}$, and $A(\eta) = \log(1 + e^\eta)$, with $\Xi = \mathbb{R}$. The mean identity checks: $A'(\eta) = e^\eta / (1 + e^\eta) = p = \mathbb{E}_p[X]$ and $A''(\eta) = e^\eta / (1 + e^\eta)^2 = p(1 - p) = \text{Var}_p(X)$.
2. *Binomial* $\text{Bin}(m, p)$ for fixed known m , $p \in (0, 1)$, counting measure on $\{0, 1, \dots, m\}$. The mass function $\binom{m}{x} p^x (1 - p)^{m-x}$ gives $h(x) = \binom{m}{x}$, $T(x) = x$, the same log-odds $\eta = \log \frac{p}{1-p}$, and $A(\eta) = m \log(1 + e^\eta)$, with $\Xi = \mathbb{R}$.^a
3. *Poisson* $\text{Poi}(\lambda)$, $\lambda > 0$, counting measure. From example 2.4.5: $h(x) = 1/x!$, $T(x) = x$, $\eta = \log \lambda$, $A(\eta) = e^\eta$, with $\Xi = \mathbb{R}$.
4. *Normal* $\mathcal{N}(\mu, \sigma^2)$, $(\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$, Lebesgue measure. Expanding the exponent of the density,

$$f_{\mu, \sigma^2}(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2 - \frac{\mu^2}{2\sigma^2} - \log \sigma\right)$$

a two-parameter family with $h(x) = (2\pi)^{-1/2}$, natural statistic $T(x) = (x, x^2)$, and natural parameters

$$\eta = (\eta_1, \eta_2) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)$$

Inverting, $\sigma^2 = -1/(2\eta_2)$ and $\mu = -\eta_1/(2\eta_2)$, so

$$A(\eta) = \frac{\mu^2}{2\sigma^2} + \log \sigma = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2} \log(-2\eta_2)$$

and $\Xi = \mathbb{R} \times (-\infty, 0)$, the half-space forced by $\eta_2 = -1/(2\sigma^2) < 0$ (equivalently, the Gaussian integral converges exactly when the coefficient of x^2 is negative). The natural statistic (x, x^2) is, for a sample, the pair $(\sum_j X_j, \sum_j X_j^2)$, equivalent to $(\bar{X}, S^2)$ of example 1.1.7: this is the sufficient statistic met in section 2.1, now recovered as the natural statistic of a full-rank two-parameter exponential family, hence sufficient and complete by theorem 2.4.3.

5. *Gamma* $\text{Gamma}(\alpha, \beta)$ with shape $\alpha > 0$ and rate $\beta > 0$, Lebesgue measure on $(0, \infty)$. The density $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ rewrites as

$$f_{\alpha, \beta}(x) = \frac{1}{x} \exp(\alpha \log x - \beta x + \alpha \log \beta - \log \Gamma(\alpha))$$

a two-parameter family with $h(x) = 1/x$, natural statistic $T(x) = (\log x, x)$, natural parameters $\eta = (\alpha, -\beta)$, and

$$A(\eta) = \log \Gamma(\alpha) - \alpha \log \beta = \log \Gamma(\eta_1) - \eta_1 \log(-\eta_2)$$

with $\Xi = (0, \infty) \times (-\infty, 0)$. Fixing α known reduces this to the one-parameter exponential family (with $T(x) = x$) that contains the $\text{Exp}(\beta) = \text{Gamma}(1, \beta)$ law.

6. *Uniform* $\text{Unif}(0, \theta)$, $\theta > 0$, Lebesgue measure. This is *not* an exponential family. Its density $f_\theta(x) = \theta^{-1} \mathbb{1}\{0 < x < \theta\}$ carries the indicator $\mathbb{1}\{0 < x < \theta\}$, whose *support* depends on θ . No factorization of the form (2.3) can reproduce this: the exponent $\sum_i \eta_i(\theta) T_i(x) - B(\theta)$ is finite for every x in the fixed set $\{h(x) > 0\}$, so $f_\theta(x) = h(x)e^{(\cdots)}$ is positive on one and the same set of x for every θ , whereas f_θ is positive exactly on $(0, \theta)$, a set that grows with θ . The parameter-dependent support is the obstruction, and it is the same feature that made $\text{Unif}(0, \theta)$ the irregular foil of example 1.1.7: the sufficient statistic is the maximum $X_{(n)}$ rather than a fixed-dimension sum of the sort produced here, the cumulant machinery of proposition 2.4.4 has no purchase, and the smoothness the later regularity theory assumes is absent.

^athe sample size m is a known constant here, not a parameter. Were m unknown the family would not be exponential.

The five regular models fit one template and the sixth refuses it, and the refusal is diagnostic rather than accidental. Wherever the support of the data moves with the parameter, the exponential form is impossible, the natural statistic is replaced by an extreme order statistic, and the derivative-based theory of the following chapters must give way to an argument tailored to the boundary. This is why $\text{Unif}(0, \theta)$ recurs as the standing counterexample.

Exercise 2.4.1

1. The exponential law $\text{Exp}(\lambda)$ has density $f_\lambda(x) = \lambda e^{-\lambda x} \mathbb{1}\{x > 0\}$ for $\lambda > 0$. Put it in canonical form: identify h , T , the natural parameter η , the cumulant function $A(\eta)$, and the natural parameter space Ξ . Verify $A'(\eta) = \mathbb{E}_\eta[X]$ and $A''(\eta) = \text{Var}_\eta(X)$ directly, and check that they equal $1/\lambda$ and $1/\lambda^2$.
2. For fixed known $r \geq 1$ the negative binomial law has mass function $f_p(x) = \binom{x+r-1}{x} (1-p)^r p^x$ on $x \in \{0, 1, 2, \dots\}$, with $p \in (0, 1)$. Show it is a one-parameter exponential family, identify its natural statistic and natural parameter, and compute $A(\eta)$. Deduce $\mathbb{E}_p[X]$ by differentiating A .
3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$. Using theorem 2.4.3, name the sufficient and complete statistic for the sample and give its law. Then, without invoking Basu directly, argue that any unbiased estimator of λ that is a function of $\sum_j X_j$ is the unique unbiased estimator among functions of $\sum_j X_j$.

4. Consider $\mathcal{N}(\mu, \mu^2)$ with a single scalar parameter $\mu \in (0, \infty)$, so that the standard deviation equals the mean. Write the density in the form (2.3) and show that the natural parameters $\eta_1(\mu), \eta_2(\mu)$ trace a curve in $\mathbb{R}^2$ rather than filling an open set. Explain, citing the full-rank hypothesis of theorem 2.4.3(2), why $(\sum_j X_j, \sum_j X_j^2)$ is sufficient but need not be complete here.
5. Give a self-contained proof that $\text{Unif}(0, \theta)$ is not an exponential family, arguing from the support directly: suppose $f_\theta(x) = h(x) \exp(\sum_i \eta_i(\theta) T_i(x) - B(\theta))$ held for all $\theta > 0$ and $x > 0$, and derive a contradiction from the fact that the set $\{x: f_\theta(x) > 0\}$ depends on θ while $\{x: h(x) > 0\}$ does not.
6. In a full-rank canonical exponential family, let $m(\eta) = \nabla A(\eta)$ be the mean of the natural statistic. Using proposition 2.4.4, show that m is injective on $\text{int } \Xi$ and that its Jacobian is $\text{Cov}_\eta(T)$, which is invertible under full rank. (This map, and its inverse, is what the maximum-likelihood estimator inverts in ??.)

Part II

Estimation

Part I set up the decision problem and reduced the data to the statistic that carries the information about θ . This Part spends both on the first of the two inferential acts: producing a value, offered as the unknown parameter itself. Three routes to an estimator run through it, and they disagree about what an estimator is for. Chapter 3 takes the decision-theoretic route as far as it goes, asking which estimator has smallest risk among those that are unbiased, and answering it exactly through the information inequality and the completeness of Part I. ?? abandons the search for an exactly optimal rule and maximizes the likelihood instead, a recipe that applies to every model and is justified, in general, only asymptotically. ?? puts a prior on the parameter and reports the posterior, which turns estimation into a computation rather than an optimization. The three converge in ??, where the maximum-likelihood estimator turns out to attain the information bound in the large-sample limit and the influence of the prior washes out.

Point Estimation and Optimality

An estimator is a decision rule whose action is a guess at θ itself, and under squared-error loss its risk is the mean squared error. Chapter 1 left the search for a uniformly best rule in a bad state: no estimator has smallest risk at every θ , because the constant rule $\delta \equiv \theta_0$ beats every competitor at θ_0 . This chapter takes the first of the two escapes named there, restricting the class of rules, and shows that on the restricted class the question becomes answerable.

Requiring an estimator to be unbiased removes the constant rules, and among what is left there is a floor on the variance that no estimator can pass. Section 3.1 defines bias and splits the mean squared error into bias and variance, which is what makes the restriction cost something. Section 3.2 builds the Fisher information and proves the information inequality, the floor itself, together with the regularity conditions it needs and the models that violate them. Section 3.3 constructs the best unbiased estimator, by conditioning on a sufficient statistic to cut variance and invoking completeness to make the answer unique. That estimator need not sit on the floor, and usually does not: the two properties are different, and section 3.4 separates them. That section also carries the bound to several parameters, where estimating one coordinate in the presence of the others can cost more than estimating it alone, though whether it does turns on the parametrization.

3.1 Unbiasedness, Bias, and Mean Squared Error

The decision-theoretic frame of chapter 1 already contains estimation, with the parameter space as the action space and, under squared-error loss, the mean squared error as the risk. Two things are missing before an optimality theory can begin. The quantity to be reported is usually not θ itself but a function of it, and no condition has yet been imposed on estimators that would cut the field of competitors down to a class in which a best element might exist. This section fixes the first and introduces the oldest answer to the second, the requirement that an estimator be correct on average, and then examines what that requirement costs.

Definition 1.2.3 called a decision rule an estimator when the action space is the parameter space itself. The estimation problems of this chapter more often concern a function of the parameter than the parameter itself. In the Gaussian model of example 1.1.7 the parameter is the pair (μ, σ^2) , while the number to be reported is often a single one of them, or the standard deviation σ , or the ratio μ/σ ; in the Bernoulli model one may want p , or the variance $p(1-p)$, or the odds $p/(1-p)$. Each of these is a real-valued function of the parameter, and each poses its own estimation problem inside the same model.

Definition 3.1.1: Estimand and Estimator

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model (definition 1.1.1) with sample space $\mathcal{X}$. An *estimand* is a function

$$\psi: \Theta \rightarrow \mathbb{R}$$

of the parameter, the real quantity the analysis is required to report. An *estimator* of ψ is a decision rule (definition 1.2.3) for the decision problem whose action space is the range $\mathcal{A} = \psi(\Theta)$, that is, a measurable map

$$\hat{\psi}: \mathcal{X} \rightarrow \psi(\Theta)$$

written $\hat{\psi} = \hat{\psi}(X)$. Its value $\hat{\psi}(x)$ at a realized data value x is an *estimate* of $\psi(\theta)$.

When a construction returns a rule whose values fall outside $\psi(\Theta)$, the action space is enlarged to $\mathcal{A} = \mathbb{R}$ and the rule is still called an estimator of ψ . Example 3.1.5 exhibits a model in which the enlargement is forced.

The estimator and the estimate are different kinds of object, and the notation keeps them apart. The estimator $\hat{\psi}$ is a rule, fixed before the data arrive and therefore a random variable with a distribution under each P_θ ; the estimate $\hat{\psi}(x)$ is the single number the rule returns once x is in hand. Every statement in this chapter is about the estimator, since it is the distribution of $\hat{\psi}$ under P_θ , and not any one of its realized values, that can be compared across competing rules.

Taking ψ to be the identity on $\Theta \subseteq \mathbb{R}$ recovers the estimator of θ itself, and the notation reverts to $\hat{\theta}$. The general case matters for two reasons. It allows a real report to be extracted from a model whose parameter is a vector, as when μ is estimated in the Gaussian model with σ^2 unknown, a configuration in which σ^2 is a nuisance parameter and the theory of section 3.4 applies. And it allows the same model to pose several inequivalent problems at once, since p and $p(1-p)$ are estimated well by quite different rules.

Naming the estimand narrows nothing by itself: any measurable map from $\mathcal{X}$ into $\psi(\Theta)$ qualifies as an estimator, including the maps that ignore the data entirely. The first condition the subject imposes asks that the estimator be centered on its target, in the sense that its average value under P_θ is the number being estimated, and that this hold at every candidate θ rather than at one.

Definition 3.1.2: Unbiased Estimator

Let $\hat{\psi}$ be an estimator of the estimand ψ with $\mathbb{E}_\theta[|\hat{\psi}|] < \infty$ for every $\theta \in \Theta$. The *bias* of $\hat{\psi}$ at θ is

$$\text{Bias}(\theta, \hat{\psi}) = \mathbb{E}_\theta[\hat{\psi}] - \psi(\theta)$$

and $\hat{\psi}$ is *unbiased* for ψ if

$$\mathbb{E}_\theta[\hat{\psi}] = \psi(\theta) \quad \text{for every } \theta \in \Theta$$

that is, if the function $\theta \mapsto \text{Bias}(\theta, \hat{\psi})$ vanishes identically on Θ . An estimand for which at least one unbiased estimator exists is called *estimable*.

For ψ the identity the bias defined here is the bias of proposition 1.2.5, so the two usages agree wherever both apply. The content of the definition is carried by the quantifier. Unbiasedness is a property of the whole function $\theta \mapsto \mathbb{E}_\theta[\hat{\psi}]$, not a property of the estimator at any single parameter value, and the pointwise version of the condition says nothing: for any prescribed θ_0 the constant rule $\hat{\psi} \equiv \psi(\theta_0)$ has $\mathbb{E}_{\theta_0}[\hat{\psi}] = \psi(\theta_0)$, so “unbiased at θ_0 ” is satisfied by a rule that never looks at the data. Requiring the equality across all of Θ is what rules such a rule out, and it is also what makes unbiasedness a restriction on the whole risk function $\theta \mapsto R(\theta, \hat{\psi})$ of definition 1.2.4 rather than a constraint at one point of it.

Example 3.1.3: Unbiased Estimators in the Three Running Models

Each of the three models of example 1.1.7 is examined with an i.i.d. sample $X_1, \dots, X_n$.

1. *The Bernoulli model.* Estimate $\psi(p) = p$ by $\bar{X}$. Since $P_p(\{1\}) = p$ and $P_p(\{0\}) = 1 - p$ (example 1.1.7(1)), the common law has mean $\mathbb{E}_p[X_1] = 1 \cdot p + 0 \cdot (1 - p) = p$, so by example 1.1.6(1) one has $\mathbb{E}_p[\bar{X}] = p$ for every $p \in [0, 1]$, and $\bar{X}$ is unbiased with $\text{Bias}(p, \bar{X}) = 0$. Its values lie in $[0, 1] = \psi(\Theta)$, so no enlargement of the action space is needed.
2. *The Gaussian model.* With $\theta = (\mu, \sigma^2)$, the sample mean $\bar{X}$ is unbiased for the estimand $\psi(\theta) = \mu$, again by example 1.1.6(1). For the estimand $\psi(\theta) = \sigma^2$, the sample variance $S^2 = (n-1)^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$ satisfies $\mathbb{E}_\theta[S^2] = \sigma^2$ for every θ (example 1.1.6(2)), so S^2 is unbiased, while the divisor- n version

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2$$

has $\mathbb{E}_\theta[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2$ by linearity of expectation [5, Foundational Properties of Expectation], hence

$$\text{Bias}(\theta, \hat{\sigma}^2) = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}$$

a bias that is negative at every θ and shrinks like $1/n$.

Unbiasedness does not pass through a nonlinear function of the estimand. The sample standard deviation $S = \sqrt{S^2}$ is not unbiased for σ , and the reason is contained in the variance identity $\text{Var}_\theta(S) = \mathbb{E}_\theta[S^2] - \mathbb{E}_\theta[S]^2$, whose terms are finite because $\mathbb{E}_\theta[S^2] = \sigma^2$ and $0 \leq S \leq 1 + S^2$ forces $\mathbb{E}_\theta[S] \leq 1 + \sigma^2$. Rearranging it gives $\mathbb{E}_\theta[S]^2 = \sigma^2 - \text{Var}_\theta(S) \leq \sigma^2$, so

$\mathbb{E}_\theta[S] \leq \sigma$, with equality only where $\text{Var}_\theta(S) = 0$. The sample standard deviation understates σ on average whenever it is not almost surely constant, and the exercises compute the exact factor for $n = 2$.

3. *The uniform model.* Estimate $\psi(\theta) = \theta$ from $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ using the sample maximum $M = X_{(n)}$. Each X_i has density $\theta^{-1} \mathbb{1}\{0 < x < \theta\}$ (example 1.1.7(3)). Independence gives $P_\theta(M \leq t) = (t/\theta)^n$ on $[0, \theta]$ and so the density $f_M(t) = nt^{n-1}/\theta^n$ there, a computation the reader has already run in the exercises of section 1.1. Both moments follow from it,

$$\mathbb{E}_\theta[M] = \int_0^\theta t \frac{nt^{n-1}}{\theta^n} dt = \frac{n}{n+1}\theta, \quad \mathbb{E}_\theta[M^2] = \int_0^\theta t^2 \frac{nt^{n-1}}{\theta^n} dt = \frac{n}{n+2}\theta^2$$

so that $\text{Bias}(\theta, M) = -\theta/(n+1)$ and

$$\text{Var}_\theta(M) = \frac{n}{n+2}\theta^2 - \left(\frac{n}{n+1}\theta\right)^2 = n\theta^2 \cdot \frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)^2} = \frac{n\theta^2}{(n+2)(n+1)^2}$$

using $(n+1)^2 - n(n+2) = 1$. The bias of M is negative for a structural reason: every observation falls in $(0, \theta)$, so $M < \theta$ always holds and the sample maximum can only understate the endpoint. Multiplying it up removes the bias exactly, since $\mathbb{E}_\theta\left[\frac{n+1}{n}M\right] = \theta$ for every θ , and $\frac{n+1}{n}M$ is the unbiased estimator this model contributes to the running examples.

The three computations show what the bias records and what it does not. In each case the bias is a function on Θ , and in each case it was removed, or could be, by an adjustment that does not depend on the unknown parameter: dividing by $n-1$ rather than n , or multiplying the maximum by $(n+1)/n$. What none of the computations addresses is whether the adjustment improves the estimator. Answering that requires the risk, and under squared-error loss the risk splits into a bias term and a variance term exactly as it did in proposition 1.2.5, now with an arbitrary real estimand in place of the parameter.

Proposition 3.1.4: Bias-Variance Decomposition for a General Estimand

Let $\psi: \Theta \rightarrow \mathbb{R}$ be an estimand, let $\hat{\psi}$ be an estimator of ψ , and let $\theta \in \Theta$ be any parameter value with $\mathbb{E}_\theta[\hat{\psi}^2] < \infty$ (the decomposition below holds at each such θ separately; no integrability at other parameter values is used). Under the squared-error loss $L(\theta, a) = (\psi(\theta) - a)^2$ the risk of $\hat{\psi}$ (definition 1.2.4) is its *mean squared error*

$$\text{MSE}(\theta, \hat{\psi}) = R(\theta, \hat{\psi}) = \mathbb{E}_\theta[(\hat{\psi} - \psi(\theta))^2]$$

and it decomposes as

$$\text{MSE}(\theta, \hat{\psi}) = \text{Var}_\theta(\hat{\psi}) + \text{Bias}(\theta, \hat{\psi})^2$$

at each such θ . In particular an estimator unbiased at θ with $\mathbb{E}_\theta[\hat{\psi}^2] < \infty$ has $\text{MSE}(\theta, \hat{\psi}) = \text{Var}_\theta(\hat{\psi})$ there.

Proof:

Fix $\theta \in \Theta$ and write $c = \psi(\theta)$, a number depending on θ but not on the data.

Step 1: the risk is the mean squared error, and it is finite. By definition 1.2.4 the risk of $\hat{\psi}$ at θ is $R(\theta, \hat{\psi}) = \mathbb{E}_\theta[L(\theta, \hat{\psi}(X))] = \mathbb{E}_\theta[(c - \hat{\psi})^2]$, which is $\text{MSE}(\theta, \hat{\psi})$ as the statement defines it. For finiteness, expand $(c - \hat{\psi})^2 = \hat{\psi}^2 - 2c\hat{\psi} + c^2$. The pointwise bound $|\hat{\psi}| \leq 1 + \hat{\psi}^2$ together with monotonicity and linearity of expectation [5, Foundational Properties of Expectation] gives $\mathbb{E}_\theta[|\hat{\psi}|] \leq 1 + \mathbb{E}_\theta[\hat{\psi}^2] < \infty$, using $\mathbb{E}_\theta[1] = 1$ because P_θ is a probability measure. Linearity now makes $\mathbb{E}_\theta[(c - \hat{\psi})^2]$ finite, and in particular $\mathbb{E}_\theta[\hat{\psi}]$ and $\text{Var}_\theta(\hat{\psi})$ are defined.

Step 2: reduction to the case already proved. Consider the statistical model $\{Q_t : t \in \mathbb{R}\}$ on $(\mathcal{X}, \mathcal{B})$ that assigns the single law $Q_t = P_\theta$ to every parameter value $t \in \mathbb{R}$, which is a statistical model in the sense of definition 1.1.1, and regard $\hat{\psi}$ as an estimator of the real parameter t in that model. Its hypothesis is discharged by Step 1: $\mathbb{E}_{Q_t}[\hat{\psi}^2] = \mathbb{E}_\theta[\hat{\psi}^2] < \infty$ for every $t \in \mathbb{R}$. Hence proposition 1.2.5 applies, and reading its conclusion (1.1) at the parameter value $t = c$ gives

$$\mathbb{E}_\theta[(\hat{\psi} - c)^2] = \text{Var}_\theta(\hat{\psi}) + (\mathbb{E}_\theta[\hat{\psi}] - c)^2$$

since $Q_c = P_\theta$. Substituting $c = \psi(\theta)$ and $\mathbb{E}_\theta[\hat{\psi}] - \psi(\theta) = \text{Bias}(\theta, \hat{\psi})$ turns this into the claimed decomposition at θ , and $\theta \in \Theta$ was arbitrary. Setting $\text{Bias}(\theta, \hat{\psi}) = 0$ gives the statement for an unbiased estimator, as we sought to show.

The decomposition is what turns a bias computation into a comparison of estimators. For an unbiased estimator the squared bias term is absent, so its entire risk is its variance, and two unbiased estimators are ordered by their variances alone. A biased competitor is under no such constraint: it carries a squared bias term at every θ where its bias is nonzero, and it has the smaller mean squared error exactly at those θ where the variance it saves exceeds that term.

The uniform model settles the comparison left open in example 3.1.3(3). With $M = X_{(n)}$ and the two moments computed there, the unbiased estimator $\hat{\theta} = \frac{n+1}{n}M$ has zero bias, so

$$\text{MSE}(\theta, \hat{\theta}) = \text{Var}_\theta(\hat{\theta}) = \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{(n+2)(n+1)^2} = \frac{\theta^2}{n(n+2)}$$

while the uncorrected maximum M , with bias $-\theta/(n+1)$, has

$$\text{MSE}(\theta, M) = \frac{n\theta^2}{(n+2)(n+1)^2} + \frac{\theta^2}{(n+1)^2} = \frac{\theta^2}{(n+1)^2} \cdot \frac{n+(n+2)}{n+2} = \frac{2\theta^2}{(n+1)(n+2)}$$

The ratio of the two is $\text{MSE}(\theta, M)/\text{MSE}(\theta, \hat{\theta}) = 2n/(n+1)$, which exceeds 1 for every $n \geq 2$ and does not depend on θ , so here the correction for bias lowers the mean squared error at every parameter value. The correction is not always the right one, and it is not even optimal among multiples of M : the estimator $\frac{n+2}{n+1}M$, whose multiplier falls below $\frac{n+1}{n}$ and so leaves a downward bias, has smaller mean squared error than $\hat{\theta}$ at every θ , as the exercises verify.

That an unbiased estimator can be beaten is the mildest of the objections to unbiasedness as a criterion. Two sharper ones are available in models no less standard than the running three.

Example 3.1.5: Two Failures of Unbiasedness

1. *An estimand with no unbiased estimator.* Take $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ as in example 1.1.7(1), with the parameter space restricted to $\Theta = (0, 1)$ so that the estimand

$$\psi(p) = \frac{1}{p}$$

is defined. Let $\delta: \{0, 1\}^n \rightarrow \mathbb{R}$ be any estimator, the action space taken as large as $\mathbb{R}$ so that the conclusion is the strongest available. By independence the joint mass function of the sample at $x = (x_1, \dots, x_n)$ is $\prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^t (1-p)^{n-t}$ with $t = \sum_i x_i$, so

$$\mathbb{E}_p[\delta(X)] = \sum_{x \in \{0,1\}^n} \delta(x) p^{t(x)} (1-p)^{n-t(x)}$$

Each summand is a polynomial in p of degree at most n , so the finite sum $q(p) = \mathbb{E}_p[\delta(X)]$ is one as well. If δ were unbiased for ψ then $q(p) = 1/p$, that is $pq(p) - 1 = 0$, for every $p \in (0, 1)$. A polynomial vanishing on the infinite set $(0, 1)$ is the zero polynomial, yet the constant term of $pq(p) - 1$ equals -1 . No unbiased estimator of $1/p$ exists, and $1/p$ is not estimable in this model.

2. *An unbiased estimator outside the range of the estimand.* Take a single observation $X \sim \text{Poi}(\lambda)$ with $\Theta = (0, \infty)$, and let

$$\psi(\lambda) = e^{-2\lambda} = (P_\lambda(X = 0))^2$$

the square of the chance that an observation comes out zero, so that $\psi(\Theta) = (0, 1)$. The estimator $\delta(X) = (-1)^X$ is unbiased, since summing the Poisson mass function against $(-1)^x$ gives

$$\mathbb{E}_\lambda[(-1)^X] = \sum_{x \geq 0} (-1)^x e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \sum_{x \geq 0} \frac{(-\lambda)^x}{x!} = e^{-\lambda} e^{-\lambda} = e^{-2\lambda}$$

and it is the only one up to almost sure equality. Writing the Poisson mass function as $f_\lambda(x) = \frac{1}{x!} \exp(x \log \lambda - \lambda)$ exhibits the family as a one-parameter exponential family in canonical form (definition 2.4.1) with natural statistic $T(x) = x$, natural parameter $\eta = \log \lambda$ and cumulant function $A(\eta) = e^\eta$. As λ runs over $(0, \infty)$ the natural parameter η runs over all of $\mathbb{R}$, which has nonempty interior; the natural statistic x is nonconstant, so no affine relation binds it and the family is of full rank, and X is complete by theorem 2.4.3(2). Any two unbiased estimators of ψ differ by a function of X with expectation zero at every λ , hence agree a.s. by definition 2.3.1.

The estimator is therefore forced, and it returns 1 when X is even and -1 when X is odd. Neither value lies in $\psi(\Theta) = (0, 1)$, and with probability $P_\lambda(X \text{ odd}) = (1 - e^{-2\lambda})/2$ the report is a negative estimate of a probability. This is the enlargement of the action space flagged in definition 3.1.1: with $\mathcal{A} = \psi(\Theta)$ the estimand has no unbiased estimator at all, and admitting one requires accepting rules that report impossible values. The rule $\mathbb{1}\{X \text{ even}\}$, which stays in the closure $[0, 1]$ and so never reports a negative probability, has smaller mean squared error at every λ , as the exercises verify.

The everyday objection to unbiasedness needs no failure at all. In the Gaussian model with both parameters unknown the unbiased S^2 is beaten, at every θ , by the biased $\hat{\sigma}^2 = \frac{n-1}{n}S^2$ of example 3.1.3(2): the trade of a bias $-\sigma^2/n$ for a variance reduction is favourable, and example 1.3.4(2) settles the comparison in a sharper form than a head-to-head would. Minimizing the risk of $c \sum_i (X_i - \bar{X})^2$ over c puts the optimum at $c^* = 1/(n+1)$, so the unbiased choice $c = 1/(n-1)$ and the maximum-likelihood choice $c = 1/n$ are both dominated (definition 1.3.1) by a third estimator that is neither, and S^2 is therefore inadmissible (definition 1.3.2).

Unbiasedness is therefore not a guarantee about risk. It is an instance of the first escape route of section 1.3, a restriction of the class of rules, and what recommends it is the effect of that restriction on the risk function. Within the unbiased class proposition 3.1.4 collapses the risk to the variance, so minimizing risk uniformly in θ becomes minimizing a single variance function of θ , and that problem, unlike the unrestricted one of proposition 1.3.3, often has a solution. What the examples above record is what the restriction costs: the class can be empty, its only member can report values the model calls impossible, and it can exclude an estimator that beats everything inside it.

Exercise 3.1.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ with $n \geq 2$ and put $T = \sum_{i=1}^n X_i$. Show that $\bar{X}^2$ is a biased estimator of $\psi(p) = p^2$ with $\text{Bias}(p, \bar{X}^2) = p(1-p)/n$, and then show that

$$\hat{\psi} = \frac{T(T-1)}{n(n-1)}$$

is unbiased for p^2 . Check that both estimators take values in $\psi(\Theta) = [0, 1]$, so that neither requires the enlarged action space of definition 3.1.1.

2. Let $X \sim \text{Bin}(n, p)$ with $\Theta = (0, 1)$, the sampling law of the total number of successes in example 1.1.7(1). Prove that an estimand $\psi: (0, 1) \rightarrow \mathbb{R}$ is estimable in the sense of definition 3.1.2 if and only if it is the restriction to $(0, 1)$ of a polynomial of degree at most n . For the converse direction, show first that $\mathbb{E}_p[X(X-1)\cdots(X-k+1)] = n(n-1)\cdots(n-k+1)p^k$ for $1 \leq k \leq n$. Deduce that neither $1/p$ nor $\sqrt{p}$ is estimable.
3. Let $X_1, X_2 \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, so $n = 2$. Show that $S = |X_1 - X_2|/\sqrt{2}$. Using that $X_1 - X_2 \sim \mathcal{N}(0, 2\sigma^2)$, compute $\mathbb{E}_\theta[S]$ exactly and confirm that S underestimates σ on average even though S^2 is unbiased for σ^2 . Find the constant c for which cS is unbiased for σ .
4. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and $M = X_{(n)}$. Using the moments computed in example 3.1.3(3), write $\text{MSE}(\theta, cM)$ as a function of the constant $c > 0$, minimize it over c , and show that the minimizer is $c^* = (n+2)/(n+1)$ with $\text{MSE}(\theta, c^*M) = \theta^2/(n+1)^2$. Conclude that the unbiased estimator $\frac{n+1}{n}M$ is inadmissible (definition 1.3.2) for every $n \geq 1$.
5. In the setting of example 3.1.5(2), let $\delta(X) = (-1)^X$ and $\delta^+(X) = \mathbb{1}\{X \text{ even}\}$, both regarded as estimators of $\psi(\lambda) = e^{-2\lambda}$. Show that for every λ and every outcome x one has $|\delta^+(x) - \psi(\lambda)| \leq |\delta(x) - \psi(\lambda)|$, with strict inequality when x is odd, and deduce that δ^+ dominates δ (definition 1.3.1). Then compute $\text{MSE}(\lambda, \delta)$ and $\text{MSE}(\lambda, \delta^+)$ in closed form and verify that the first exceeds twice the second.
6. Let $X \sim \text{Ber}(p)$ be a single observation with $\Theta = [0, 1]$, and take $\psi(p) = p^2$. Show that $\mathbb{E}_p[\delta(X)]$ is an affine function of p for every estimator δ , and deduce that p^2 is not estimable in this model. Then fix $p_0 \in (0, 1)$ and exhibit infinitely many estimators δ satisfying $\mathbb{E}_{p_0}[\delta(X)] = p_0^2$.

State what the two computations together show about the quantifier in definition 3.1.2.

3.2 Fisher Information and the Cramér-Rao Bound

An unbiased estimator has no bias to trade, so proposition 3.1.4 collapses its mean squared error onto its variance, and the search for a best unbiased estimator becomes a search for the smallest variance. That search has a floor, and the floor belongs to the model rather than to any estimator: it is set by how fast the densities f_θ move as θ moves. A family whose densities at θ and at $\theta + h$ are nearly indistinguishable cannot support a precise estimate of θ from any amount of data, while a family whose densities separate quickly can.

The derivative of the log-likelihood in θ measures that separation at a single data point, and its variance aggregates the measurement over the sample space. The lower bound then follows from one application of the Cauchy-Schwarz inequality to the covariance between an estimator and that derivative. The mathematical cost lies almost entirely in the hypotheses that make the covariance computable, chief among them the interchange of a θ -derivative with an integral in x ; a model that fails that interchange can and does beat the bound.

Throughout, $\{P_\theta: \theta \in \Theta\}$ is dominated by a σ -finite measure μ with densities $f_\theta = dP_\theta/d\mu$, the setting of chapter 2, and $\Theta \subseteq \mathbb{R}$. We write $\ell(\theta; x) = \log f_\theta(x)$ for the log-likelihood at the data point x , a function of θ for each fixed x . The likelihood function in its own right is the subject of ??.

Definition 3.2.1: The Score Function

Fix $\theta \in \Theta$ and let x be a point with $f_\theta(x) > 0$ at which the derivative $\partial_\theta f_\theta(x)$ exists. The score at θ is

$$s_\theta(x) = \partial_\theta \ell(\theta; x) = \frac{\partial_\theta f_\theta(x)}{f_\theta(x)}$$

the second expression being the chain rule applied to $\log f_\theta(x)$, valid because $f_\theta(x) > 0$.

The score is a relative rate of change: it reports the proportional increase in the density at the observed x produced by an increase in θ . Two of the running models of example 1.1.7 make this concrete at once. For a single Bernoulli observation $X \sim \text{Ber}(p)$ on $\{0, 1\}$ the density with respect to counting measure is $f_p(x) = p^x(1-p)^{1-x}$, so $\ell(p; x) = x \log p + (1-x) \log(1-p)$ and

$$s_p(x) = \frac{x}{p} - \frac{1-x}{1-p} = \frac{x(1-p) - p(1-x)}{p(1-p)} = \frac{x-p}{p(1-p)}$$

For a single Gaussian observation $X \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 known, $\ell(\mu; x) = -\frac{1}{2} \log(2\pi\sigma^2) - (x - \mu)^2/(2\sigma^2)$ and

$$s_\mu(x) = \frac{x - \mu}{\sigma^2}$$

In both cases the score is the observation minus its mean under P_θ , rescaled, so it is positive exactly when the observation exceeds that mean, which is exactly when raising θ raises the density at the observed point. Setting the score to zero is the equation the maximum-likelihood method solves (??).

To do anything with the score beyond computing it, one needs to move a θ -derivative across an integral in x . That move is not free: it fails already for $\text{Unif}(0, \theta)$, whose density is positive exactly on a set that grows with θ .

Definition 3.2.2: The Regularity Conditions

The family $\{P_\theta : \theta \in \Theta\}$ is *regular* if the following hold.

- (R1) Θ is an open interval in $\mathbb{R}$.
- (R2) (*Common support*) The set $\mathcal{S} = \{x \mid f_\theta(x) > 0\}$ is the same for every $\theta \in \Theta$.
- (R3) (*Differentiability*) For every $x \in \mathcal{S}$ the map $\theta \mapsto f_\theta(x)$ is absolutely continuous on every compact subinterval of Θ . The derivative $\partial_\theta f_\theta(x)$ then exists for Lebesgue-almost every θ , and the fundamental theorem of calculus recovers $f_\theta(x)$ from it, so the score $s_\theta(x)$ of definition 3.2.1 is defined at almost every θ for each fixed x . Absolute continuity rather than differentiability is what the integrals below use; save for that change, (R1) to (R3) are the assumptions (5.12) of [11, §2.5].^a
- (R4) (*Nondegeneracy*) $0 < \mathbb{E}_\theta[s_\theta(X)^2] < \infty$ for every $\theta \in \Theta$.
- (R5) (*Interchange*) For every $\theta \in \Theta$,

$$\partial_\theta \int_{\mathcal{S}} f_\theta(x) d\mu(x) = \int_{\mathcal{S}} \partial_\theta f_\theta(x) d\mu(x)$$

Given a statistic h with $\mathbb{E}_\theta|h(X)| < \infty$ for every θ , the family is *regular for h* if in addition

$$\partial_\theta \int_{\mathcal{S}} h(x) f_\theta(x) d\mu(x) = \int_{\mathcal{S}} h(x) \partial_\theta f_\theta(x) d\mu(x) \quad \text{for every } \theta \in \Theta$$

^aa pointwise derivative at every θ is the more familiar statement, but it excludes the Laplace location family met after proposition 3.2.3, whose density is not differentiable at $\theta = x$. Absolute continuity admits it. Weakening instead to “differentiable at almost every x ” would be a mistake: it admits the density of the location uniform $\text{Unif}(\theta, \theta + 1)$ (example 2.3.5), whose θ -derivative vanishes off the two endpoints, and a formal computation would then return information 0 where the truth is ∞ .

Conditions (R1) to (R3) are bookkeeping: they say that θ has room to move, that the set of possible data does not move with it, and that the densities are differentiable where they are positive. Condition (R4) rules out a degenerate score. Condition (R5) is the substantive one, and it is where (R2) does its work. The left-hand side differentiates $\int_{\mathcal{S}} f_\theta d\mu = 1$, a constant, and returns 0, so (R5) asserts that the integral of $\partial_\theta f_\theta$ over the fixed set $\mathcal{S}$ vanishes. If instead the region of positivity grew with θ , differentiating the integral would pick up a contribution from the moving boundary, and the two sides would differ by exactly that contribution. Example 3.2.9 carries this out on $\text{Unif}(0, \theta)$.

A dominating function delivers (R5) and its variant for a statistic h , and the criterion is the standard one rather than anything special to this setting. Suppose there are a $\delta > 0$, a μ -null set $N \subseteq \mathcal{S}$ and a $g \in L^1(\mu)$ such that for every $x \in \mathcal{S} \setminus N$ the map $u \mapsto f_u(x)$ is differentiable at every u with $|u - \theta| < \delta$ and satisfies $|h(x) \partial_u f_u(x)| \leq g(x)$ there. Differentiability along a whole interval of u at one fixed x is more than (R3) asks, whose absolute continuity yields a derivative only at almost every u , so it has to be checked separately, which is immediate in every family to which the criterion is applied below, where $\theta \mapsto f_\theta(x)$ is smooth for each fixed x . Applying *Measure Theory* [4, Relating Derivative and Integral], part (2), to $(x, u) \mapsto h(x) f_u(x)$ on $(\mathcal{S} \setminus N) \times [\theta - \delta/2, \theta + \delta/2]$, whose

hypotheses are that differentiability and the integrable dominating function just named, gives

$$\partial_\theta \int_{\mathcal{S}} h(x) f_\theta(x) d\mu(x) = \int_{\mathcal{S}} h(x) \partial_\theta f_\theta(x) d\mu(x) \quad (3.1)$$

which is the asserted interchange, the null set N changing neither side. Taking $h \equiv 1$ gives (R5). Every verification below, save for the Laplace location family's, produces such a g and then appeals to (3.1).

The two models above illustrate both halves. The Gaussian location family has support $\mathbb{R}$ for every μ , so (R2) holds, and for μ ranging over a bounded interval J the bound $(x - \mu)^2 \geq d(x, J)^2$, where $d(x, J)$ is the distance from x to J , gives

$$|\partial_\mu f_\mu(x)| = \frac{|x - \mu|}{\sigma^2} f_\mu(x) \leq C(1 + |x|) e^{-d(x, J)^2/(2\sigma^2)}$$

with C depending only on σ^2 and J . The right-hand side is Lebesgue integrable, since $d(x, J) \geq |x| - c$ for a constant c and a polynomial times a Gaussian is integrable, and the same estimate with an extra polynomial factor covers any statistic of polynomial growth. The Bernoulli family needs no analysis at all: the integral over $\mathcal{S} = \{0, 1\}$ is a two-term sum, and a finite sum of differentiable functions may always be differentiated term by term.

With (R5) in hand the score can be integrated, and the first consequence is that it is centred. This is what makes its second moment a variance and the whole construction below a covariance calculation.

Proposition 3.2.3: The Score Has Mean Zero

Let $\{P_\theta : \theta \in \Theta\}$ be regular (definition 3.2.2). Then for every $\theta \in \Theta$:

1. $\mathbb{E}_\theta[s_\theta(X)] = 0$.
2. If moreover $\theta \mapsto f_\theta(x)$ is twice differentiable on Θ for every $x \in \mathcal{S}$, if $\int_{\mathcal{S}} |\partial_\theta^2 f_\theta| d\mu < \infty$, and if the second-order interchange $\partial_\theta^2 \int_{\mathcal{S}} f_\theta d\mu = \int_{\mathcal{S}} \partial_\theta^2 f_\theta d\mu$ holds, then

$$\mathbb{E}_\theta[\partial_\theta^2 \ell(\theta; X)] = -\mathbb{E}_\theta[s_\theta(X)^2]$$

Proof:

Part (1). First, $s_\theta(X)$ is integrable. The Cauchy-Schwarz inequality [5, The Cauchy-Schwarz Inequality] applied to $s_\theta(X)$ and the constant function 1, both of which lie in $L^2(P_\theta)$ by (R4) and because P_θ is a probability measure, gives $\mathbb{E}_\theta|s_\theta(X)| \leq (\mathbb{E}_\theta[s_\theta(X)^2])^{1/2} < \infty$.

Because f_θ vanishes off $\mathcal{S}$, one has $P_\theta(\mathcal{S}) = 1$ and every expectation under P_θ is an integral over $\mathcal{S}$ against $f_\theta d\mu$. On $\mathcal{S}$ the density is strictly positive, so definition 3.2.1 gives $s_\theta(x)f_\theta(x) = \partial_\theta f_\theta(x)$ there, and

$$\mathbb{E}_\theta[s_\theta(X)] = \int_{\mathcal{S}} s_\theta(x) f_\theta(x) d\mu(x) = \int_{\mathcal{S}} \partial_\theta f_\theta(x) d\mu(x) = \partial_\theta \int_{\mathcal{S}} f_\theta(x) d\mu(x) = \partial_\theta 1 = 0$$

the third equality being (R5) and the fourth using that $\int_{\mathcal{S}} f_\theta d\mu = P_\theta(\mathcal{S}) = 1$ for every θ , which is where the θ -independence of $\mathcal{S}$ from (R2) is used.

Part (2). Differentiating $s_\theta = \partial_\theta f_\theta / f_\theta$ once more in θ , by the quotient rule and on the set $\mathcal{S}$ where the denominator is positive,

$$\partial_\theta^2 \ell(\theta; x) = \frac{\partial_\theta^2 f_\theta(x)}{f_\theta(x)} - \left(\frac{\partial_\theta f_\theta(x)}{f_\theta(x)} \right)^2 = \frac{\partial_\theta^2 f_\theta(x)}{f_\theta(x)} - s_\theta(x)^2$$

Both terms on the right are P_θ -integrable, the first because $\int_{\mathcal{S}} |\partial_\theta^2 f_\theta| d\mu < \infty$ by hypothesis and the second by (R4), so the expectation of the difference is the difference of the expectations:

$$\mathbb{E}_\theta[\partial_\theta^2 \ell(\theta; X)] = \int_{\mathcal{S}} \partial_\theta^2 f_\theta(x) d\mu(x) - \mathbb{E}_\theta[s_\theta(X)^2] = \partial_\theta^2 \int_{\mathcal{S}} f_\theta(x) d\mu(x) - \mathbb{E}_\theta[s_\theta(X)^2]$$

using the assumed second-order interchange. The remaining integral is again the constant 1, whose second derivative is 0, leaving $\mathbb{E}_\theta[\partial_\theta^2 \ell(\theta; X)] = -\mathbb{E}_\theta[s_\theta(X)^2]$, as we sought to show.

Part (2) is stated with hypotheses of its own because the first-order interchange does not imply the second-order one. In the Laplace location family $f_\theta(x) = \frac{1}{2}e^{-|x-\theta|}$ on $\mathbb{R}$ the support is all of $\mathbb{R}$, so (R2) holds, and for each fixed x the map $\theta \mapsto \frac{1}{2}e^{-|x-\theta|}$ is Lipschitz, hence absolutely continuous, so (R3) holds. It is the family the condition is stated in that form for: the pointwise derivative fails at the single parameter value $\theta = x$. Off the null set $\{x = \theta\}$ the score is $\text{sgn}(x - \theta)$, with mean 0 by symmetry and second moment 1, and (R5) holds because $\partial_\theta f_\theta$ is odd about $x = \theta$. Yet $\partial_\theta^2 \ell(\theta; x) = 0$ at every $x \neq \theta$, so the expected second derivative is 0 rather than -1 . Where both interchanges do hold, part (2) is the computational route usually taken, since a second derivative of an explicit log-density is easier to average than the square of a first derivative.

Definition 3.2.4: Fisher Information

Let $\{P_\theta : \theta \in \Theta\}$ be regular. The *Fisher information* of the family at θ is the variance of the score,

$$\mathcal{I}(\theta) = \text{Var}_\theta(s_\theta(X))$$

By proposition 3.2.3(1) the score has mean zero, so $\mathcal{I}(\theta) = \mathbb{E}_\theta[s_\theta(X)^2]$, and under the additional hypotheses of proposition 3.2.3(2) also $\mathcal{I}(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \ell(\theta; X)]$. Condition (R4) says exactly that $0 < \mathcal{I}(\theta) < \infty$. When the amount of data matters, write $\mathcal{I}_1(\theta)$ for the information carried by one observation and $\mathcal{I}_n(\theta)$ for that carried by a sample of size n .

The three expressions read differently. The first says that information is the variability of the score across the sample space: a model is informative at θ when the direction in which θ should be nudged varies sharply with the data, so that the data has something to say about θ . The third says that information is the average curvature of the log-likelihood at θ , with a sign flip to make it positive at a maximum. A sharply peaked log-likelihood pins θ down; a flat one does not.

Example 3.2.5: Information in the Bernoulli and Gaussian Models

Both computations use the scores already found after definition 3.2.1, and each is checked twice, once as a variance and once as an expected second derivative.

1. *Bernoulli*. Let $X \sim \text{Ber}(p)$ with $p \in \Theta = (0, 1)$. The support $\mathcal{S} = \{0, 1\}$ is free of p , and $f_u(x) = u^x(1-u)^{1-x}$ is affine in u with $\partial_u f_u(x) = 2x - 1$, so for every statistic h the quantity $|h(x) \partial_u f_u(x)| = |h(x)|$ is a u -free bound, summable over the two points. By (3.1) the family

is regular and regular for every statistic. From $s_p(x) = (x - p)/(p(1 - p))$,

$$\mathcal{I}_1(p) = \text{Var}_p\left(\frac{X - p}{p(1 - p)}\right) = \frac{\text{Var}_p(X)}{p^2(1 - p)^2} = \frac{p(1 - p)}{p^2(1 - p)^2} = \frac{1}{p(1 - p)}$$

For the second form, $\partial_p^2 \ell(p; x) = -x/p^2 - (1 - x)/(1 - p)^2$, whose expectation under P_p is $-p/p^2 - (1 - p)/(1 - p)^2 = -1/p - 1/(1 - p) = -1/(p(1 - p))$, and negating recovers $\mathcal{I}_1(p) = 1/(p(1 - p))$.

The information is smallest at $p = 1/2$ and grows without bound as p approaches either endpoint, which matches the behaviour of the model. A coin with p near 0 produces heads so rarely that a single head is strong evidence about p , whereas near $p = 1/2$ the two outcomes stay nearly equally likely under all nearby values of p .

2. *Gaussian location.* Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 known and $\mu \in \Theta = \mathbb{R}$. The support $\mathbb{R}$ is free of μ , and $\partial_u f_u$ is dominated, locally uniformly in u , by a fixed integrable function, so the family is regular and regular for every statistic of polynomial growth. From $s_\mu(x) = (x - \mu)/\sigma^2$,

$$\mathcal{I}_1(\mu) = \frac{\text{Var}_\mu(X)}{\sigma^4} = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}$$

and $\partial_\mu^2 \ell(\mu; x) = -1/\sigma^2$ is already constant, so its negated expectation is $1/\sigma^2$ as well. The information does not depend on μ , as it must for a location family, since shifting the data and the parameter together changes nothing.

In both models the sample mean $\bar{X}$ attains the bound of theorem 3.2.7 below. For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ the sample information is $\mathcal{I}_n(p) = n/(p(1 - p))$ by proposition 3.2.6, the estimand $\psi(p) = p$ has $\psi' \equiv 1$, and the bound $\psi'(p)^2/\mathcal{I}_n(p) = p(1 - p)/n$ equals $\text{Var}_p(\bar{X})$, so the unbiased estimator $\bar{X}$ of example 3.1.3 attains it. For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known, $\mathcal{I}_n(\mu) = n/\sigma^2$ and the bound is $\sigma^2/n = \text{Var}_\mu(\bar{X})$, again attained by $\bar{X}$. In each case the sample score is an affine function of $\bar{X}$: summing the one-observation scores gives $n(\bar{X} - p)/(p(1 - p))$ and $n(\bar{X} - \mu)/\sigma^2$ respectively, which is the equality condition of corollary 3.2.8(1) with $a(p) = p(1 - p)/n$ and $a(\mu) = \sigma^2/n$.

The claim that information adds over an independent sample was used in the preceding paragraph, and it is what makes the bound decay at a definite rate in n . It follows from the score of a product model being the sum of the coordinate scores, together with the vanishing of the cross terms.

Proposition 3.2.6: Information Is Additive Over an I.I.D. Sample

Let $\{P_\theta: \theta \in \Theta\}$ be a regular family with densities f_θ relative to μ , and let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta$, so that $X = (X_1, \dots, X_n)$ has density $f_\theta^{\otimes n}(x) = \prod_{j=1}^n f_\theta(x_j)$ relative to $\mu^{\otimes n}$. Then the sample model is regular and

$$\mathcal{I}_n(\theta) = n \mathcal{I}_1(\theta)$$

Suppose moreover that the one-observation family is regular for every statistic h of polynomial growth by way of the criterion (3.1): for each $\theta_0 \in \Theta$ and each such h there are a $\delta > 0$, a μ -null $N \subseteq \mathcal{S}$ and a $g \in L^1(\mu)$ such that for every $y \in \mathcal{S} \setminus N$ the map $\theta \mapsto f_\theta(y)$ is differentiable on $(\theta_0 - \delta, \theta_0 + \delta)$ and $|h(y) \partial_\theta f_\theta(y)| \leq g(y)$ there. Then the sample model is regular for every statistic of polynomial growth on $\mathcal{S}^n$, so the applications below may invoke theorem 3.2.7 for estimators such as $\bar{X}$ without checking the sample-space interchange again.

Proof:

The support of $f_\theta^{\otimes n}$ is $\mathcal{S}^n$, free of θ because $\mathcal{S}$ is, so (R2) holds for the sample model, and (R1) is a condition on Θ alone. On $\mathcal{S}^n$ the log-likelihood is the sum $\ell_n(\theta; x) = \sum_{j=1}^n \ell(\theta; x_j)$. Fix $x \in \mathcal{S}^n$. Each factor $\theta \mapsto f_\theta(x_j)$ is absolutely continuous on a compact subinterval by (R3), and is bounded there, so the finite product $\theta \mapsto f_\theta^{\otimes n}(x)$ is absolutely continuous on it as well. This is (R3) for the sample model, and term-by-term differentiation at the almost every θ where the factors are differentiable gives

$$s_\theta^{(n)}(x) = \sum_{j=1}^n s_\theta(x_j)$$

Step 1: the cross terms vanish. Fix $i \neq j$. Under $P_\theta^{\otimes n}$ the pair (X_i, X_j) has density $f_\theta(x_i)f_\theta(x_j)$ relative to $\mu \otimes \mu$, since integrating out any one of the remaining coordinates contributes a factor $\int_{\mathcal{S}} f_\theta d\mu = 1$. The function $(x_i, x_j) \mapsto s_\theta(x_i)s_\theta(x_j)f_\theta(x_i)f_\theta(x_j)$ is a product of a function of x_i and a function of x_j , and its absolute value integrates to

$$\int_{\mathcal{S}} \int_{\mathcal{S}} |s_\theta(x_i)| |s_\theta(x_j)| f_\theta(x_i)f_\theta(x_j) d\mu(x_i) d\mu(x_j) = (\mathbb{E}_\theta |s_\theta(X_1)|)^2 < \infty$$

by Tonelli's half of [4, Fubini-Tonelli Theorem], whose hypotheses hold because the integrand is nonnegative and μ is σ -finite, the finiteness of $\mathbb{E}_\theta |s_\theta(X_1)|$ being the integrability established at the start of the proof of proposition 3.2.3. The integrand is therefore absolutely integrable, so Fubini's half of the same theorem factors the integral:

$$\mathbb{E}_\theta [s_\theta(X_i)s_\theta(X_j)] = \left(\int_{\mathcal{S}} s_\theta f_\theta d\mu \right)^2 = (\mathbb{E}_\theta [s_\theta(X_1)])^2 = 0$$

the last equality by proposition 3.2.3(1).

Step 2: assembling. Each summand $s_\theta(X_j)$ has mean 0 by proposition 3.2.3(1), so $s_\theta^{(n)}(X)$ has mean 0 and its variance is its second moment. Expanding the square of the sum,

$$\mathcal{I}_n(\theta) = \mathbb{E}_\theta \left[\left(\sum_{j=1}^n s_\theta(X_j) \right)^2 \right] = \sum_{j=1}^n \mathbb{E}_\theta [s_\theta(X_j)^2] + \sum_{i \neq j} \mathbb{E}_\theta [s_\theta(X_i)s_\theta(X_j)] = n \mathcal{I}_1(\theta)$$

the diagonal terms each equalling $\mathcal{I}_1(\theta)$ because X_j has law P_θ , and the off-diagonal terms vanishing by Step 1. In particular $0 < \mathcal{I}_n(\theta) < \infty$, which is (R4) for the sample model.

Finally (R5) holds for the sample. Differentiating the product $\prod_j f_\theta(x_j)$ of strictly positive factors gives $\partial_\theta f_\theta^{\otimes n} = s_\theta^{(n)} f_\theta^{\otimes n}$ at every $x \in \mathcal{S}^n$ off the null set named above, so

$$\int_{\mathcal{S}^n} \partial_\theta f_\theta^{\otimes n} d\mu^{\otimes n} = \mathbb{E}_\theta[s_\theta^{(n)}(X)] = 0 = \partial_\theta 1 = \partial_\theta \int_{\mathcal{S}^n} f_\theta^{\otimes n} d\mu^{\otimes n}$$

which is the required identity.

For the last claim, fix $\theta_0 \in \Theta$ and a statistic h of polynomial growth on $\mathcal{S}^n$, so that $|h(x)| \leq A(1 + \sum_{l=1}^n |x_l|)^m$ for some $A > 0$ and some integer $m \geq 1$. Since $1 + \sum_l |x_l| \leq \prod_l (1 + |x_l|)$ and $(1 + t)^m \leq 2^{m-1}(1 + t^m)$ for $t \geq 0$, this bound factors across the coordinates:

$$|h(x)| \leq C \prod_{l=1}^n h_k(x_l), \quad h_k(y) = 1 + |y|^k, \quad k = m, \quad C = A 2^{(m-1)n}$$

Now h_k is a statistic of polynomial growth on $\mathcal{S}$, so the hypothesis gives a $\delta > 0$, a μ -null N and a $g \in L^1(\mu)$ for it at θ_0 , and regularity for h_k presupposes $\mathbb{E}_\theta[h_k(X_1)] < \infty$ at every θ . That finiteness and the display give $\mathbb{E}_\theta|h(X)| \leq C(\mathbb{E}_\theta[h_k(X_1)])^n < \infty$, the integrability that definition 3.2.2 asks of a statistic before regularity for it can be discussed. Since $h_k \geq 1$, the same g dominates $|\partial_\theta f_\theta|$ on $\mathcal{S} \setminus N$.

The step the argument turns on is that a bound on the derivative also bounds the density itself, uniformly in θ . For $y \in \mathcal{S} \setminus N$ the map $\theta \mapsto f_\theta(y)$ is differentiable on $(\theta_0 - \delta, \theta_0 + \delta)$, so the mean value theorem gives $h_k(y) |f_\theta(y) - f_{\theta_0}(y)| \leq \delta g(y)$ there. Setting $G = h_k f_{\theta_0} + \delta g$, this reads

$$h_k(y) f_\theta(y) \leq G(y) \quad \text{for } y \in \mathcal{S} \setminus N \text{ and } |\theta - \theta_0| < \delta$$

and $G \in L^1(\mu)$, its first term because $\int_{\mathcal{S}} h_k f_{\theta_0} d\mu = \mathbb{E}_{\theta_0}[h_k(X_1)]$ is finite and its second because g is. Both g and G are free of θ , which is what a dominating function has to be and what the θ -dependent factors $f_\theta(x_l)$ are not.

Now expand the derivative of the product. For $x \in \mathcal{S}^n$ with no coordinate in N and for $|\theta - \theta_0| < \delta$,

$$|h(x) \partial_\theta f_\theta^{\otimes n}(x)| \leq C \sum_{j=1}^n h_k(x_j) |\partial_\theta f_\theta(x_j)| \prod_{l \neq j} h_k(x_l) f_\theta(x_l) \leq C \sum_{j=1}^n g(x_j) \prod_{l \neq j} G(x_l)$$

the first inequality by $\partial_\theta f_\theta^{\otimes n}(x) = \sum_j (\partial_\theta f_\theta(x_j)) \prod_{l \neq j} f_\theta(x_l)$ together with the factored bound on $|h|$, the second because $h_k |\partial_\theta f_\theta| \leq g$ by the choice of g and $h_k f_\theta \leq G$ by the display above. The right-hand side does not involve θ , and Tonelli's half of [4, Fubini-Tonelli Theorem] integrates it to $Cn(\int_{\mathcal{S}} g d\mu)(\int_{\mathcal{S}} G d\mu)^{n-1} < \infty$, so it lies in $L^1(\mu^{\otimes n})$. The set of $x \in \mathcal{S}^n$ having a coordinate in N is $\mu^{\otimes n}$ -null, so (3.1) applies on $\mathcal{S}^n$ with the measure $\mu^{\otimes n}$, the statistic h and this dominating function, and yields the interchange. The sample model is therefore regular for h , completing the proof.

Additivity is why a parametric model improves at the rate it does: n independent observations carry n times the information of one, so the bound below shrinks like $1/n$ and the standard deviation of an unbiased estimator that attains it like $n^{-1/2}$.

The inequality says that an unbiased estimator's variance cannot fall below a ratio whose numerator measures how fast the estimand moves with θ and whose denominator measures how fast the model does.

Theorem 3.2.7: The Cramér-Rao Inequality

Let $\{P_\theta: \theta \in \Theta\}$ be regular (definition 3.2.2), with score s_θ and Fisher information $\mathcal{I}(\theta)$. Let $\psi: \Theta \rightarrow \mathbb{R}$ be differentiable and let $\hat{\psi} = \hat{\psi}(X)$ be an estimator (definition 3.1.1) that is unbiased for ψ (definition 3.1.2), with $\text{Var}_\theta(\hat{\psi}) < \infty$ for every θ . Assume the family is regular for $\hat{\psi}$. Then for every $\theta \in \Theta$,

$$\text{Var}_\theta(\hat{\psi}) \geq \frac{\psi'(\theta)^2}{\mathcal{I}(\theta)} \quad (3.2)$$

Proof:

Step 1: the covariance of the estimator with the score is $\psi'(\theta)$. Both $\hat{\psi}(X)$ and $s_\theta(X)$ lie in $L^2(P_\theta)$, the former because $\text{Var}_\theta(\hat{\psi}) < \infty$ and $\mathbb{E}_\theta[\hat{\psi}] = \psi(\theta)$ is finite, the latter by (R4). Their product is therefore P_θ -integrable by the Cauchy-Schwarz inequality [5, The Cauchy-Schwarz Inequality], and $\text{Cov}_\theta(\hat{\psi}, s_\theta)$ is defined. On $\mathcal{S}$ the identity $s_\theta f_\theta = \partial_\theta f_\theta$ of definition 3.2.1 holds, so

$$\mathbb{E}_\theta[\hat{\psi}(X)s_\theta(X)] = \int_{\mathcal{S}} \hat{\psi}(x)s_\theta(x)f_\theta(x) d\mu(x) = \int_{\mathcal{S}} \hat{\psi}(x) \partial_\theta f_\theta(x) d\mu(x)$$

Because the family is regular for $\hat{\psi}$ by hypothesis, the last integral equals

$$\partial_\theta \int_{\mathcal{S}} \hat{\psi}(x)f_\theta(x) d\mu(x) = \partial_\theta \mathbb{E}_\theta[\hat{\psi}(X)] = \partial_\theta \psi(\theta) = \psi'(\theta)$$

the middle equality being unbiasedness. Since $\mathbb{E}_\theta[s_\theta(X)] = 0$ by proposition 3.2.3(1), whose hypotheses are the regularity assumed here,

$$\text{Cov}_\theta(\hat{\psi}, s_\theta) = \mathbb{E}_\theta[\hat{\psi}s_\theta] - \mathbb{E}_\theta[\hat{\psi}]\mathbb{E}_\theta[s_\theta] = \psi'(\theta)$$

Step 2: Cauchy-Schwarz. Set $U = \hat{\psi}(X) - \psi(\theta)$ and $V = s_\theta(X)$. Both are centred elements of $L^2(P_\theta)$ with $\mathbb{E}_\theta[U^2] = \text{Var}_\theta(\hat{\psi})$ and $\mathbb{E}_\theta[V^2] = \mathcal{I}(\theta)$, and $\mathbb{E}_\theta[UV] = \text{Cov}_\theta(\hat{\psi}, s_\theta) = \psi'(\theta)$ by Step 1. The Cauchy-Schwarz inequality [5, The Cauchy-Schwarz Inequality] applied to U and V gives

$$\psi'(\theta)^2 = (\mathbb{E}_\theta[UV])^2 \leq \mathbb{E}_\theta[U^2]\mathbb{E}_\theta[V^2] = \text{Var}_\theta(\hat{\psi})\mathcal{I}(\theta) \quad (3.3)$$

By (R4) the factor $\mathcal{I}(\theta)$ is strictly positive and finite, so dividing through by it yields $\text{Var}_\theta(\hat{\psi}) \geq \psi'(\theta)^2/\mathcal{I}(\theta)$, which is (3.2), as we sought to show.

The bound separates into a part belonging to the estimand and a part belonging to the model. The numerator $\psi'(\theta)^2$ grows when small changes in θ produce large changes in the quantity being

estimated, which makes that quantity harder to pin down; the denominator $\mathcal{I}(\theta)$ grows when small changes in θ produce large changes in the law of the data, which makes θ easier to pin down. For an i.i.d. sample proposition 3.2.6 replaces $\mathcal{I}(\theta)$ by $n\mathcal{I}_1(\theta)$, so (3.2) reads $\text{Var}_\theta(\hat{\psi}) \geq \psi'(\theta)^2/(n\mathcal{I}_1(\theta))$ and no unbiased estimator has standard deviation smaller than a constant times $n^{-1/2}$. The several-parameter form of the same inequality, in which the reciprocal becomes a matrix inverse and estimating one coordinate in the presence of the others can cost more than estimating it alone, is section 3.4. Whether it does cost more turns on the parametrization, and there are models where it does not.

The theorem leaves open when the bound is met, and by which estimator. Both are settled by the case of equality in (3.3), and the answer forces the model into the shape studied in section 2.4.

Corollary 3.2.8: Attainment and One-Parameter Exponential Families

Parts (1) and (2) assume the hypotheses of theorem 3.2.7; part (3) stands on the hypotheses stated in it.

1. Fix $\theta \in \Theta$. Equality holds in (3.2) at θ if and only if there is a constant $a(\theta) \in \mathbb{R}$ with

$$\hat{\psi}(X) - \psi(\theta) = a(\theta) s_\theta(X) \quad P_\theta\text{-a.s.}$$

and in that case $a(\theta) = \psi'(\theta)/\mathcal{I}(\theta)$.

2. Suppose equality holds in (3.2) at every $\theta \in \Theta$, that ψ' is continuous and nowhere zero on Θ , that $\mathcal{I}$ is continuous on Θ , and that $\theta \mapsto \partial_\theta f_\theta(x)$ is continuous on Θ for every $x \in \mathcal{S}$. Fix $\theta_0 \in \Theta$. Then there are a measurable $h \geq 0$, a continuously differentiable strictly monotone $\eta: \Theta \rightarrow \mathbb{R}$, and $B: \Theta \rightarrow \mathbb{R}$ such that

$$f_\theta(x) = h(x) \exp(\eta(\theta)\hat{\psi}(x) - B(\theta)) \quad \text{for every } \theta \in \Theta \text{ and } \mu\text{-a.e. } x$$

so the family is a one-parameter exponential family (definition 2.4.1) whose natural statistic is the estimator $\hat{\psi}$ itself.

3. Conversely, let $\{P_\eta: \eta \in \Xi\}$ be a canonical one-parameter exponential family with natural statistic T and cumulant function A , restrict the indexing to the interior of Ξ (at a boundary point of Ξ the statistic T need not even be integrable, so no mean identity is available there), let η be interior to Ξ , and suppose $\text{Var}_\eta(T) > 0$. Then T is unbiased over that restricted family for $\psi(\eta) = A'(\eta)$, one has $\mathcal{I}(\eta) = A''(\eta)$, and

$$\text{Var}_\eta(T) = \frac{\psi'(\eta)^2}{\mathcal{I}(\eta)}$$

so the two sides of (3.2) agree.

Proof:

1. Keep $U = \hat{\psi}(X) - \psi(\theta)$ and $V = s_\theta(X)$ from Step 2 of the proof of theorem 3.2.7, and recall that equality in (3.2) is the same as equality in (3.3). Since $\mathbb{E}_\theta[V^2] = \mathcal{I}(\theta) > 0$ by

(R4), the number $c = \mathbb{E}_\theta[UV]/\mathbb{E}_\theta[V^2]$ is defined, and expanding the square,

$$0 \leq \mathbb{E}_\theta[(U - cV)^2] = \mathbb{E}_\theta[U^2] - 2c\mathbb{E}_\theta[UV] + c^2\mathbb{E}_\theta[V^2] = \mathbb{E}_\theta[U^2] - \frac{(\mathbb{E}_\theta[UV])^2}{\mathbb{E}_\theta[V^2]}$$

Multiplying by $\mathbb{E}_\theta[V^2] > 0$ shows that equality in (3.3) holds if and only if $\mathbb{E}_\theta[(U - cV)^2] = 0$, that is if and only if $U = cV$ P_θ -a.s. This is the asserted identity with $a(\theta) = c$, and by Step 1 of that proof $c = \mathbb{E}_\theta[UV]/\mathbb{E}_\theta[V^2] = \psi'(\theta)/\mathcal{I}(\theta)$.

2. *Step 1: one null set serves every θ .* Write $a = \psi'/\mathcal{I}$, continuous on Θ by hypothesis and nowhere zero, since ψ' is nowhere zero and $0 < \mathcal{I} < \infty$. For each θ let $N_\theta \subseteq \mathcal{S}$ be the set where the identity of part (1) fails, so $P_\theta(N_\theta) = 0$, that is $\int_{N_\theta} f_\theta d\mu = 0$. As $f_\theta > 0$ on $\mathcal{S}$, this forces $\mu(N_\theta) = 0$. Choose a countable dense subset $D \subseteq \Theta$ and put $N = \bigcup_{\theta \in D} N_\theta$, a μ -null subset of $\mathcal{S}$.

Fix $x \in \mathcal{S} \setminus N$. For every $\theta \in D$,

$$s_\theta(x) = \frac{\hat{\psi}(x) - \psi(\theta)}{a(\theta)}$$

The right-hand side is continuous in θ , because ψ is differentiable and hence continuous and a is continuous and nonvanishing. The left-hand side is continuous in θ as well, being the quotient $\partial_\theta f_\theta(x)/f_\theta(x)$ of a numerator continuous in θ by hypothesis and a denominator that is continuous in θ and nonvanishing on $\mathcal{S}$. Continuity of the *numerator* is what does not come from (R3), which gives absolute continuity in θ and so continuity of the denominator $f_\theta(x)$ itself but nothing about $\partial_\theta f_\theta(x)$: it comes from this part's hypothesis that $\theta \mapsto \partial_\theta f_\theta(x)$ is defined and continuous on Θ for every $x \in \mathcal{S}$, which makes $\theta \mapsto f_\theta(x)$ continuously differentiable. Two continuous functions agreeing on the dense set D agree on all of Θ , so the displayed identity holds for every $\theta \in \Theta$.

Step 2: integrating in θ . Set

$$\eta(\theta) = \int_{\theta_0}^{\theta} \frac{du}{a(u)}, \quad B(\theta) = \int_{\theta_0}^{\theta} \frac{\psi(u)}{a(u)} du$$

Both integrands are continuous on the interval Θ (R1), so both functions are defined and continuously differentiable with $\eta' = 1/a$ and $B' = \psi/a$. For $x \in \mathcal{S} \setminus N$ the map $u \mapsto \ell(u; x)$ is differentiable on Θ with derivative $s_u(x) = \hat{\psi}(x)/a(u) - \psi(u)/a(u)$, continuous in u by Step 1, so the fundamental theorem of calculus applies to it and gives

$$\ell(\theta; x) - \ell(\theta_0; x) = \hat{\psi}(x) \eta(\theta) - B(\theta)$$

Exponentiating, $f_\theta(x) = f_{\theta_0}(x) \exp(\eta(\theta)\hat{\psi}(x) - B(\theta))$. Set $h = f_{\theta_0} \mathbb{1}_{\mathcal{S}}$. For $x \notin \mathcal{S}$ both sides vanish, since $f_\theta(x) = 0$ for every θ by (R2) and $h(x) = 0$ by construction, so the identity holds for every $\theta \in \Theta$ and every x outside the μ -null set N . Finally $\eta' = 1/a$ is continuous and nowhere zero on the interval Θ , hence of constant sign, so η is strictly monotone.

3. Write $f_\eta(x) = h(x)e^{\eta T(x) - A(\eta)}$. The support $\{h > 0\}$ does not depend on η , and on it $\ell(\eta; x) = \log h(x) + \eta T(x) - A(\eta)$, which is differentiable in η with

$$s_\eta(x) = T(x) - A'(\eta)$$

Since η is interior to Ξ , the identities (2.7) of proposition 2.4.4, in the one-dimensional case where the gradient is A' and the Hessian is A'' , give $A'(\eta) = \mathbb{E}_\eta[T]$ and $A''(\eta) = \text{Var}_\eta(T)$. Hence $\mathbb{E}_\eta[s_\eta(X)] = \mathbb{E}_\eta[T] - A'(\eta) = 0$ and

$$\mathcal{I}(\eta) = \text{Var}_\eta(s_\eta(X)) = \text{Var}_\eta(T) = A''(\eta)$$

which is strictly positive by the assumption $\text{Var}_\eta(T) > 0$ and finite because T has a variance. Put $\psi(\eta) = A'(\eta) = \mathbb{E}_\eta[T]$, so that T is unbiased for ψ , and $\psi'(\eta) = A''(\eta)$ because A is infinitely differentiable at interior points by proposition 2.4.4. Therefore

$$\frac{\psi'(\eta)^2}{\mathcal{I}(\eta)} = \frac{A''(\eta)^2}{A''(\eta)} = A''(\eta) = \text{Var}_\eta(T)$$

which is the asserted equality. It agrees with part (1), since here $T - \psi(\eta) = T - A'(\eta) = s_\eta$ and so $a(\eta) = 1 = \psi'(\eta)/\mathcal{I}(\eta)$. Step 1 of the proof of theorem 3.2.7 has been verified rather than assumed in this case, because $\text{Cov}_\eta(T, s_\eta) = \text{Var}_\eta(T) = A''(\eta) = \psi'(\eta)$ directly, completing the proof.

Parts (2) and (3) together say that attaining the bound at every parameter value is a structural property of the model. Under the continuity hypotheses of part (2), attainment at every θ forces the family to be a one-parameter exponential family and forces the estimand to be the mean of the natural statistic, whatever parametrization the family is written in; part (3) is the converse. Within the reach of those hypotheses, then, attainment everywhere is confined to a single class of models, which is why (3.2) is used more often to record a floor on what any unbiased estimator can achieve than to search for one. Only where the floor is attained does it also certify a particular estimator as optimal, and the construction of good unbiased estimators proceeds by a different route (section 3.3). The two attainments in example 3.2.5 are instances. The Bernoulli family is canonical with natural statistic x and natural parameter the log-odds, by example 2.4.6(1). The Gaussian family at known σ^2 is canonical with carrier $h(x) = (2\pi\sigma^2)^{-1/2}e^{-x^2/(2\sigma^2)}$, natural statistic $T(x) = x$, natural parameter $\eta = \mu/\sigma^2$ and $A(\eta) = \sigma^2\eta^2/2$, the one-parameter reduction of the two-parameter form in example 2.4.6(4). In each case the estimand attaining the bound is $\mathbb{E}[T]$, which is p and μ respectively.

One of the running models fails (R2), and it fails badly enough that its natural estimator has variance below what (3.2) would allow.

Example 3.2.9: The Uniform Model Violates the Support Condition

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ with $\theta \in (0, \infty)$, so $f_\theta(x) = \theta^{-1} \mathbb{1}\{0 < x < \theta\}$ relative to Lebesgue measure. The set where the density is positive is $(0, \theta)$, which grows with θ , so (R2) fails. This is the same feature that keeps the family out of the exponential class in example 2.4.6(6).

The conclusions fail, one by one. For $0 < x < \theta$ the log-density is $\ell(\theta; x) = -\log \theta$, so the score is

the constant $s_\theta(x) = -1/\theta$. Then

$$\mathbb{E}_\theta[s_\theta(X)] = -\frac{1}{\theta} \neq 0$$

so the conclusion of proposition 3.2.3(1) fails. The reason is visible in the interchange:

$$\int_0^\theta \partial_\theta f_\theta(x) dx = \int_0^\theta \left(-\frac{1}{\theta^2}\right) dx = -\frac{1}{\theta}, \quad \partial_\theta \int_0^\theta f_\theta(x) dx = \partial_\theta 1 = 0$$

so (R5) fails, and the two sides differ by $1/\theta$, which is the value $f_\theta(\theta^-)$ of the density at the endpoint that moves. Because the score is not centred, the two expressions in definition 3.2.4 come apart: $\text{Var}_\theta(s_\theta(X)) = 0$, the score being constant, while $\mathbb{E}_\theta[s_\theta(X)^2] = 1/\theta^2$.

The bound is beaten. Take the second expression, which is the one yielding a finite positive number, so that the sample information would read n/θ^2 and (3.2) would predict $\text{Var}_\theta(\hat{\theta}) \geq \theta^2/n$ for any unbiased $\hat{\theta}$. Now $\hat{\theta} = \frac{n+1}{n} X_{(n)}$ is unbiased for θ by example 3.1.3(3), which also computes the law of the maximum and its first two moments, giving $\text{Var}_\theta(X_{(n)}) = n\theta^2/((n+2)(n+1)^2)$. Scaling by $\frac{n+1}{n}$ multiplies the variance by $(n+1)^2/n^2$, giving

$$\text{Var}_\theta(\hat{\theta}) = \frac{\theta^2}{n(n+2)}$$

smaller than θ^2/n for every $n \geq 1$, and of order n^{-2} rather than n^{-1} .

Where the proof breaks. The sample score is the constant $-n/\theta$, so $\text{Cov}_\theta(\hat{\theta}, s_\theta^{(n)}) = 0$, whereas Step 1 of the proof of theorem 3.2.7 concluded that this covariance equals $\psi'(\theta) = 1$. The step that fails is the interchange, and the two sides can be compared directly: $\int \hat{\theta} \partial_\theta f_\theta^{\otimes n} dx = -\frac{n}{\theta} \mathbb{E}_\theta[\hat{\theta}] = -n$, while $\partial_\theta \mathbb{E}_\theta[\hat{\theta}] = \partial_\theta \theta = 1$.

Nothing here contradicts theorem 3.2.7, whose hypotheses (R2) and (R5) both fail in this model. What the example shows is that the bound measures regularity rather than difficulty. The uniform parameter is estimated better than the regular rate allows, not worse: $\theta - X_{(n)}$ has mean $\theta/(n+1)$, so the maximum sits within order n^{-1} of θ , where a regular model concentrates only at order $n^{-1/2}$. The extra precision comes from the endpoint of the support, which is θ itself, and which no derivative of the density at an interior point involves. Whenever the support moves with the parameter, $n^{-1/2}$ is the wrong benchmark and the analysis has to be carried out at the boundary.

Exercise 3.2.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ with $\lambda \in (0, \infty)$. Compute the score of a single observation, and obtain $\mathcal{I}_1(\lambda)$ twice, once as $\text{Var}_\lambda(s_\lambda(X))$ and once as $-\mathbb{E}_\lambda[\partial_\lambda^2 \ell(\lambda; X)]$. Write down the resulting bound (3.2) for unbiased estimators of λ , verify that $\bar{X}$ attains it, and identify the constant $a(\lambda)$ of corollary 3.2.8(1).
2. Let $g: \Theta \rightarrow H$ be a continuously differentiable bijection with $g'(\theta) \neq 0$ everywhere, and rewrite a regular family in the parameter $\eta = g(\theta)$, so that $\tilde{f}_\eta = f_{g^{-1}(\eta)}$. Prove that the Fisher information transforms as $\mathcal{I}(\theta) = g'(\theta)^2 \tilde{\mathcal{I}}(g(\theta))$, and deduce that the right-hand side

of (3.2) is unchanged by the reparametrization when the estimand is. Then check the rule on the Bernoulli model: taking g to be the log-odds map of example 2.4.6(1), confirm that the information $\tilde{\mathcal{I}}(\eta) = A''(\eta)$ given by corollary 3.2.8(3) reproduces $\mathcal{I}(p) = 1/(p(1-p))$.

3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, v)$ with the mean known to be 0 and the variance $v \in (0, \infty)$ unknown. Show that $\mathcal{I}_1(v) = 1/(2v^2)$, so that every unbiased estimator of v has variance at least $2v^2/n$, and show that $\hat{v} = \frac{1}{n} \sum_{j=1}^n X_j^2$ attains this bound. (The fourth moment of a centred Gaussian follows from one integration by parts.)
4. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_\theta$ where $f_\theta(x) = e^{-(x-\theta)} \mathbb{1}\{x > \theta\}$ and $\theta \in \mathbb{R}$. Say which conditions of definition 3.2.2 fail (more than one does). Compute $\mathbb{E}_\theta[s_\theta(X)]$ and $\mathbb{E}_\theta[s_\theta(X)^2]$ for a single observation, and record the bound that (3.2) would give for unbiased estimators of θ if the second quantity were used as the one-observation information. Then show that $\bar{X}_{(1)} - 1/n$ is unbiased for θ with variance $1/n^2$, strictly smaller than that bound for $n \geq 2$.
5. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known, and take $\psi(\mu) = \mu^2$. Show that $\bar{X}^2 - \sigma^2/n$ is unbiased for ψ and that its variance exceeds the bound (3.2) by exactly $2\sigma^4/n^2$. Then prove that no unbiased estimator of μ^2 attains the bound at every μ , by examining what attainment at $\mu = 0$ would force.
6. Let $f_\theta(x) = \frac{1}{2}e^{-|x-\theta|}$ on $\mathbb{R}$ with $\theta \in \mathbb{R}$. Verify (R2), verify (R3) by showing that $\theta \mapsto f_\theta(x)$ is Lipschitz at each fixed x , and verify (R5) by computing both of its sides. Then compute $\mathbb{E}_\theta[\partial_\theta^2 \ell(\theta; X)]$, compare it with $-\mathcal{I}_1(\theta)$, and show that the second-order interchange hypothesis of proposition 3.2.3(2) fails as well by computing both of its sides. Say which of that part's three hypotheses does hold. Finally, compare the bound for unbiased estimators of θ with $\text{Var}_\theta(\bar{X})$, and use corollary 3.2.8(1) to explain why $\bar{X}$ cannot attain it.

3.3 UMVU Estimation: Rao-Blackwell and Lehmann-Scheffé

The information inequality bounds the variance of an unbiased estimator from below, and by corollary 3.2.8(2) the bound is met at every θ at once only in a one-parameter exponential family, and there only when the estimand is the mean of the natural statistic, provided ψ' is continuous and nowhere zero, $\mathcal{I}$ is continuous, and $\theta \mapsto \partial_\theta f_\theta(x)$ is continuous for each x in the common support. The constraint is on the estimand, not on the parametrization: both sides of (3.2) transform the same way under a smooth change of parameter, so no choice of coordinates can create or destroy equality. Within the reach of those hypotheses a model outside that class has no unbiased estimator meeting the bound at every θ , and corollary 3.2.8(1) leaves equality possible at some parameter values and not at others, so the bound neither certifies any particular candidate as optimal nor says how to build a better one. What is wanted instead is a construction that takes an unbiased estimator and returns one at least as good, together with a criterion that recognizes when the output cannot be improved again.

Both were prepared in chapter 2. The construction is conditioning on a sufficient statistic, which produces a legitimate estimator precisely because the conditional law of the data given a sufficient statistic is free of θ (definition 2.1.2). The criterion is completeness (definition 2.3.1), which forces an unbiased function of the statistic to be the only one. In a model carrying a complete sufficient statistic the two together single out one estimator that is best in the unbiased class at every θ at once.

Proposition 1.3.3 showed that under squared-error loss no rule is uniformly best in the class of all rules, because a constant rule already achieves risk zero at its own parameter value, and it named restricting the class of rules as one of the two responses. Unbiasedness is such a restriction, and it is a convenient one under squared-error loss: the bias term of proposition 3.1.4 vanishes on the unbiased class, so comparing unbiased estimators means comparing their variance functions, and the question of a uniformly best member is at least well posed.

Definition 3.3.1: Uniformly Minimum Variance Unbiased Estimator

Let $\psi: \Theta \rightarrow \mathbb{R}$ be an estimand and let $\hat{\psi} = \hat{\psi}(X)$ be an unbiased estimator of ψ (definition 3.1.2) with $\text{Var}_\theta(\hat{\psi}) < \infty$ for every $\theta \in \Theta$. Then $\hat{\psi}$ is a *uniformly minimum variance unbiased* estimator of ψ , abbreviated *UMVU*, if

$$\text{Var}_\theta(\hat{\psi}) \leq \text{Var}_\theta(\tilde{\psi}) \quad \text{for every } \theta \in \Theta$$

holds for every unbiased estimator $\tilde{\psi}$ of ψ .

The quantifiers are what make the definition restrictive. The competitor $\tilde{\psi}$ ranges over all unbiased estimators, and the inequality is asked at every θ simultaneously, so a UMVU estimator is a single rule that wins the whole comparison rather than one that wins on part of Θ and loses elsewhere. Since an unbiased estimator has $\text{MSE}(\theta, \hat{\psi}) = \text{Var}_\theta(\hat{\psi})$ by proposition 3.1.4, a UMVU estimator is exactly a rule of uniformly smallest squared-error risk within the unbiased class.

A restricted version of the comparison can be settled by hand, and it shows both what is being asked and why the definition needs a tool. Take $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known, and consider the linear estimators $\hat{\mu}_a = \sum_{i=1}^n a_i X_i$. Such an estimator has $\mathbb{E}_\mu[\hat{\mu}_a] = \mu \sum_i a_i$, so it is unbiased for μ exactly when $\sum_i a_i = 1$, and by independence its variance is $\text{Var}_\mu(\hat{\mu}_a) = \sigma^2 \sum_i a_i^2$. Under the constraint $\sum_i a_i = 1$,

$$\sum_{i=1}^n \left(a_i - \frac{1}{n}\right)^2 = \sum_{i=1}^n a_i^2 - \frac{2}{n} \sum_{i=1}^n a_i + n \cdot \frac{1}{n^2} = \sum_{i=1}^n a_i^2 - \frac{1}{n}$$

so $\sum_i a_i^2 \geq 1/n$ with equality exactly when every $a_i = 1/n$. The sample mean $\bar{X}$ therefore has strictly smaller variance than every other linear unbiased estimator, at every μ . What the computation does not address is the nonlinear competitors, and there are uncountably many of them. They cannot be checked one at a time, so the argument has to come from the structure of the model rather than from the algebra of a candidate.

Sufficiency provides that structure. Proposition 2.1.5 already showed that a rule using the whole sample is matched, risk for risk and under any loss, by a randomized rule that reads only a sufficient statistic T : draw an auxiliary $\tilde{X}$ from the θ -free conditional law $P(X \in \cdot | T)$ and apply the original rule to it. That result preserves risk and stops there. Replacing the auxiliary draw by its conditional average removes the extra randomness, and when the loss is convex the removal cannot hurt.

Theorem 3.3.2: Rao-Blackwell

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model, let $T = T(X)$ be sufficient for it (definition 2.1.2), and let $\delta : \mathcal{X} \rightarrow \mathbb{R}$ be a real-valued estimator with $\mathbb{E}_\theta|\delta(X)| < \infty$ for every $\theta \in \Theta$. Define

$$\eta(t) = \int \delta(x) P(dx|T=t)$$

the average of δ against the common conditional law of definition 2.1.2. Then the following hold.

1. The function η does not depend on θ , so $\eta(T)$ is an estimator, and $\mathbb{E}_\theta[\eta(T)] = \mathbb{E}_\theta[\delta(X)]$ for every $\theta \in \Theta$. In particular, if δ is unbiased for $\psi(\theta)$ then so is $\eta(T)$.
2. Let $L : \Theta \times \mathbb{R} \rightarrow [0, \infty)$ be a loss (definition 1.2.1) with action space $\mathcal{A} = \mathbb{R}$ such that $a \mapsto L(\theta, a)$ is convex for each θ . Then, for every θ with $R(\theta, \delta) < \infty$,

$$R(\theta, \eta) \leq R(\theta, \delta)$$

3. Fix θ and suppose $\mathbb{E}_\theta[\delta(X)^2] < \infty$. Under the squared-error loss $L(\theta, a) = (a - \psi(\theta))^2$ the two risks differ by exactly

$$R(\theta, \delta) - R(\theta, \eta) = \mathbb{E}_\theta[(\delta(X) - \eta(T))^2]$$

so the inequality in (2) is strict at θ unless $\delta(X) = \eta(T)$ holds P_θ -almost surely.

Proof:

Step 1: $\eta(T)$ is an estimator. By definition 2.1.2 there is a single regular conditional distribution $P(X \in \cdot | T = t)$ that serves as the conditional law of X given $T = t$ for every $\theta \in \Theta$ at once. The defining integral for $\eta(t)$ is taken against that one fixed law, so the function $t \mapsto \eta(t)$ carries no dependence on θ : its value at t is computable by a statistician who knows the model and the observed value of T , which is what definition 1.2.3 asks of a rule. The hypothesis $\mathbb{E}_\theta|\delta(X)| < \infty$ makes the conditional expectation of $\delta(X)$ defined under P_θ , and because $P(\cdot | T = t)$ is a version of the conditional law of X given $T = t$ under P_θ , integrating δ against it computes that conditional expectation (*Probability* [5, Existence of Regular Conditional Distributions], whose integral formula covers every g with $g(X)$ integrable, not only indicators; the cited theorem states the real-valued case, and the vector-valued kernel used here is exactly what the standing standard-Borel convention of box 2.1.1 gives):

$$\eta(T) = \mathbb{E}_\theta[\delta(X)|T] \quad P_\theta\text{-a.s.}$$

for every $\theta \in \Theta$. Taking expectations and applying the tower property [5, Tower Property] with the trivial σ -algebra in the outer slot,

$$\mathbb{E}_\theta[\eta(T)] = \mathbb{E}_\theta[\mathbb{E}_\theta[\delta(X)|T]] = \mathbb{E}_\theta[\delta(X)]$$

If δ is unbiased for ψ , the right-hand side is $\psi(\theta)$ for every θ , which is unbiasedness of $\eta(T)$ (definition 3.1.2).

Step 2: convex loss cannot punish the averaging. Fix θ with $R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))] < \infty$

and write $\varphi(a) = L(\theta, a)$, a convex function $\mathbb{R} \rightarrow [0, \infty)$ by hypothesis. The variable $\delta(X)$ is P_θ -integrable by assumption, and $\varphi(\delta(X))$ is P_θ -integrable because $R(\theta, \delta) < \infty$, so the conditional Jensen inequality [5, Conditional Jensen Inequality] applies with $\mathcal{G} = \sigma(T)$ and gives

$$\varphi(\mathbb{E}_\theta[\delta(X)|T]) \leq \mathbb{E}_\theta[\varphi(\delta(X))|T] \quad P_\theta\text{-a.s.}$$

By Step 1 the left-hand side is $\varphi(\eta(T)) = L(\theta, \eta(T))$. Both sides are nonnegative, so taking $\mathbb{E}_\theta$ preserves the inequality, and the tower property [5, Tower Property] evaluates the right-hand side:

$$R(\theta, \eta) = \mathbb{E}_\theta[L(\theta, \eta(T))] \leq \mathbb{E}_\theta[\mathbb{E}_\theta[L(\theta, \delta(X))|T]] = \mathbb{E}_\theta[L(\theta, \delta(X))] = R(\theta, \delta)$$

Step 3: the exact gap under squared error. Now take $L(\theta, a) = (a - \psi(\theta))^2$ and fix θ with $\mathbb{E}_\theta[\delta(X)^2] < \infty$. Applying the conditional Jensen inequality [5, Conditional Jensen Inequality] to the convex map $a \mapsto a^2$ and using Step 1 gives $\eta(T)^2 \leq \mathbb{E}_\theta[\delta(X)^2|T]$ almost surely, so $\mathbb{E}_\theta[\eta(T)^2] \leq \mathbb{E}_\theta[\delta(X)^2] < \infty$ and $\eta(T)$ is square integrable as well. Abbreviate $\psi = \psi(\theta)$ and split the error of δ around $\eta(T)$:

$$(\delta(X) - \psi)^2 = ((\delta(X) - \eta(T)) + (\eta(T) - \psi))^2 = (\delta(X) - \eta(T))^2 + 2(\delta(X) - \eta(T))(\eta(T) - \psi) + (\eta(T) - \psi)^2$$

Every term on the right is P_θ -integrable, both variables being square integrable, so expectations may be taken term by term. The cross term vanishes. Indeed $\eta(T) = \mathbb{E}_\theta[\delta(X)|T]$ is the orthogonal projection of $\delta(X)$ onto the square-integrable functions of T (*Probability* [5, Conditional Expectation as Orthogonal Projection]), so the residual $\delta(X) - \eta(T)$ is orthogonal to every square-integrable function of T , and $\eta(T) - \psi$ is such a function since $\eta(T)$ is square integrable and ψ is a constant. Hence

$$\mathbb{E}_\theta[(\delta(X) - \eta(T))(\eta(T) - \psi)] = 0$$

and taking $\mathbb{E}_\theta$ of the expansion leaves

$$R(\theta, \delta) = \mathbb{E}_\theta[(\delta(X) - \eta(T))^2] + \mathbb{E}_\theta[(\eta(T) - \psi)^2] = \mathbb{E}_\theta[(\delta(X) - \eta(T))^2] + R(\theta, \eta)$$

which is the stated identity after rearrangement. The subtracted term is the expectation of a nonnegative variable, so it is zero exactly when $\delta(X) = \eta(T)$ holds P_θ -almost surely, and is strictly positive otherwise, completing the proof.

Part (2) sharpens proposition 2.1.5 in two directions. That proposition matched the risk of δ under an arbitrary loss by a rule that still needed a coin toss; the theorem replaces the coin toss by an average, so the improved rule is nonrandomized, and it delivers an inequality where the proposition delivered an equality. The added hypothesis is convexity of the loss in the action, which the equality statement did not need. Part (3) says how much is gained under squared error: the improvement is the mean squared distance between the original estimator and its conditional average, so the operation fails to help only when δ was already a function of T up to a null set. Passing from δ to $\eta(T)$ is called *Rao-Blackwellization*, and by part (1) it never disturbs unbiasedness, which is what lets it be applied inside the unbiased class.

The construction is also a computation, and the following case runs it in full. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ with $\lambda \in (0, \infty)$, and take as estimand the probability of an empty count, $\psi(\lambda) = P_\lambda(X_1 = 0) = e^{-\lambda}$.

The indicator $\delta(X) = \mathbb{1}\{X_1 = 0\}$ is unbiased for ψ , and it ignores $X_2, \dots, X_n$ entirely, returning only the values 0 and 1. By example 2.4.6(3) the Poisson model is a one-parameter exponential family whose natural statistic for a single observation is x itself, with natural parameter space $\Xi = \mathbb{R}$; the statistic x is nonconstant, so both full-rank clauses hold. Its natural statistic for the sample is the total $T = \sum_{i=1}^n X_i$, sufficient and complete by theorem 2.4.3. To Rao-Blackwellize, compute the conditional law of the data given $T = t$ from the joint mass function $P_\lambda(X = x) = e^{-n\lambda} \lambda^{\sum_i x_i} / \prod_i x_i!$ on $\{0, 1, 2, \dots\}^n$: for x with $\sum_i x_i = t$,

$$P_\lambda(X = x | T = t) = \frac{e^{-n\lambda} \lambda^t / \prod_i x_i!}{\sum_{y: \sum_i y_i = t} e^{-n\lambda} \lambda^t / \prod_i y_i!} = \frac{1 / \prod_i x_i!}{\sum_{y: \sum_i y_i = t} 1 / \prod_i y_i!} = \frac{t!}{\prod_i x_i!} n^{-t}$$

the last equality because the multinomial theorem gives $\sum_{y: \sum_i y_i = t} t! / \prod_i y_i! = (1 + \dots + 1)^t = n^t$. The conditional law is free of λ , which re-derives the sufficiency of T here, and it is the multinomial law spreading t occurrences over n equally likely cells. Summing that law over the vectors with $x_1 = 0$ and applying the multinomial theorem again, now with $n - 1$ cells,

$$\eta(t) = P(X_1 = 0 | T = t) = n^{-t} \sum_{\substack{x_2, \dots, x_n \geq 0 \\ x_2 + \dots + x_n = t}} \frac{t!}{x_2! \cdots x_n!} = n^{-t} (n-1)^t = \left(\frac{n-1}{n}\right)^t$$

So Rao-Blackwellizing the crude indicator produces $\eta(T) = \left(1 - \frac{1}{n}\right)^T$, an unbiased estimator of $e^{-\lambda}$ by part (1) and one of strictly smaller variance by part (3), since $\mathbb{1}\{X_1 = 0\}$ is not a function of T for $n \geq 2$.

Rao-Blackwellization improves an estimator without certifying the result. Starting from two different unbiased estimators and conditioning each on the same sufficient statistic could in principle land on two different unbiased functions of T , and then neither would have a claim to being best. Completeness is exactly the hypothesis that forbids this, since by definition 2.3.1 the only function of T whose expectation vanishes identically is the zero function.

Theorem 3.3.3: Lehmann-Scheffé

Let $T = T(X)$ be sufficient (definition 2.1.2) and complete (definition 2.3.1) for the model $\{P_\theta : \theta \in \Theta\}$, and let $g(T)$ be an unbiased estimator of $\psi(\theta)$ with $\text{Var}_\theta(g(T)) < \infty$ for every $\theta \in \Theta$. Then $g(T)$ is UMVU for ψ (definition 3.3.1). It is moreover the only one: if $\tilde{\psi}$ is any UMVU estimator of ψ , then $\tilde{\psi} = g(T)$ holds P_θ -almost surely for every $\theta \in \Theta$.

Proof:

Step 1: at most one unbiased function of T . Suppose $g_1(T)$ and $g_2(T)$ are both unbiased for ψ , so that $\mathbb{E}_\theta[g_1(T)] = \mathbb{E}_\theta[g_2(T)] = \psi(\theta)$ for every θ . Their difference $h = g_1 - g_2$ satisfies $\mathbb{E}_\theta[h(T)] = 0$ for every $\theta \in \Theta$, and completeness of T (definition 2.3.1) forces $h(T) = 0$ to hold P_θ -a.s. for every θ . Hence $g_1(T) = g_2(T)$ P_θ -a.s. for every θ .

Step 2: $g(T)$ beats every unbiased competitor. Let $\tilde{\psi} = \tilde{\psi}(X)$ be an unbiased estimator of ψ and fix $\theta_0 \in \Theta$. If $\text{Var}_{\theta_0}(\tilde{\psi}) = \infty$ the required inequality $\text{Var}_{\theta_0}(g(T)) \leq \text{Var}_{\theta_0}(\tilde{\psi})$ is immediate, so assume $\mathbb{E}_{\theta_0}[\tilde{\psi}^2] < \infty$. Unbiasedness gives $\mathbb{E}_\theta[\tilde{\psi}] < \infty$ for every θ , so theorem 3.3.2 applies to $\tilde{\psi}$: the conditional average $\zeta(T) = \mathbb{E}[\tilde{\psi}(X) | T]$ is an estimator, free of θ by sufficiency, and it is

unbiased for ψ by part (1). Applying Step 1 to the two unbiased functions ζ and g of T ,

$$\zeta(T) = g(T) \quad P_\theta\text{-a.s. for every } \theta \in \Theta$$

Now invoke theorem 3.3.2(3) at θ_0 under the squared-error loss $L(\theta, a) = (a - \psi(\theta))^2$, whose hypothesis $\mathbb{E}_{\theta_0}[\tilde{\psi}^2] < \infty$ has just been assumed:

$$R(\theta_0, \tilde{\psi}) - R(\theta_0, \zeta) = \mathbb{E}_{\theta_0}[(\tilde{\psi} - \zeta(T))^2] \geq 0$$

Both $\tilde{\psi}$ and $\zeta(T)$ are unbiased for ψ , so each of these two risks is the corresponding variance by proposition 3.1.4, and $\text{Var}_{\theta_0}(\zeta(T)) = \text{Var}_{\theta_0}(g(T))$ because the two variables agree P_{θ_0} -a.s. Therefore $\text{Var}_{\theta_0}(g(T)) \leq \text{Var}_{\theta_0}(\tilde{\psi})$. As θ_0 was arbitrary, $g(T)$ is UMVU.

Step 3: uniqueness. Let $\tilde{\psi}$ be any UMVU estimator of ψ and fix $\theta \in \Theta$. Being UMVU, $\tilde{\psi}$ satisfies $\text{Var}_\theta(\tilde{\psi}) \leq \text{Var}_\theta(g(T)) < \infty$, since $g(T)$ is one of the unbiased competitors it is required to beat, and Step 2 gives the reverse inequality $\text{Var}_\theta(g(T)) \leq \text{Var}_\theta(\tilde{\psi})$. The two variances are therefore equal and finite. Running the computation of Step 2 with this $\tilde{\psi}$ at the parameter value θ , the risk difference displayed there is $\text{Var}_\theta(\tilde{\psi}) - \text{Var}_\theta(\zeta(T)) = 0$, so

$$\mathbb{E}_\theta[(\tilde{\psi} - \zeta(T))^2] = 0$$

which forces $\tilde{\psi} = \zeta(T)$ P_θ -a.s., and $\zeta(T) = g(T)$ P_θ -a.s. by Step 2. Hence $\tilde{\psi} = g(T)$ P_θ -a.s. for every θ , as we sought to show.

The theorem turns the search for a UMVU estimator into a finite task. Once a complete sufficient statistic T has been identified, there are two routes to the answer and they meet at the same place. One route solves the equation $\mathbb{E}_\theta[g(T)] = \psi(\theta)$ directly for an unbiased g , a moment calculation in the law of T . The other takes any unbiased estimator at all, however crude, and conditions it on T . Step 1 says the two cannot disagree. Completeness is what makes the search terminate, since the difference of two unbiased functions of T is an unbiased estimator of zero and completeness admits only the trivial one.

Example 3.3.4: UMVU Estimators in the Three Running Models

Throughout, $X_1, \dots, X_n$ is a random sample from the model of example 1.1.7.

1. *Bernoulli.* Let $X_i \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ with $p \in (0, 1)$ and $T = \sum_{i=1}^n X_i$. By example 2.4.6(1) the Bernoulli model is a one-parameter exponential family whose natural statistic for a single observation is x itself, with natural parameter space $\Xi = \mathbb{R}$; the statistic x is nonconstant, so both full-rank clauses hold. Its natural statistic for the sample is the count T , which is therefore sufficient and complete by theorem 2.4.3. The completeness is also available as the direct polynomial argument of example 2.3.2. Since $\mathbb{E}_p[\bar{X}] = p$ and $\bar{X} = T/n$ is a function of T , theorem 3.3.3 makes $\bar{X}$ the UMVU estimator of p , with variance $p(1-p)/n$. For the nonlinear estimand $\psi(p) = p(1-p)$, use $T \sim \text{Bin}(n, p)$ and hence $\mathbb{E}_p[T] = np$ and $\text{Var}_p(T) = np(1-p)$ (example 1.1.7) to get $\mathbb{E}_p[T^2] = np(1-p) + n^2p^2$, whence

$$\mathbb{E}_p[T(n-T)] = n\mathbb{E}_p[T] - \mathbb{E}_p[T^2] = n^2p - np(1-p) - n^2p^2 = n(n-1)p(1-p)$$

So $T(n-T)/(n(n-1))$ is an unbiased function of T for $n \geq 2$, and it is the UMVU estimator of $p(1-p)$.

2. *Gaussian.* Let $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $n \geq 2$ and both parameters unknown. By example 2.4.6(4) this is a full-rank two-parameter exponential family whose natural statistic for the sample is $U = (\sum_j X_j, \sum_j X_j^2)$, so U is sufficient and complete by theorem 2.4.3. The pair $(\bar{X}, S^2)$ is a measurable bijective function of U : the identities $\bar{X} = n^{-1} \sum_j X_j$ and $S^2 = (\sum_j X_j^2 - n\bar{X}^2)/(n-1)$ recover it from U , while $\sum_j X_j = n\bar{X}$ and $\sum_j X_j^2 = (n-1)S^2 + n\bar{X}^2$ recover U from it. Consequently the functions of $(\bar{X}, S^2)$ are exactly the functions of U , and the two statistics generate the same σ -algebra, so conditioning on one is conditioning on the other and $(\bar{X}, S^2)$ is itself sufficient and complete. Both coordinates are unbiased for their targets by example 3.1.3, so theorem 3.3.3 makes $\bar{X}$ the UMVU estimator of μ and S^2 the UMVU estimator of σ^2 . Here $\bar{X}$ beats every unbiased estimator of μ , not only the linear ones. This does not subsume the computation opening this section, which fixes $\sigma^2 = \sigma_0^2$: unbiasedness in that submodel is a weaker requirement, so its competitor class is larger, and $\bar{X} + a(S^2 - \sigma_0^2)$ belongs to it for every a while failing to be unbiased here. The submodel needs its own application of theorem 3.3.3, which it gets, since with σ^2 known $\sum_i X_i$ is the complete sufficient natural statistic of a full-rank one-parameter exponential family.

3. *Uniform.* Let $X_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ with $\theta \in (0, \infty)$ and let $M = X_{(n)} = \max_i X_i$. The joint density with respect to Lebesgue measure on $\mathbb{R}^n$ is

$$\theta^{-n} \prod_{i=1}^n \mathbb{1}\{0 < x_i < \theta\} = \theta^{-n} \mathbb{1}\{\max_i x_i < \theta\} \cdot \prod_{i=1}^n \mathbb{1}\{x_i > 0\}$$

a factor depending on x only through $\max_i x_i$ times a factor free of θ , so M is sufficient by theorem 2.1.3. Its distribution function is $P_\theta(M \leq m) = \prod_i P_\theta(X_i \leq m) = (m/\theta)^n$ for $0 < m < \theta$, by independence, giving the density $f_\theta(m) = nm^{n-1}\theta^{-n}$ on $(0, \theta)$ and the moments

$$\mathbb{E}_\theta[M] = \int_0^\theta m \frac{nm^{n-1}}{\theta^n} dm = \frac{n}{n+1} \theta, \quad \mathbb{E}_\theta[M^2] = \int_0^\theta m^2 \frac{nm^{n-1}}{\theta^n} dm = \frac{n}{n+2} \theta^2$$

So $\frac{n+1}{n}M$ is unbiased for θ .

The statistic M is also complete. Let g satisfy $\mathbb{E}_\theta[g(M)] = 0$ for every $\theta > 0$, the expectation being defined, so that $\int_0^\theta |g(m)|m^{n-1} dm < \infty$ for each θ and $h(m) = g(m)m^{n-1}$ is locally integrable on $(0, \infty)$. Multiplying the hypothesis by θ^n/n turns it into

$$\int_0^\theta h(m) dm = 0 \quad \text{for every } \theta > 0$$

Fix $x > 0$ and $r > 0$. Subtracting the identity at $\theta = x$ from the identity at $\theta = x+r$ gives $\int_x^{x+r} h(m) dm = 0$, so every average $r^{-1} \int_x^{x+r} h$ is zero. The intervals $(x, x+r)$ shrink nicely to x as $r \downarrow 0$, being contained in $(x-r, x+r)$ and of half its length, so the Lebesgue differentiation theorem [4, The Lebesgue Differentiation Theorem] identifies the limit of those averages with $h(x)$ for Lebesgue-almost every $x > 0$. Hence $h = 0$ almost everywhere,

and since $m^{n-1} > 0$ on $(0, \infty)$ also $g = 0$ almost everywhere. As M has a density under every P_θ , this gives $g(M) = 0$ P_θ -a.s. for every θ , which is completeness. By theorem 3.3.3 the estimator $\frac{n+1}{n}M$ is UMVU for θ , and its variance is

$$\text{Var}_\theta\left(\frac{n+1}{n}M\right) = \frac{(n+1)^2}{n^2} \left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2} \right) \theta^2 = \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{(n+2)(n+1)^2} = \frac{\theta^2}{n(n+2)}$$

The uniform model is the one to keep in view, because it fails the regularity conditions of definition 3.2.2 and yet the theory of this section applies to it untouched. Rao-Blackwell and Lehmann-Scheffé ask for sufficiency and completeness, and neither asks for a differentiable likelihood or for a support free of θ . The variance $\theta^2/(n(n+2))$ of the UMVU estimator decays like n^{-2} , against the n^{-1} rate of the Gaussian and Bernoulli means, which is the phenomenon recorded in example 3.2.9 seen from the other side.

Completeness carries the uniqueness, and dropping it removes the conclusion rather than weakening it. The next example returns to the model of example 2.3.5, where a sufficient statistic is available but the unbiased estimators of θ include a one-parameter family of competitors, and computes the variances to show that the unbiased class has no best member.

Example 3.3.5: No UMVU Estimator Without Completeness

Take a single observation X from $\text{Unif}(\theta, \theta + 1)$ with $\theta \in \mathbb{R}$, the model of example 2.3.5. The statistic $T = X$ is the whole data, hence sufficient, so Rao-Blackwellization has nothing to condition away and returns every estimator unchanged. It is not complete: $\mathbb{E}_\theta[\sin(2\pi X)] = 0$ for every θ , as computed there.

Since $\mathbb{E}_\theta[X] = \theta + \frac{1}{2}$, the estimator $\hat{\theta}_0 = X - \frac{1}{2}$ is unbiased for θ , and so is

$$\hat{\theta}_c = X - \frac{1}{2} + c \sin(2\pi X)$$

for every $c \in \mathbb{R}$. Write $X = \theta + U$ with $U \sim \text{Unif}(0, 1)$, a law not involving θ , and put $a = \cos(2\pi\theta)$, $b = \sin(2\pi\theta)$, $P = \sin(2\pi U)$ and $Q = \cos(2\pi U)$, so that $\sin(2\pi X) = aP + bQ$ by the addition formula. The moments needed are

$$\mathbb{E}[P] = \mathbb{E}[Q] = 0, \quad \mathbb{E}[P^2] = \mathbb{E}[Q^2] = \frac{1}{2}, \quad \mathbb{E}[PQ] = 0, \quad \mathbb{E}[UP] = -\frac{1}{2\pi}, \quad \mathbb{E}[UQ] = 0$$

each an elementary integral over $(0, 1)$: for instance $\mathbb{E}[P^2] = \int_0^1 \sin^2(2\pi u) du = \int_0^1 \frac{1}{2}(1 - \cos(4\pi u)) du = \frac{1}{2}$, and integrating by parts, $\mathbb{E}[UP] = \left[-u \cos(2\pi u)/2\pi \right]_0^1 + \frac{1}{2\pi} \int_0^1 \cos(2\pi u) du = -\frac{1}{2\pi}$. Since $a^2 + b^2 = 1$, these give $\text{Var}(aP + bQ) = \frac{1}{2}(a^2 + b^2) = \frac{1}{2}$ and $\text{Cov}(U, aP + bQ) = -a/2\pi$, so

$$\text{Var}_\theta(\hat{\theta}_c) = \text{Var}(U) + c^2 \text{Var}(aP + bQ) + 2c \text{Cov}(U, aP + bQ) = \frac{1}{12} + \frac{c^2}{2} - \frac{c \cos(2\pi\theta)}{\pi}$$

No member of this family is UMVU. Given $c \neq 0$, choose θ_c with $\cos(2\pi\theta_c) = -1$ when $c > 0$ and $\cos(2\pi\theta_c) = 1$ when $c < 0$, which is possible since the cosine attains both values. At that parameter value the third term above equals $+|c|/\pi$, so

$$\text{Var}_{\theta_c}(\hat{\theta}_c) = \frac{1}{12} + \frac{c^2}{2} + \frac{|c|}{\pi} > \frac{1}{12} = \text{Var}_{\theta_0}(\hat{\theta}_0)$$

and $\hat{\theta}_c$ is beaten by $\hat{\theta}_0$ there. The remaining candidate $\hat{\theta}_0$ is beaten in turn: at $\theta = 0$ the variance of $\hat{\theta}_c$ is $\frac{1}{12} + \frac{c^2}{2} - \frac{c}{\pi}$, which at $c = 1/\pi$ equals $\frac{1}{12} - \frac{1}{2\pi^2} < \frac{1}{12}$.

That settles the family and not the unbiased class, of which $\{\hat{\theta}_c : c \in \mathbb{R}\}$ is only a one-parameter slice. An estimator $\delta(X)$ with $\mathbb{E}_\theta|\delta| < \infty$ for every θ is unbiased for θ exactly when $\int_\theta^{\theta+1} \delta(x) dx = \theta$ for every θ , and writing $\delta(x) = x - \frac{1}{2} + h(x)$ turns this into

$$\int_\theta^{\theta+1} h(x) dx = 0 \quad \text{for every } \theta \in \mathbb{R}$$

because $\hat{\theta}_0$ is itself unbiased. Put $G(t) = \int_0^t h$, absolutely continuous with $G' = h$ almost everywhere because h is locally integrable. The condition reads $G(\theta + 1) = G(\theta)$, so G is 1-periodic, hence $h(x + 1) = h(x)$ for almost every x and $\int_0^1 h = G(1) - G(0) = 0$. Conversely any such h has $\int_\theta^{\theta+1} h = \int_0^1 h = 0$. The unbiased estimators of θ are therefore the rules $X - \frac{1}{2} + h(X)$ with h 1-periodic of mean zero over a period, an infinite-dimensional class in which $\hat{\theta}_c$ is the case $h(x) = c \sin(2\pi x)$.

Averaging the variance over one period disposes of that class at once. Fix an unbiased $\delta = X - \frac{1}{2} + h(X)$. If h is not square integrable over a period then neither is $u \mapsto u - \frac{1}{2} + h(\theta + u)$, so $\text{Var}_\theta(\delta) = \infty$ at every θ and the finiteness required by definition 3.3.1 fails. Take $\int_0^1 h^2 < \infty$, write $X = \theta + U$ as above, and expand the square:

$$\text{Var}_\theta(\delta) = \frac{1}{12} + 2 \int_0^1 \left(u - \frac{1}{2}\right) h(\theta + u) du + \int_0^1 h(u)^2 du$$

the third term because $\int_0^1 h(\theta + u)^2 du = \int_\theta^{\theta+1} h^2$ and h^2 is 1-periodic as well. Now integrate in θ over $[0, 1]$ and exchange the two integrations, which the integrability of h over a period permits. The second term drops out, since $\int_0^1 h(\theta + u) d\theta = \int_u^{u+1} h = 0$ for each u , leaving

$$\int_0^1 \text{Var}_\theta(\delta) d\theta = \frac{1}{12} + \int_0^1 h(u)^2 du$$

A UMVU estimator has $\text{Var}_\theta(\delta) \leq \text{Var}_\theta(\hat{\theta}_0) = \frac{1}{12}$ at every θ , so its average over $[0, 1]$ is at most $\frac{1}{12}$, forcing $\int_0^1 h^2 = 0$ and $\delta = \hat{\theta}_0$ almost surely under every P_θ . That estimator is beaten at $\theta = 0$ by $\hat{\theta}_{1/\pi}$, so the unbiased class here contains no uniformly best member.

The obstruction is located precisely. Sufficiency was present and gave nothing, because X was already the whole data; completeness was absent, and with it went the step in theorem 3.3.3 that collapsed all unbiased functions of the statistic to one. The variance functions $\theta \mapsto \text{Var}_\theta(\hat{\theta}_c)$ cross each other as c varies, which is the situation proposition 1.3.3 described for the class of all rules, now reproduced inside the unbiased class.

The theory has two limits worth stating alongside its conclusions. The first is existence: a UMVU estimator of ψ requires an unbiased estimator of ψ to exist at all, and example 3.1.5 exhibits an estimand for which none does. The second is that being best among unbiased estimators is not the same as being good, since proposition 3.1.4 allows an estimator to trade a little bias for a large

reduction in variance and the trade can pay uniformly in θ . The shrinkage estimators of ?? do exactly this, dominating the UMVU estimator of a Gaussian mean vector in dimension three and above.

Exercise 3.3.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ with $\lambda \in \Theta = (0, \infty)$, and let $T = \sum_{i=1}^n X_i$. Show that T is sufficient and complete, and find the UMVU estimators of λ and of λ^2 . (For the second, express λ^2 through $\mathbb{E}_\lambda[T]$ and $\text{Var}_\lambda(T)$.)
2. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ with $n \geq 2$ and $p \in (0, 1)$, and let $T = \sum_{i=1}^n X_i$. Check that $\delta(X) = X_1 X_2$ is unbiased for p^2 , compute its Rao-Blackwellization $\mathbb{E}[\delta(X)|T = t]$, and identify the UMVU estimator of p^2 . Then verify that adding it to the UMVU estimator of $p(1-p)$ from example 3.3.4(1) returns the UMVU estimator of p , as theorem 3.3.3 requires.
3. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with $n \geq 2$ and both parameters unknown. Using example 3.3.4(2), find the UMVU estimator of μ^2 , and show that it takes negative values with positive probability under every (μ, σ^2) . Which of the two defects catalogued in example 3.1.5 does this illustrate?
4. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$. Verify that $2\bar{X}$ is unbiased for θ and compute its variance. Compare it with the variance of the UMVU estimator $\frac{n+1}{n}X_{(n)}$ from example 3.3.4(3), and say for which n the two estimators coincide.
5. Let T be sufficient for $\{P_\theta : \theta \in \Theta\}$. Show that Rao-Blackwellizing an estimator that is already a function of T returns it unchanged, and deduce that the operation $\delta \mapsto \eta(T)$ of theorem 3.3.2 is idempotent. Then show that sufficiency of the conditioning statistic cannot be dropped: for $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Ber}(p)$, verify that $U = X_1$ is not sufficient and that $\mathbb{E}_p[\bar{X}|U]$ depends on p , so it is not an estimator.
6. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and $M = X_{(n)}$. Exercise 3.1.1 (4) has already found the constant $c^* = \frac{n+2}{n+1}$ minimizing $\mathbb{E}_\theta[(cM - \theta)^2]$ and shown that c^*M beats the UMVU estimator $\frac{n+1}{n}M$ at every θ . Explain why this does not contradict theorem 3.3.3, and say precisely which of the theorem's hypotheses c^*M fails.

3.4 The Information Bound in Several Parameters

The bound of theorem 3.2.7 governs one unknown parameter. Most models carry several, and even when a single coordinate is the object of interest the others are rarely known: the Gaussian model has both μ and σ^2 , and an experimenter interested in μ alone must still estimate or eliminate σ^2 . What replaces the number $\mathcal{I}(\theta)$ is a matrix, what replaces the inequality between variances is an inequality between covariance matrices, and the bound on a single coordinate becomes a diagonal entry of the inverse of that matrix rather than the reciprocal of a diagonal entry. The difference between those two quantities is what the remaining coordinates cost when they are unknown, and this section computes it.

Throughout, $\Theta \subseteq \mathbb{R}^k$ is open and connected, the family $\{P_\theta : \theta \in \Theta\}$ is dominated by a σ -finite measure μ with densities $f_\theta = dP_\theta/d\mu$, and the model is assumed regular in the sense of definition 3.2.2 along each coordinate direction: for each index j and each fixed value of the remaining coordinates, the one-parameter subfamily $t \mapsto P_{\theta+te_j}$ satisfies those conditions, where $e_1, \dots, e_k$ is the standard

basis of $\mathbb{R}^k$. This gives, for each index j , that the support $\{x: f_\theta(x) > 0\}$ is constant along every axis-parallel segment in Θ , hence (an open connected subset of $\mathbb{R}^k$ being polygonally connected by axis-parallel segments) one and the same set for every θ , that $t \mapsto f_{\theta+te_j}(x)$ is absolutely continuous at each fixed x , and that $\partial_{\theta_j} \int f_\theta d\mu = \int \partial_{\theta_j} f_\theta d\mu$. Coordinate regularity yields those k partial derivatives and no more: a function of several variables can have every partial derivative at a point and still fail to be continuous at it, so joint differentiability does not follow from them. It is assumed separately: $\theta \mapsto \ell(\theta; x)$ is continuously differentiable on Θ in all k coordinates at once for almost every x in the common support, which is what the chain rule that produces (3.8) requires. The moving support of $\text{Unif}(0, \theta)$ fails the common-support requirement already in one parameter (example 3.2.9), so that model stays outside everything proved here.

Definition 3.4.1: Fisher Information Matrix

Let $\{P_\theta: \theta \in \Theta\}$ be regular along each coordinate, with $\Theta \subseteq \mathbb{R}^k$ open. The *score vector* at θ is the gradient of the log-likelihood in the parameter,

$$s_\theta(x) = \nabla_\theta \ell(\theta; x) = \left(\frac{\partial \ell}{\partial \theta_1}(\theta; x), \dots, \frac{\partial \ell}{\partial \theta_k}(\theta; x) \right)^\top \in \mathbb{R}^k$$

whose coordinates are the scores of definition 3.2.1 for the k coordinate subfamilies. Assuming each coordinate of $s_\theta(X)$ is square-integrable under P_θ , the *Fisher information matrix* is the $k \times k$ matrix

$$\mathcal{I}(\theta) = \mathbb{E}_\theta [s_\theta(X) s_\theta(X)^\top], \quad \mathcal{I}_{ij}(\theta) = \mathbb{E}_\theta \left[\frac{\partial \ell}{\partial \theta_i}(\theta; X) \frac{\partial \ell}{\partial \theta_j}(\theta; X) \right] \quad (3.4)$$

It is symmetric and positive semidefinite.

For symmetric $k \times k$ matrices A and B , write $A \succeq B$, and say A dominates B in the *Loewner order*, when $A - B$ is positive semidefinite, and $A \succ B$ when $A - B$ is positive definite. In this notation the last sentence reads $\mathcal{I}(\theta) \succeq 0$.

Symmetry holds because $\mathcal{I}_{ij}$ and $\mathcal{I}_{ji}$ are, by (3.4), expectations of the same product. Positive semidefiniteness is the one-line computation

$$a^\top \mathcal{I}(\theta) a = \mathbb{E}_\theta [a^\top s_\theta(X) s_\theta(X)^\top a] = \mathbb{E}_\theta [(a^\top s_\theta(X))^2] \geq 0$$

valid for every $a \in \mathbb{R}^k$ because the integrand is a square. When $k = 1$ the matrix is the number $\mathbb{E}_\theta [s_\theta(X)^2]$, which is the variance of definition 3.2.4 once the score is known to have mean zero.

That mean-zero property carries over coordinate by coordinate. Fix θ and an index j . Because Θ is open, the one-parameter family $t \mapsto P_{\theta+te_j}$ is defined for t in a neighbourhood of 0 and is regular by the standing hypotheses, and its score at $t = 0$ is $\partial_{\theta_j} \ell(\theta; X)$, the j -th coordinate of $s_\theta(X)$. Applying proposition 3.2.3 to that subfamily gives $\mathbb{E}_\theta [\partial_{\theta_j} \ell(\theta; X)] = 0$ for each j , hence $\mathbb{E}_\theta [s_\theta(X)] = 0$ and

$$\mathcal{I}(\theta) = \text{Cov}_\theta(s_\theta(X)) \quad (3.5)$$

so the information matrix is the covariance matrix of the score. This is the reading used in every argument below: the entry $\mathcal{I}_{ij}$ is a covariance between two score coordinates, and its vanishing is an uncorrelatedness statement.

The second-derivative form also survives, at the price of joint smoothness and one more interchange. Assume that $\theta \mapsto f_\theta(x)$ is twice continuously differentiable on Θ for almost every x in the common support. Coordinate regularity does not give this: proposition 3.2.3(2) does not produce second derivatives, it assumes one along a single coordinate direction and returns the identity relating it to the information, so applying it in each coordinate separately says nothing about the mixed partials. Where $f_\theta > 0$, the quotient rule applied to $\partial_{\theta_j} \ell = (\partial_{\theta_j} f_\theta) / f_\theta$ gives

$$\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j}(\theta; x) = \frac{1}{f_\theta(x)} \frac{\partial^2 f_\theta}{\partial \theta_i \partial \theta_j}(x) - \frac{\partial \ell}{\partial \theta_i}(\theta; x) \frac{\partial \ell}{\partial \theta_j}(\theta; x)$$

Taking expectations under P_θ and using (3.4),

$$\mathbb{E}_\theta \left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j} \right] = \int_{\mathcal{X}} \frac{\partial^2 f_\theta}{\partial \theta_i \partial \theta_j} d\mu - \mathcal{I}_{ij}(\theta)$$

If $\int |\partial_i \partial_j f_\theta| d\mu < \infty$ (the same integrability its one-parameter counterpart proposition 3.2.3(2) requires, needed for the term-by-term expectation above) and the family admits the second-order interchange $\int \partial_i \partial_j f_\theta d\mu = \partial_i \partial_j \int f_\theta d\mu$, the remaining integral is $\partial_i \partial_j 1 = 0$ and

$$\mathcal{I}_{ij}(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j}(\theta; X) \right] \quad (3.6)$$

the multiparameter form of the identity in proposition 3.2.3. As there, the interchange is a hypothesis and not a theorem, justified in a given model by exhibiting a dominating function and appealing to [4, Dominated Convergence Theorem (DCT)]. Every model treated below admits it. Formula (3.6) is what one computes with, because differentiating a log-density twice is easier than squaring a gradient and integrating.

Information from an independent sample adds, as in the one-parameter case. If $X = (X_1, \dots, X_n)$ with $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta$, then $\ell(\theta; x) = \sum_{j=1}^n \ell_1(\theta; x_j)$ for the single-observation log-likelihood ℓ_1 , so the sample score is the sum $s_\theta(X) = \sum_{j=1}^n s_\theta^{(1)}(X_j)$ of independent copies of the single-observation score. Each has mean zero by proposition 3.2.3, applied to the one-observation family, so for $j \neq l$ independence gives $\mathbb{E}_\theta[s_\theta^{(1)}(X_j) s_\theta^{(1)}(X_l)^\top] = \mathbb{E}_\theta[s_\theta^{(1)}(X_j)] \mathbb{E}_\theta[s_\theta^{(1)}(X_l)]^\top = 0$, and expanding the square leaves only the n diagonal terms:

$$\mathcal{I}_n(\theta) = n \mathcal{I}_1(\theta) \quad (3.7)$$

the matrix form of proposition 3.2.6. Everything the bound says about a sample of size n therefore differs from what it says about one observation by the single factor n .

One further identity is needed before the examples, because the information matrix is attached to a parametrization rather than to the family of laws. It is a consequence of the chain rule rather than a result in its own right, and is recorded here as a display because three of the examples and exercises below use it. Let g be a diffeomorphism from an open $\Phi \subseteq \mathbb{R}^k$ onto Θ , write $\theta = g(\varphi)$, and let $J(\varphi) = \partial \theta / \partial \varphi$ be its Jacobian matrix, invertible at every φ . The log-likelihood in the new coordinates is $\ell(g(\varphi); x)$, so the chain rule, available because ℓ was assumed jointly differentiable in θ , gives $\nabla_\varphi \ell(g(\varphi); x) = J(\varphi)^\top s_{g(\varphi)}(x)$, and the information matrix in the φ -parametrization, written $\tilde{\mathcal{I}}(\varphi)$, is

$$\tilde{\mathcal{I}}(\varphi) = \mathbb{E}[J^\top s s^\top J] = J(\varphi)^\top \mathcal{I}(g(\varphi)) J(\varphi) \quad (3.8)$$

with $J^\top$ and J pulled outside the expectation because they do not depend on the data. Inverting, $\tilde{\mathcal{I}}(\varphi)^{-1} = J(\varphi)^{-1} \mathcal{I}(g(\varphi))^{-1} J(\varphi)^{-\top}$ whenever $\mathcal{I}(g(\varphi))$ is invertible. A diagonal information matrix need not stay diagonal under (3.8), and the running Gaussian model shows both possibilities at once.

Example 3.4.2: The Information Matrix of the Gaussian Model

Take one observation $X \sim \mathcal{N}(\mu, \sigma^2)$ from the Gaussian location-scale model of example 1.1.7, with both coordinates unknown, so $\theta = (\mu, \sigma^2) \in \Theta = \mathbb{R} \times (0, \infty)$ and $k = 2$. The log-likelihood is

$$\ell(\mu, \sigma^2; x) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma^2 - \frac{(x - \mu)^2}{2\sigma^2}$$

which is smooth in (μ, σ^2) on all of Θ , with support $\mathbb{R}$ free of the parameter. Its first derivatives are

$$\frac{\partial \ell}{\partial \mu} = \frac{x - \mu}{\sigma^2}, \quad \frac{\partial \ell}{\partial \sigma^2} = -\frac{1}{2\sigma^2} + \frac{(x - \mu)^2}{2\sigma^4}$$

and its second derivatives are

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{1}{\sigma^2}, \quad \frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} = -\frac{x - \mu}{\sigma^4}, \quad \frac{\partial^2 \ell}{\partial (\sigma^2)^2} = \frac{1}{2\sigma^4} - \frac{(x - \mu)^2}{\sigma^6}$$

Taking expectations under $\mathcal{N}(\mu, \sigma^2)$, where $\mathbb{E}[X - \mu] = 0$ and $\mathbb{E}[(X - \mu)^2] = \sigma^2$, and applying (3.6),

$$\mathcal{I}_1(\mu, \sigma^2) = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{pmatrix}, \quad \mathcal{I}_n(\mu, \sigma^2) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix} \quad (3.9)$$

the second by (3.7). The off-diagonal entry is $-\mathbb{E}[(X - \mu)/\sigma^4] = 0$, and the $(2, 2)$ entry is $-(\frac{1}{2\sigma^4} - \frac{\sigma^2}{\sigma^6}) = \frac{1}{2\sigma^4}$. Both matrices are diagonal with positive entries, so both are invertible, and

$$\mathcal{I}_n(\mu, \sigma^2)^{-1} = \begin{pmatrix} \sigma^2/n & 0 \\ 0 & 2\sigma^4/n \end{pmatrix}$$

The same family is not orthogonal in every parametrization. Put it in the canonical coordinates of example 2.4.6(4), namely $\eta_1 = \mu/\sigma^2$ and $\eta_2 = -1/(2\sigma^2)$, whose inverse is $\mu = -\eta_1/(2\eta_2)$ and $\sigma^2 = -1/(2\eta_2)$. Differentiating the inverse map,

$$J = \frac{\partial(\mu, \sigma^2)}{\partial(\eta_1, \eta_2)} = \begin{pmatrix} -\frac{1}{2\eta_2} & \frac{\eta_1}{2\eta_2^2} \\ 0 & \frac{1}{2\eta_2^2} \end{pmatrix} = \begin{pmatrix} \sigma^2 & 2\mu\sigma^2 \\ 0 & 2\sigma^4 \end{pmatrix}$$

where the second form substitutes $\eta_2 = -1/(2\sigma^2)$ and $\eta_1 = \mu/\sigma^2$. Now (3.8) with $\mathcal{I}_1(\mu, \sigma^2)$ from (3.9) gives first

$$\mathcal{I}_1(\mu, \sigma^2) J = \begin{pmatrix} 1 & 2\mu \\ 0 & 1 \end{pmatrix}, \quad \text{hence} \quad \mathcal{I}_1(\eta) = J^\top \begin{pmatrix} 1 & 2\mu \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \sigma^2 & 2\mu\sigma^2 \\ 2\mu\sigma^2 & 2\sigma^4 + 4\mu^2\sigma^2 \end{pmatrix}$$

The off-diagonal entry $2\mu\sigma^2$ vanishes only at $\mu = 0$. Diagonality of the information matrix is therefore a property of the coordinates chosen on the family, not of the family itself, and the two computations above describe the same family of laws in two ways.

A parametrization in which $\mathcal{I}(\theta)$ is diagonal is called *orthogonal*, because by (3.5) diagonality says

the score coordinates are uncorrelated under P_θ . Proposition 3.4.4 below turns that uncorrelatedness into a statement about what one coordinate costs another. First the bound itself, which is the whole matrix version of theorem 3.2.7 and is proved by the same device: the variance of a well-chosen linear combination cannot be negative.

Theorem 3.4.3: The Multiparameter Information Inequality

Let $\{P_\theta : \theta \in \Theta\}$ be regular along each coordinate, with $\Theta \subseteq \mathbb{R}^k$ open, and suppose $\mathcal{I}(\theta)$ has finite entries and is invertible at θ . Let $\hat{\theta} = \hat{\theta}(X)$ be an estimator of θ with $\mathbb{E}_{\theta'}[\hat{\theta}] = \theta'$ for every $\theta' \in \Theta$, with every coordinate square-integrable under P_θ , and satisfying the interchange

$$\frac{\partial}{\partial \theta_j} \int_{\mathcal{X}} \hat{\theta}_i(x) f_\theta(x) d\mu(x) = \int_{\mathcal{X}} \hat{\theta}_i(x) \frac{\partial f_\theta}{\partial \theta_j}(x) d\mu(x) \quad \text{for all } i, j \quad (3.10)$$

Then

$$\text{Cov}_\theta(\hat{\theta}) \succeq \mathcal{I}(\theta)^{-1}$$

that is, $\text{Cov}_\theta(\hat{\theta}) - \mathcal{I}(\theta)^{-1}$ is positive semidefinite.

Proof:

Write $C = \text{Cov}_\theta(\hat{\theta})$ and $\mathcal{I} = \mathcal{I}(\theta)$, and note that $\mathcal{I}$ is symmetric, so $\mathcal{I}^{-1}$ is symmetric as well.

Step 1: the score is centred, and $\mathcal{I}$ is its covariance. Fix an index j . Since Θ is open, the one-parameter family $t \mapsto P_{\theta + te_j}$ is defined near $t = 0$ and is regular in the sense of definition 3.2.2 by the standing hypotheses of this section, and its score at $t = 0$ is the j -th coordinate $\partial_{\theta_j} \ell(\theta; X)$ of $s_\theta(X)$. Hence $\mathbb{E}_\theta[\partial_{\theta_j} \ell(\theta; X)] = 0$ by proposition 3.2.3, whose interchange hypothesis is part of that regularity. As j was arbitrary, $\mathbb{E}_\theta[s_\theta(X)] = 0$, and therefore $\mathcal{I} = \mathbb{E}_\theta[s_\theta s_\theta^\top] = \text{Cov}_\theta(s_\theta)$, which is (3.5).

Step 2: the estimator and the score have covariance I_k . Fix indices i and j and differentiate the unbiasedness identity $\theta_i = \mathbb{E}_\theta[\hat{\theta}_i] = \int \hat{\theta}_i f_\theta d\mu$ with respect to θ_j . The left-hand side gives the Kronecker symbol δ_{ij} , and the right-hand side may be differentiated inside the integral by the hypothesis (3.10), so

$$\delta_{ij} = \int_{\mathcal{X}} \hat{\theta}_i(x) \frac{\partial f_\theta}{\partial \theta_j}(x) d\mu(x)$$

Split the integral over the common support $\mathcal{S} = \{x : f_\theta(x) > 0\}$ and its complement. Off $\mathcal{S}$ the density vanishes for *every* value of the parameter, since the support does not move, so $\partial_{\theta_j} f_\theta(x) = 0$ there and the complement contributes nothing. On $\mathcal{S}$ one may divide and multiply by $f_\theta(x)$, and $(\partial_{\theta_j} f_\theta)/f_\theta = \partial_{\theta_j} \log f_\theta$ is the j -th score coordinate, so

$$\delta_{ij} = \int_{\mathcal{S}} \hat{\theta}_i(x) \frac{\partial \ell}{\partial \theta_j}(\theta; x) f_\theta(x) d\mu(x) = \mathbb{E}_\theta \left[\hat{\theta}_i \frac{\partial \ell}{\partial \theta_j} \right]$$

Because the score coordinate has mean zero by Step 1, this expectation of a product is a covariance, $\text{Cov}_\theta(\hat{\theta}_i, \partial_{\theta_j} \ell) = \delta_{ij}$. Assembling the k^2 identities,

$$\text{Cov}_\theta(\hat{\theta}, s_\theta(X)) = I_k \quad (3.11)$$

Step 3: a variance that cannot be negative. Fix $a \in \mathbb{R}^k$ and set

$$Z = a^\top \hat{\theta} - a^\top \mathcal{I}^{-1} s_\theta(X)$$

a real random variable, square-integrable because it is a linear combination of the coordinates of $\hat{\theta}$ and of $s_\theta(X)$, all of which are square-integrable by hypothesis. Expanding its variance by the bilinearity of covariance, which is linearity of expectation [5, Foundational Properties of Expectation] applied to the defining products,

$$\text{Var}_\theta(Z) = \text{Var}_\theta(a^\top \hat{\theta}) - 2\text{Cov}_\theta(a^\top \hat{\theta}, (\mathcal{I}^{-1}a)^\top s_\theta) + \text{Var}_\theta((\mathcal{I}^{-1}a)^\top s_\theta)$$

The three terms are computed in turn. The first is $a^\top C a$ by the definition of C . The second is $a^\top \text{Cov}_\theta(\hat{\theta}, s_\theta) \mathcal{I}^{-1} a = a^\top I_k \mathcal{I}^{-1} a = a^\top \mathcal{I}^{-1} a$ by (3.11) and the symmetry of $\mathcal{I}^{-1}$. The third is $(\mathcal{I}^{-1}a)^\top \text{Cov}_\theta(s_\theta)(\mathcal{I}^{-1}a) = a^\top \mathcal{I}^{-1} \mathcal{I} \mathcal{I}^{-1} a = a^\top \mathcal{I}^{-1} a$ by Step 1. Hence

$$0 \leq \text{Var}_\theta(Z) = a^\top C a - 2a^\top \mathcal{I}^{-1} a + a^\top \mathcal{I}^{-1} a = a^\top (C - \mathcal{I}^{-1}) a$$

Since $a \in \mathbb{R}^k$ was arbitrary, $C - \mathcal{I}^{-1}$ is positive semidefinite, as we sought to show.

The inequality carries more than the k separate statements about the coordinates. Taking $a = e_i$ recovers $\text{Var}_\theta(\hat{\theta}_i) \geq (\mathcal{I}(\theta)^{-1})_{ii}$, and taking a general a bounds the variance of every linear combination $a^\top \hat{\theta}$, including contrasts such as $\hat{\theta}_1 - \hat{\theta}_2$ that no list of coordinate bounds controls. The proof also identifies the case of equality: $\text{Var}_\theta(a^\top \hat{\theta})$ equals $a^\top \mathcal{I}(\theta)^{-1} a$ exactly when the variable Z of Step 3 is P_θ -a.s. constant, so exactly when $a^\top \hat{\theta} = a^\top \theta + a^\top \mathcal{I}(\theta)^{-1} s_\theta(X)$ almost surely. This is the multiparameter form of the affine relation between estimator and score in corollary 3.2.8, and it is as restrictive here as it is there.

The quantity $(\mathcal{I}(\theta)^{-1})_{ii}$ is the diagonal entry of the *inverse*, and there is no reason for it to equal $1/\mathcal{I}_{ii}(\theta)$, the reciprocal of the diagonal entry. The gap between the two is the entire content of having unknown companions. Call θ_1 the parameter of interest and $\theta_2, \dots, \theta_k$ the *nuisance parameters*, unknown but not themselves the object of the inference. When they are known, the model is the one-parameter subfamily $t \mapsto P_{(t, \theta_2, \dots, \theta_k)}$, whose Fisher information is $\mathcal{I}_{11}(\theta)$ and whose bound is therefore $1/\mathcal{I}_{11}(\theta)$ by theorem 3.2.7. When they are unknown the bound is $(\mathcal{I}(\theta)^{-1})_{11}$, and the next result says which of the two is larger and by how much.

Proposition 3.4.4: The Bound for One Coordinate is a Schur Complement

Let $\mathcal{I} = \mathcal{I}(\theta)$ be positive definite, partitioned as

$$\mathcal{I} = \begin{pmatrix} \mathcal{I}_{11} & \mathcal{I}_{12} \\ \mathcal{I}_{12}^\top & \mathcal{I}_{22} \end{pmatrix}$$

where $\mathcal{I}_{11}$ is the scalar $(1, 1)$ entry, $\mathcal{I}_{12}$ is the $1 \times (k-1)$ row vector of the remaining entries of the first row, and $\mathcal{I}_{22}$ is the lower-right $(k-1) \times (k-1)$ block. Then:

1. $\mathcal{I}_{22}$ is positive definite, the *Schur complement* $\mathcal{I}_{11.2} = \mathcal{I}_{11} - \mathcal{I}_{12}\mathcal{I}_{22}^{-1}\mathcal{I}_{12}^\top$ is a positive number, and

$$(\mathcal{I}^{-1})_{11} = \frac{1}{\mathcal{I}_{11.2}}$$

2. $(\mathcal{I}^{-1})_{11} \geq 1/\mathcal{I}_{11}$, with equality if and only if $\mathcal{I}_{12} = 0$.

3. Consequently, for any $\hat{\theta}$ satisfying the hypotheses of theorem 3.4.3,

$$\text{Var}_{\theta}(\hat{\theta}_1) \geq (\mathcal{I}^{-1})_{11} \geq \frac{1}{\mathcal{I}_{11}}$$

where $1/\mathcal{I}_{11}$ is the bound of theorem 3.2.7 for θ_1 in the model with $\theta_2, \dots, \theta_k$ known.

Proof:

1. For a nonzero $v \in \mathbb{R}^{k-1}$ the vector $(0, v) \in \mathbb{R}^k$ is nonzero, and $(0, v)^\top \mathcal{I} (0, v) = v^\top \mathcal{I}_{22} v > 0$ because $\mathcal{I}$ is positive definite, so $\mathcal{I}_{22}$ is positive definite and in particular invertible, with $\mathcal{I}_{22}^{-1}$ positive definite as well. Set $D = \mathcal{I}_{11.2}$ and

$$L = \begin{pmatrix} 1 & \mathcal{I}_{12}\mathcal{I}_{22}^{-1} \\ 0 & I_{k-1} \end{pmatrix}, \quad \Lambda = \begin{pmatrix} D & 0 \\ 0 & \mathcal{I}_{22} \end{pmatrix}$$

Since $\mathcal{I}_{22}$ is symmetric, so is $\mathcal{I}_{22}^{-1}$, and therefore $L^\top$ has lower-left block $\mathcal{I}_{22}^{-1}\mathcal{I}_{12}^\top$. Multiplying out,

$$\Lambda L^\top = \begin{pmatrix} D & 0 \\ \mathcal{I}_{12}^\top & \mathcal{I}_{22} \end{pmatrix}, \quad L \Lambda L^\top = \begin{pmatrix} D + \mathcal{I}_{12}\mathcal{I}_{22}^{-1}\mathcal{I}_{12}^\top & \mathcal{I}_{12} \\ \mathcal{I}_{12}^\top & \mathcal{I}_{22} \end{pmatrix} = \mathcal{I}$$

the last equality because $D + \mathcal{I}_{12}\mathcal{I}_{22}^{-1}\mathcal{I}_{12}^\top = \mathcal{I}_{11}$ by the definition of D . The matrix L is unitriangular, hence invertible with

$$L^{-1} = \begin{pmatrix} 1 & -\mathcal{I}_{12}\mathcal{I}_{22}^{-1} \\ 0 & I_{k-1} \end{pmatrix}$$

as multiplying the two displays confirms. Taking determinants in $\mathcal{I} = L \Lambda L^\top$ gives $\det \mathcal{I} = D \det \mathcal{I}_{22}$, and both determinants are positive because $\mathcal{I}$ and $\mathcal{I}_{22}$ are positive definite, so $D > 0$ and Λ is invertible. Inverting the factorization,

$$\mathcal{I}^{-1} = (L^\top)^{-1} \Lambda^{-1} L^{-1} = (L^{-1})^\top \Lambda^{-1} L^{-1}$$

and hence

$$(\mathcal{I}^{-1})_{11} = e_1^\top \mathcal{I}^{-1} e_1 = (L^{-1} e_1)^\top \Lambda^{-1} (L^{-1} e_1) = e_1^\top \Lambda^{-1} e_1 = \frac{1}{D}$$

using that the first column of L^{-1} is e_1 , so $L^{-1} e_1 = e_1$, and that Λ^{-1} has $(1, 1)$ entry $1/D$.

2. Write $b = \mathcal{I}_{12}^\top \in \mathbb{R}^{k-1}$. Then $\mathcal{I}_{12} \mathcal{I}_{22}^{-1} \mathcal{I}_{12}^\top = b^\top \mathcal{I}_{22}^{-1} b \geq 0$ because $\mathcal{I}_{22}^{-1}$ is positive definite, and this quantity is zero if and only if $b = 0$, again by positive definiteness. Hence $0 < D \leq \mathcal{I}_{11}$, with equality exactly when $\mathcal{I}_{12} = 0$, and taking reciprocals reverses the inequality between the two positive numbers: $(\mathcal{I}^{-1})_{11} = 1/D \geq 1/\mathcal{I}_{11}$, with the same equality condition.
3. The first inequality is theorem 3.4.3 with $a = e_1$, which gives $\text{Var}_\theta(\hat{\theta}_1) = e_1^\top \text{Cov}_\theta(\hat{\theta}) e_1 \geq e_1^\top \mathcal{I}^{-1} e_1 = (\mathcal{I}^{-1})_{11}$. The second is part (2). For the identification of $1/\mathcal{I}_{11}$, fix $\theta_2, \dots, \theta_k$ and consider the one-parameter family $t \mapsto P_{(t, \theta_2, \dots, \theta_k)}$, which is regular in the sense of definition 3.2.2 by the standing hypotheses. Its score at $t = \theta_1$ is $\partial_{\theta_1} \ell(\theta; X)$, which has mean zero by proposition 3.2.3, so its Fisher information (definition 3.2.4) is $\text{Var}_\theta(\partial_{\theta_1} \ell) = \mathbb{E}_\theta[(\partial_{\theta_1} \ell)^2] = \mathcal{I}_{11}(\theta)$. Applying theorem 3.2.7 to that family with estimand $\psi(t) = t$, so that $\psi' = 1$, gives the bound $1/\mathcal{I}_{11}$, completing the proof.

Nuisance parameters cost precision, and part (2) measures the cost exactly: the achievable variance for θ_1 is inflated by the factor $\mathcal{I}_{11}/\mathcal{I}_{11.2}$, which is at least one and is strictly greater than one whenever the first score coordinate is correlated with any of the others. The Schur complement says which part of $\mathcal{I}_{11}$ is subtracted. Write s_1 for the first score coordinate and $s_{-1} = (s_2, \dots, s_k)^\top$ for the rest, so that $\text{Var}_\theta(s_1) = \mathcal{I}_{11}$, $\text{Cov}_\theta(s_{-1}) = \mathcal{I}_{22}$ and $\text{Cov}_\theta(s_{-1}, s_1) = \mathcal{I}_{12}^\top$ by (3.5). For the linear combination $c^\top s_{-1}$ with $c^\top = \mathcal{I}_{12} \mathcal{I}_{22}^{-1}$,

$$\text{Var}_\theta(s_1 - c^\top s_{-1}) = \mathcal{I}_{11} - 2\mathcal{I}_{12} \mathcal{I}_{22}^{-1} \mathcal{I}_{12}^\top + \mathcal{I}_{12} \mathcal{I}_{22}^{-1} \mathcal{I}_{22} \mathcal{I}_{22}^{-1} \mathcal{I}_{12}^\top = \mathcal{I}_{11} - \mathcal{I}_{12} \mathcal{I}_{22}^{-1} \mathcal{I}_{12}^\top = \mathcal{I}_{11.2}$$

so the Schur complement is the variance left in s_1 after the best linear approximation to it by the other score coordinates has been removed. What is subtracted is the part of the sensitivity of the likelihood to θ_1 that moving the nuisance parameters could have produced instead, and only the remainder is information the nuisance parameters cannot imitate. The equality case says the cost vanishes in an orthogonal parametrization, where the score coordinates are uncorrelated and nothing is subtracted, and (3.9) exhibits one: in the Gaussian model an unknown σ^2 leaves the bound for μ unchanged.

With a bound in hand for each coordinate, the natural way to grade an unbiased estimator is by how close it comes to that bound.

Definition 3.4.5: Efficiency

Let $\hat{\theta}$ satisfy the hypotheses of theorem 3.4.3, and fix a coordinate i with $\text{Var}_\theta(\hat{\theta}_i) > 0$. The *efficiency* of $\hat{\theta}_i$ at θ is the ratio of the information bound to the attained variance,

$$e_\theta(\hat{\theta}_i) = \frac{(\mathcal{I}(\theta)^{-1})_{ii}}{\text{Var}_\theta(\hat{\theta}_i)}$$

By theorem 3.4.3 this lies in $(0, 1]$. The estimator $\hat{\theta}_i$ is *efficient* if $e_\theta(\hat{\theta}_i) = 1$ for every $\theta \in \Theta$, so that it attains the bound throughout the model.

Efficiency is a ratio of two variances and so is free of the units of θ_i , which makes it comparable across estimators and across models. Its reciprocal has a concrete reading for an independent sample: since $\mathcal{I}_n = n\mathcal{I}_1$ by (3.7), an unbiased estimator of efficiency e based on n observations has the variance that an efficient estimator would attain from en observations, so a value of $e = 0.9$ says that one tenth of the sample has been wasted.

A family of unbiased estimators of the Gaussian mean makes the numbers explicit. The definition is stated for a coordinate of an estimator of the whole vector, so to speak of the efficiency of an estimator $\tilde{\mu}$ of μ alone is to speak of the first coordinate of the pair $(\tilde{\mu}, S^2)$, which is unbiased for (μ, σ^2) and satisfies (3.10), which is the condition definition 3.2.2 calls being regular for a statistic, applied to each coordinate. Both coordinates are polynomials in the data, so (3.1) gives it in each of the two one-parameter subfamilies, which is what the hypotheses of theorem 3.4.3 ask for. Take $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, fix weights $w_1, \dots, w_n$ with $\sum_i w_i = 1$, and set $\tilde{\mu} = \sum_i w_i X_i$. Then $\mathbb{E}_\theta[\tilde{\mu}] = \mu \sum_i w_i = \mu$, so $\tilde{\mu}$ is unbiased, and by independence $\text{Var}_\theta(\tilde{\mu}) = \sigma^2 \sum_i w_i^2$. The bound from (3.9) is σ^2/n , so

$$e_\theta(\tilde{\mu}) = \frac{\sigma^2/n}{\sigma^2 \sum_i w_i^2} = \frac{1}{n \sum_i w_i^2}$$

independently of μ and σ^2 . The optimality algebra is the one already run in section 3.3: expanding $\sum_i (w_i - 1/n)^2 \geq 0$ gives $\sum_i w_i^2 \geq 1/n$ with equality only for $w_i \equiv 1/n$. The only change here is that σ^2 has passed from known to unknown, which the diagonal form of (3.9) makes immaterial. So the efficiency never exceeds one and equals one only for $\bar{X}$, which is therefore efficient. Discarding the last observation, that is taking $w_1 = \dots = w_{n-1} = 1/(n-1)$ and $w_n = 0$, gives $\sum_i w_i^2 = 1/(n-1)$ and efficiency $(n-1)/n$, the fraction of the sample retained.

The exponential families of section 2.4 compute their own information matrix, because the log-likelihood in canonical coordinates is a linear function of the data minus the cumulant function, and proposition 2.4.4 has already differentiated that function twice.

Example 3.4.6: The Information Matrix of a Canonical Exponential Family

Let $\{P_\eta : \eta \in \Xi\}$ be a k -parameter exponential family of full rank in canonical form (definition 2.4.1), with natural statistic T , cumulant function A , and a single observation X . For $\eta \in \text{int } \Xi$ the log-likelihood is

$$\ell(\eta; x) = \log h(x) + \eta^\top T(x) - A(\eta)$$

so the score vector is $s_\eta(x) = T(x) - \nabla A(\eta)$, and the Hessian in η is $-\nabla^2 A(\eta)$, which does not involve x at all. Since $\nabla A(\eta) = \mathbb{E}_\eta[T]$ by (2.7), the score has mean zero without any further hypothesis, and

$$\mathcal{I}(\eta) = \text{Cov}_\eta(T(X) - \nabla A(\eta)) = \text{Cov}_\eta(T) = \nabla^2 A(\eta) \quad (3.12)$$

the last equality being the second identity of (2.7). The same value comes from (3.6), since the Hessian is deterministic and its expectation is itself. Under full rank no nonzero linear combination $a^\top T$ is P_η -a.s. constant, so $\text{Var}_\eta(a^\top T) > 0$ for every $a \neq 0$, which makes $\nabla^2 A(\eta) = \text{Cov}_\eta(T)$ positive definite rather than merely semidefinite (proposition 2.4.4 gives the identity and the semidefiniteness; positive definiteness is this extra line). So $\mathcal{I}(\eta)$ is invertible and theorem 3.4.3 reads $\text{Cov}_\eta(\hat{\eta}) \succeq (\nabla^2 A(\eta))^{-1}$ for every unbiased $\hat{\eta}$ meeting its hypotheses.

For the Bernoulli family of example 2.4.6(1), where $k = 1$ and $A(\eta) = \log(1 + e^\eta)$, this gives $\mathcal{I}(\eta) = A''(\eta) = e^\eta/(1 + e^\eta)^2 = p(1 - p)$. That is the reciprocal of $\mathcal{I}(p) = 1/(p(1 - p))$

from example 3.2.5, and the two agree with (3.8): the Jacobian of $\eta \mapsto p = e^\eta/(1 + e^\eta)$ is $dp/d\eta = e^\eta/(1 + e^\eta)^2 = p(1 - p)$, so $\mathcal{I}(\eta) = (p(1 - p))^2 \cdot (p(1 - p))^{-1} = p(1 - p)$.

For the Gaussian family of example 2.4.6(4), where $T(x) = (x, x^2)$, formula (3.12) says $\mathcal{I}(\eta) = \text{Cov}_\eta(X, X^2)$. Writing $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$, the odd moments $\mathbb{E}[Z]$ and $\mathbb{E}[Z^3]$ vanish by symmetry of the standard normal density, while integrating by parts against $\varphi'(z) = -z\varphi(z)$ gives $\mathbb{E}[Z^4] = \int z^3(-\varphi'(z)) dz = 3 \int z^2\varphi(z) dz = 3$. In particular $\text{Var}(Z) = 1$, $\text{Cov}(Z, Z^2) = \mathbb{E}[Z^3] - \mathbb{E}[Z]\mathbb{E}[Z^2] = 0$ and $\text{Var}(Z^2) = \mathbb{E}[Z^4] - \mathbb{E}[Z^2]^2 = 2$. Expanding $X^2 = \mu^2 + 2\mu\sigma Z + \sigma^2 Z^2$ and using bilinearity of covariance,

$$\text{Var}(X) = \sigma^2 \text{Var}(Z) = \sigma^2, \quad \text{Cov}(X, X^2) = 2\mu\sigma^2 \text{Var}(Z) + \sigma^3 \text{Cov}(Z, Z^2) = 2\mu\sigma^2$$

$$\text{Var}(X^2) = 4\mu^2\sigma^2 \text{Var}(Z) + 4\mu\sigma^3 \text{Cov}(Z, Z^2) + \sigma^4 \text{Var}(Z^2) = 4\mu^2\sigma^2 + 2\sigma^4$$

This reproduces the matrix

$$\mathcal{I}(\eta) = \begin{pmatrix} \sigma^2 & 2\mu\sigma^2 \\ 2\mu\sigma^2 & 2\sigma^4 + 4\mu^2\sigma^2 \end{pmatrix}$$

already obtained by reparametrization in example 3.4.2. Applying proposition 3.4.4 to it, the Schur complement is

$$\mathcal{I}_{11 \cdot 2} = \sigma^2 - \frac{4\mu^2\sigma^4}{2\sigma^4 + 4\mu^2\sigma^2} = \sigma^2 - \frac{2\mu^2\sigma^2}{\sigma^2 + 2\mu^2} = \frac{\sigma^4}{\sigma^2 + 2\mu^2}$$

so the bound for η_1 with η_2 unknown is $(\mathcal{I}^{-1})_{11} = (\sigma^2 + 2\mu^2)/\sigma^4$ against $1/\sigma^2$ with η_2 known. The inflation factor is $1 + 2\mu^2/\sigma^2$, which is one exactly at $\mu = 0$, where the two natural coordinates become orthogonal.

Exercise 3.4.1

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown. Using (3.9), write down $\mathcal{I}_n(\mu, \sigma^2)^{-1}$ and state the information bound for an unbiased estimator of μ and for an unbiased estimator of σ^2 . Show that $\bar{X}$ attains the first of these. Then explain, citing proposition 3.4.4, why the bound for μ is the same whether or not σ^2 is known.
2. Let $\mathcal{I}$ be a positive definite 2×2 information matrix with entries $\mathcal{I}_{11} = a$, $\mathcal{I}_{12} = \mathcal{I}_{21} = b$ and $\mathcal{I}_{22} = d$. Invert it directly and confirm proposition 3.4.4(1) in this case. Show that the inflation factor $\mathcal{I}_{11}(\mathcal{I}^{-1})_{11}$ equals $1/(1 - \varrho^2)$, where $\varrho = b/\sqrt{ad}$ is the correlation between the two coordinates of the score under P_θ , and verify that the value obtained in example 3.4.6 for the Gaussian family in canonical coordinates agrees with this formula.
3. Let $\hat{\theta}$ satisfy the hypotheses of theorem 3.4.3 and fix $a \in \mathbb{R}^k$.
 1. Show that $\text{Var}_\theta(a^\top \hat{\theta}) = a^\top \mathcal{I}(\theta)^{-1} a$ holds if and only if $a^\top \hat{\theta}(X) = a^\top \theta + a^\top \mathcal{I}(\theta)^{-1} s_\theta(X)$ holds P_θ -a.s.
 2. In the Gaussian sample model with both parameters unknown, verify this identity for $a = e_1$ and $\hat{\theta}_1 = \bar{X}$.
 3. Show that no estimator of σ^2 that is unbiased for every (μ, σ^2) can attain the bound $2\sigma^4/n$ at every parameter value. (Take $a = e_2$ in part (a), and compare the resulting formula at two values of μ with the same σ^2 .)

4. Let $\{P_\eta\}$ be a full-rank canonical exponential family with natural statistic T and cumulant function A , and let $X_1, \dots, X_n$ be an i.i.d. sample. Reparametrize by the mean $m = \nabla A(\eta)$ of the natural statistic, which is a diffeomorphism of $\text{int } \Xi$ onto its image (strict convexity of A under full rank makes ∇A injective, and its Jacobian $\nabla^2 A = \text{Cov}_\eta(T)$ is invertible there). Show that the information matrix in the m -parametrization is $n \text{Cov}_\eta(T)^{-1}$, so that the bound for an unbiased estimator of m is $\text{Cov}_\eta(T)/n$, and show that $\bar{T}_n = n^{-1} \sum_{j=1}^n T(X_j)$ attains it. Identify m and $\bar{T}_n$ explicitly for the Gaussian family of example 2.4.6(4).
5. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poi}(\lambda)$ and, independently, $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Poi}(\nu)$, with $\lambda, \nu > 0$ both unknown. Compute $\mathcal{I}(\lambda, \nu)$ for the combined data and check that it is diagonal. Now reparametrize by (ρ, ν) with $\rho = \lambda/\nu$, the ratio being the parameter of interest. Compute $\mathcal{I}(\rho, \nu)$ using (3.8), show that it is not diagonal, and use proposition 3.4.4 to show that the bound for an unbiased estimator of ρ is $\rho(\rho + 1)/(n\nu)$ when ν is unknown against $\rho/(n\nu)$ when it is known.
6. Let $\mathcal{U}$ be the class of estimators of θ that are unbiased at every $\theta \in \Theta$ and satisfy the hypotheses of theorem 3.4.3.
 1. Show that if $\hat{\theta} \in \mathcal{U}$ and its i -th coordinate is efficient in the sense of definition 3.4.5, then $\text{Var}_\theta(\hat{\theta}_i) \leq \text{Var}_\theta(\tilde{\theta}_i)$ for every $\tilde{\theta} \in \mathcal{U}$ and every θ .
 2. Deduce from exercise 3.4.1 (3)(c) that in the Gaussian model with both parameters unknown, no member of $\mathcal{U}$ has an efficient second coordinate. Contrast this with the fact that a UMVU estimator (definition 3.3.1) of σ^2 does exist, and say which of the two properties, efficiency or uniformly minimum variance unbiasedness, is the stronger.

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